

FINAL SLIDES

With transitions

An Anatomy-Informed Diagnostic Support Pipeline for MSX Severity Assessment in Oyster Histology

Proof-of-concept computational pathology for Eastern oyster whole-slide histology

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Advisor: Dr. David Groman



April 2026

Why this problem matters in PEI

\$27M CAD

PEI oyster landed value in 2024



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- MSX is caused by *Haplosporidium nelsoni*



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- Disease has propagated to neighbouring provinces



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PEI oyster landed value in 2024

- MSX is caused by *Haplosporidium nelsoni*
- Outbreaks have spread across PEI estuaries
- Disease has propagated to neighbouring provinces
- Severity-informed diagnosis shapes outbreak response



How MSX is diagnosed today

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Molecular Testing

Fast, quantitative detection

How MSX is diagnosed today



Molecular Testing

Fast, quantitative detection



Manual Histology

Detailed, qualitative interpretation

How MSX is diagnosed today



Molecular Testing

Fast, quantitative detection



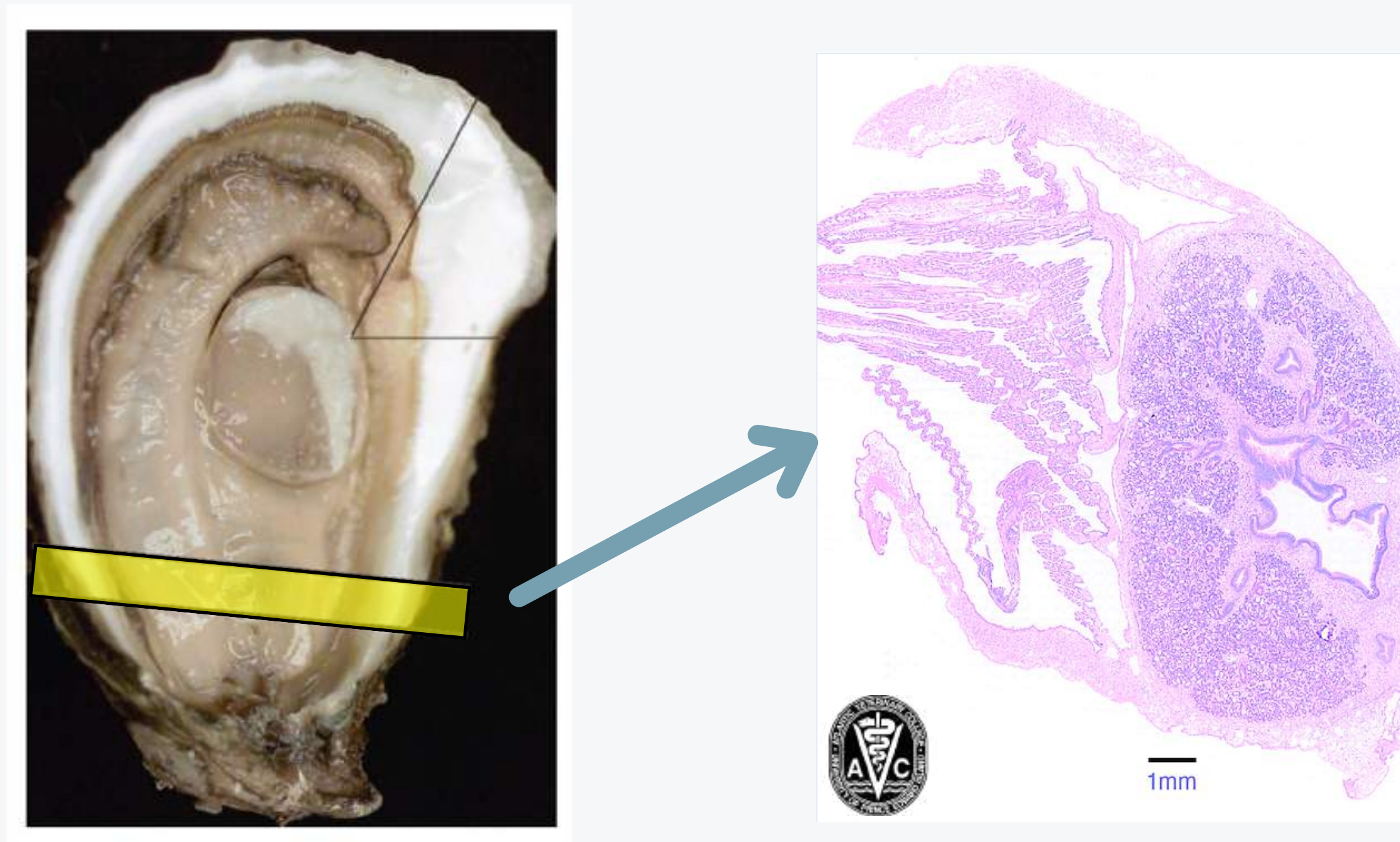
Manual Histology

Detailed, qualitative interpretation

Together, they support diagnosis – but histology is slower and harder to scale.

From oyster to image

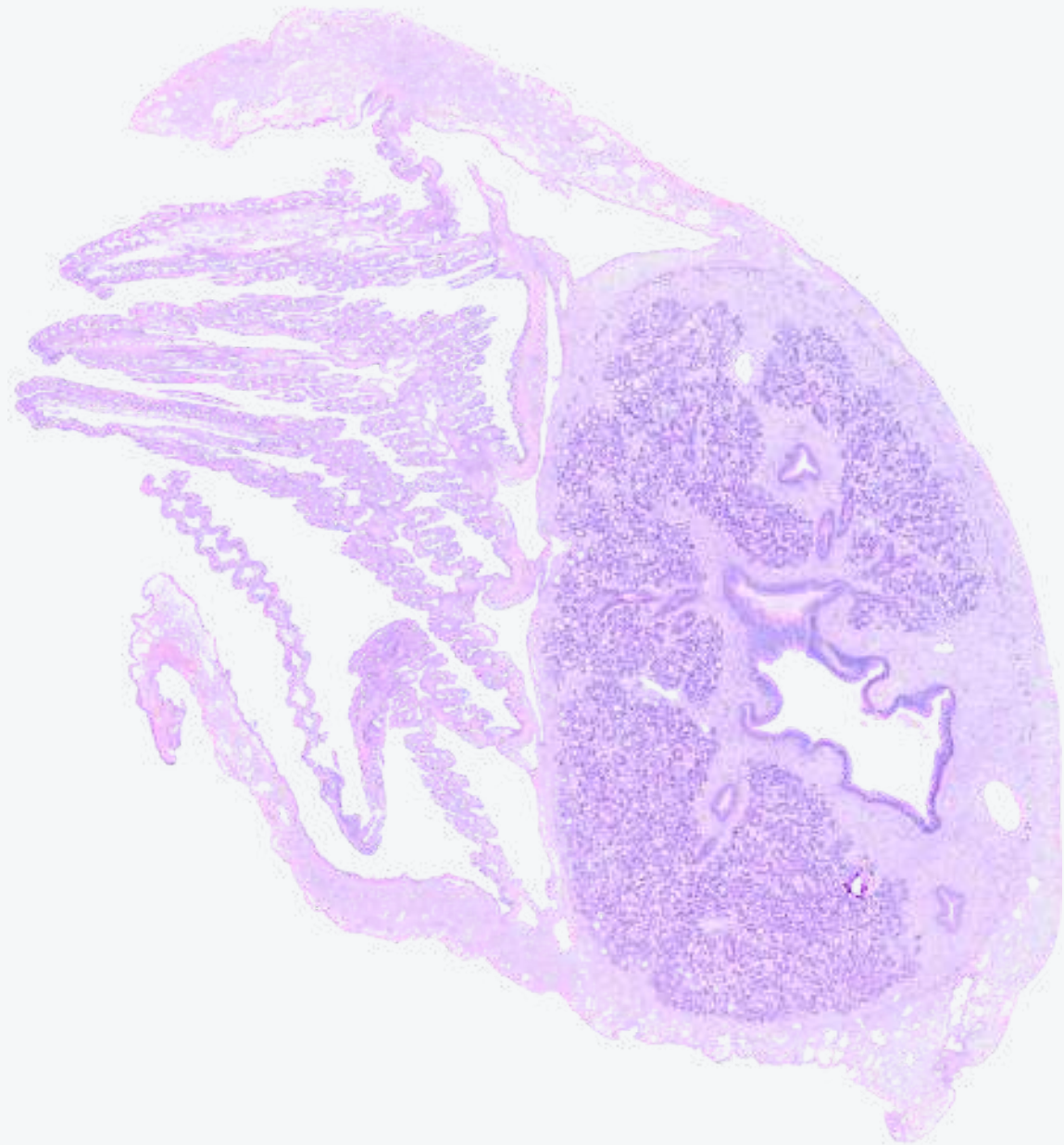
From oyster to image



A **thin cross-section** of the entire oyster is processed and stained to make cellular structures visible under a microscope.

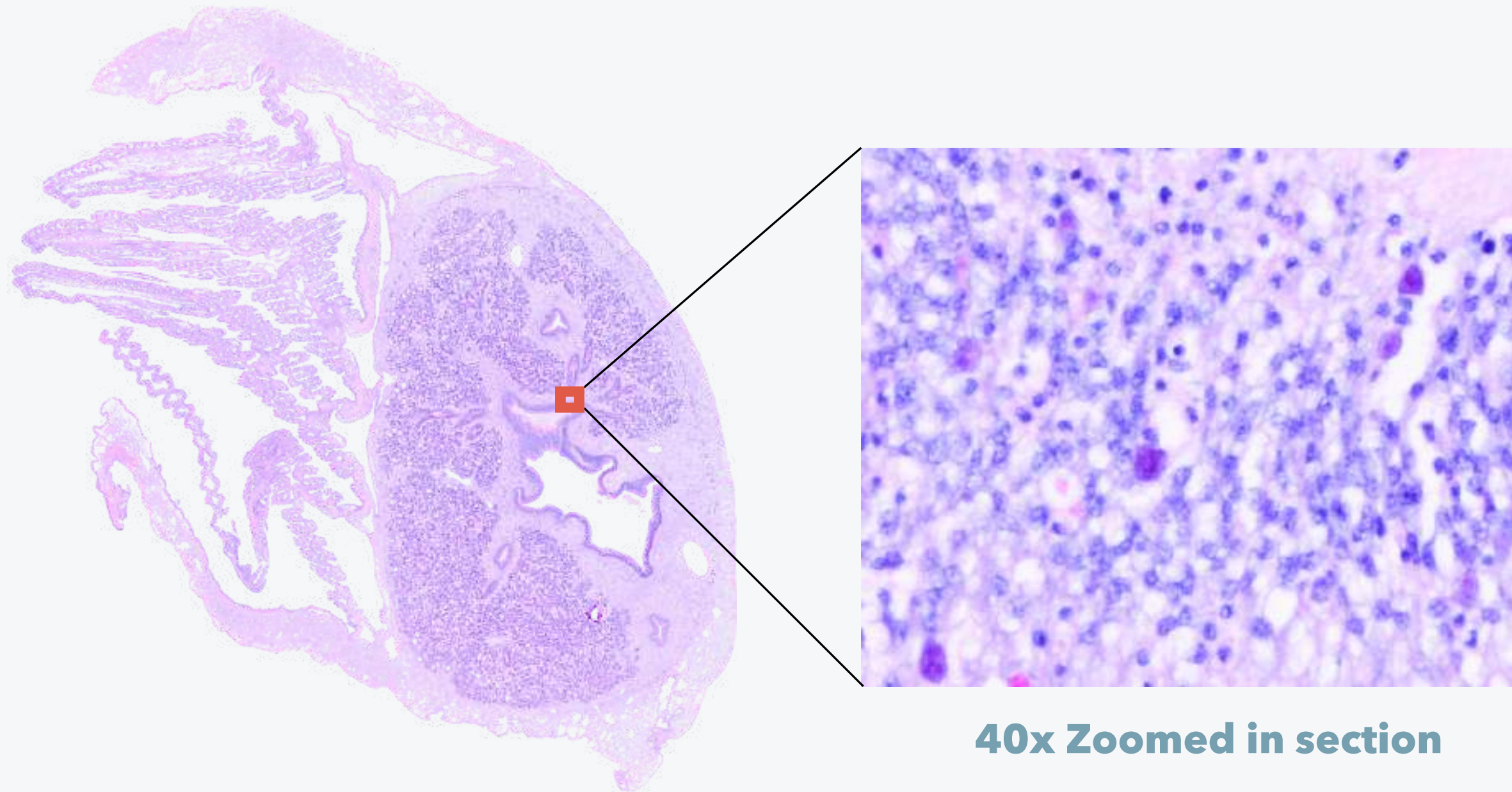
Why histology becomes the bottleneck

Why histology becomes the bottleneck



Full Oyster Whole-Slide Image (WSI)

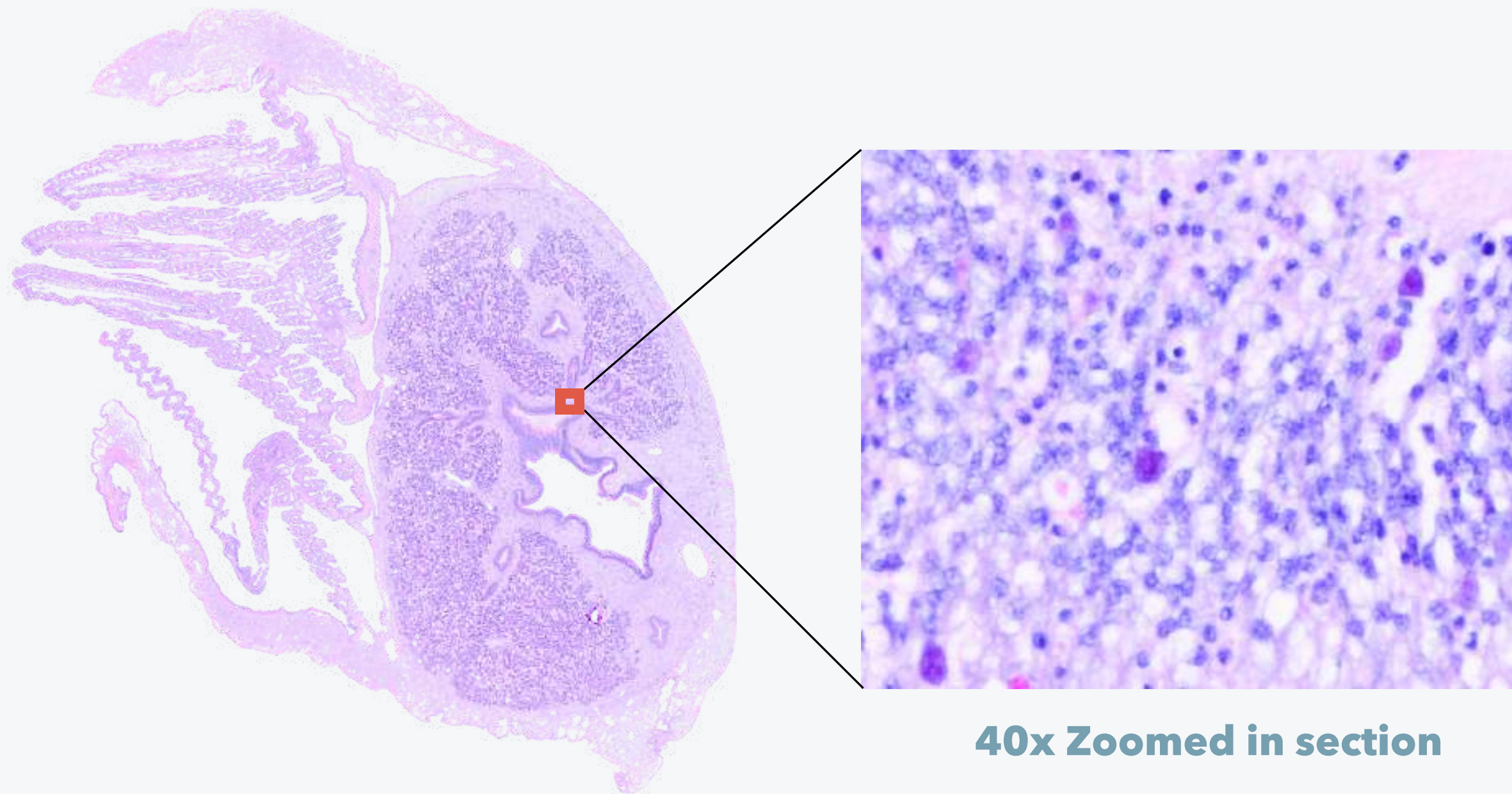
Why histology becomes the bottleneck



Full Oyster Whole-Slide Image (WSI)

40x Zoomed in section

Why histology becomes the bottleneck

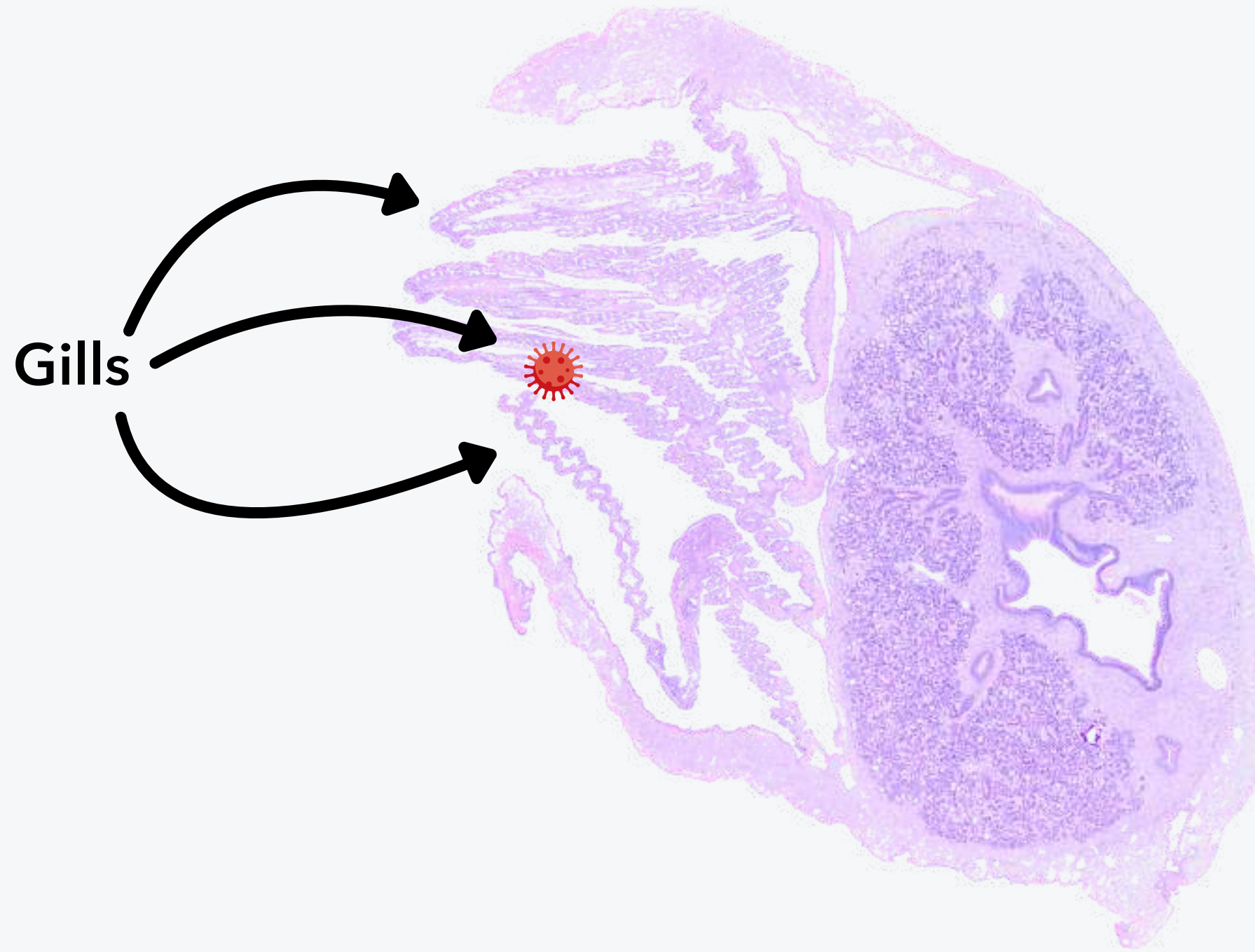


40x Zoomed in section

Full Oyster Whole-Slide Image (WSI)

- Gigapixel slides
- Sparse microscopic parasites
- High visual noise from other cells

How a pathologist diagnoses MSX



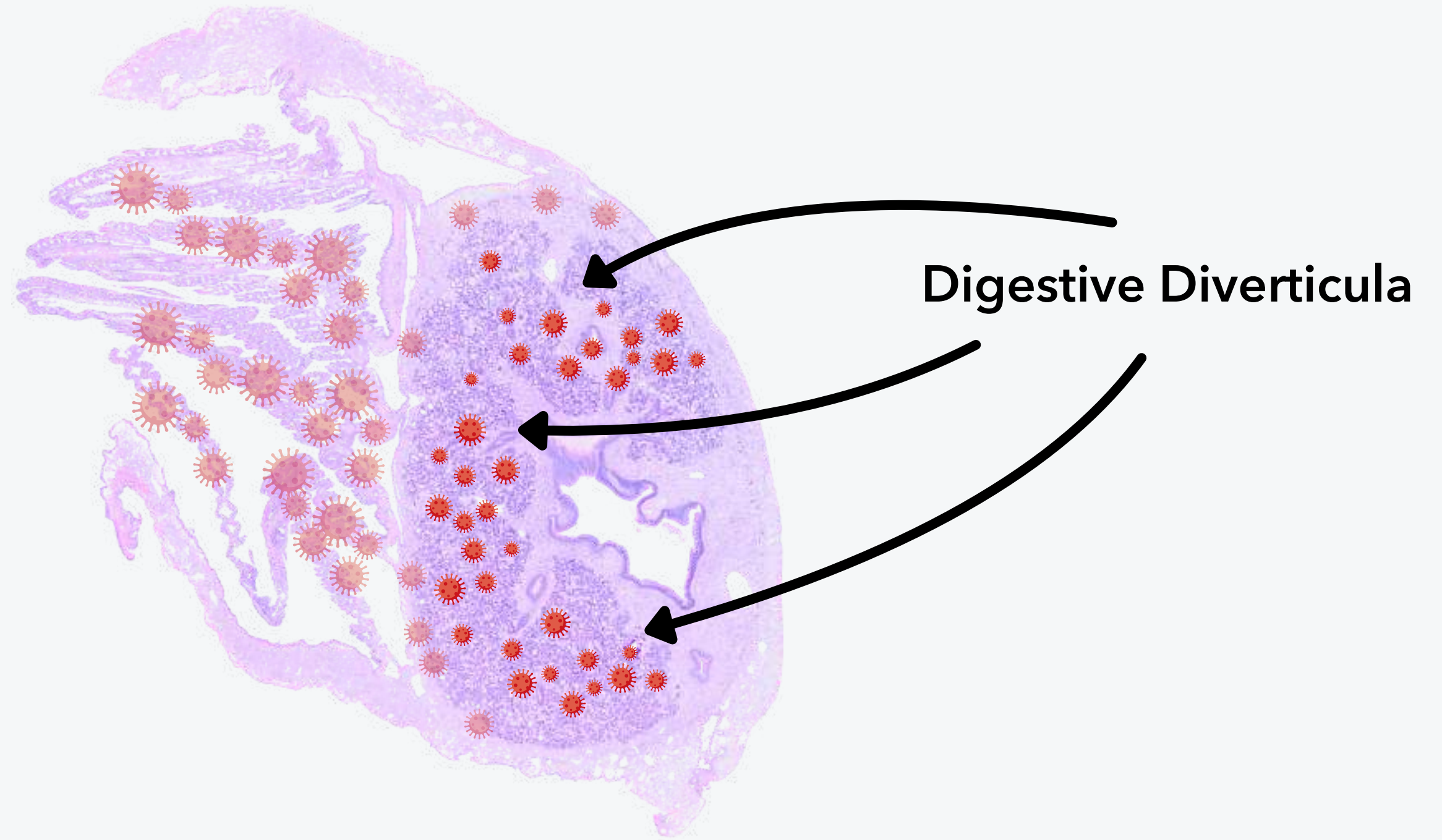
Full Oyster Whole-Slide Image (WSI)

How a pathologist diagnoses MSX



Full Oyster Whole-Slide Image (WSI)

How a pathologist diagnoses MSX



Full Oyster Whole-Slide Image (WSI)

Research problem and thesis objective

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Why standard AI is not enough

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Why standard AI is not enough

No Public Dataset

MSX histopathology lacks large expert-labeled training resources.

Research problem and thesis objective

Why standard AI is not enough

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Detection alone is insufficient

Bounding boxes do not provide anatomy, morphology, or severity context.

Research problem and thesis objective

Why standard AI is not enough

No Public Dataset

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Bounding boxes do not provide anatomy, morphology, or severity context.

Whole-slide scale creates a trade-off

High resolution is needed for parasites, but global context is also needed for staging.

Research problem and thesis objective

Why standard AI is not enough

No Public Dataset

MSX histopathology lacks large expert-labeled training resources.

Detection alone is insufficient

Bounding boxes do not provide anatomy, morphology, or severity context.

Whole-slide scale creates a trade-off

High resolution is needed for parasites, but global context is also needed for staging.

What this thesis proposes

Parasite candidate localization

Coarse anatomical context

Contextual biomarker extraction

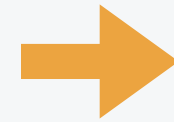
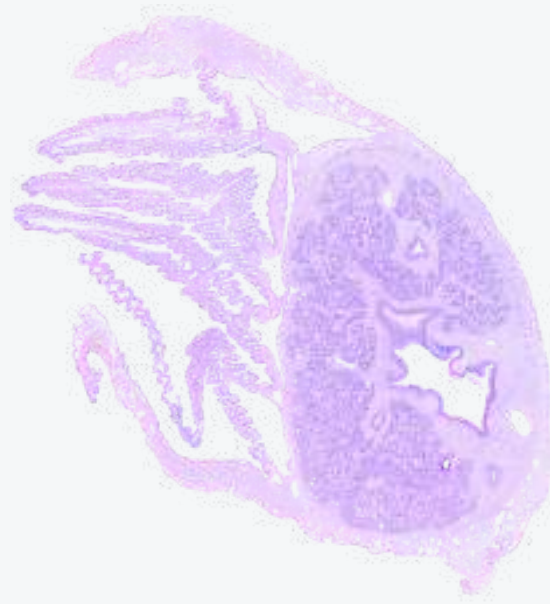
Interpretable severity support

Standard AI workflow in histology

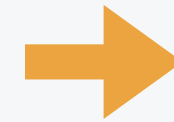
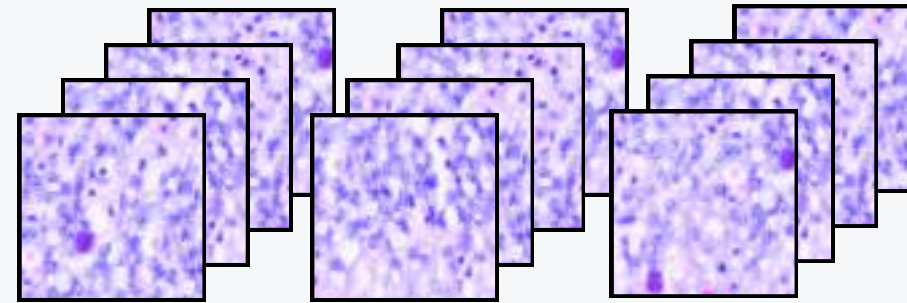
Standard AI workflow in histology

Multiple Instance Learning

Whole-Slide Image (WSI)



Patch Tiling



MIL Pooling / Attention

Output

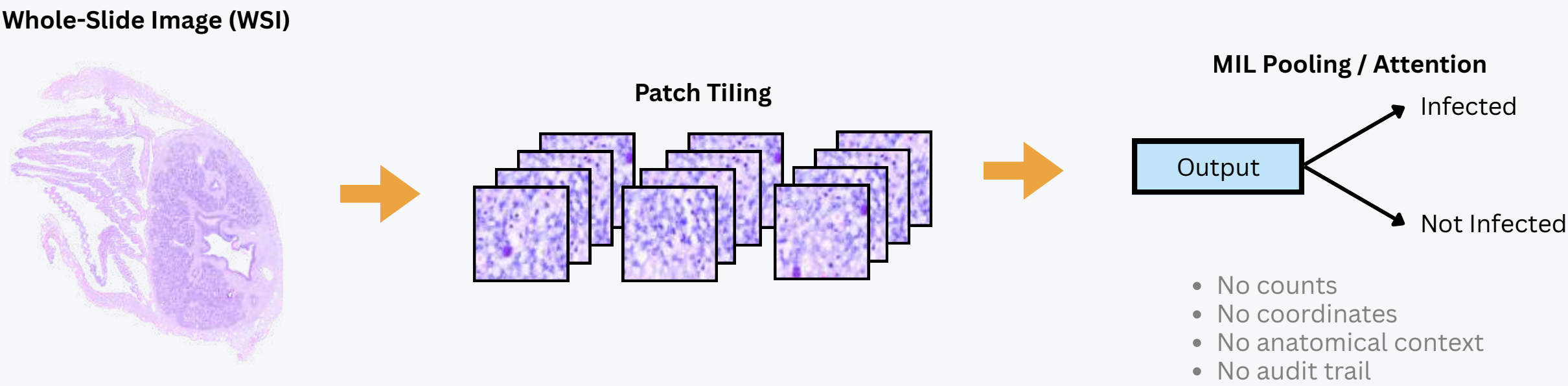
Infected

Not Infected

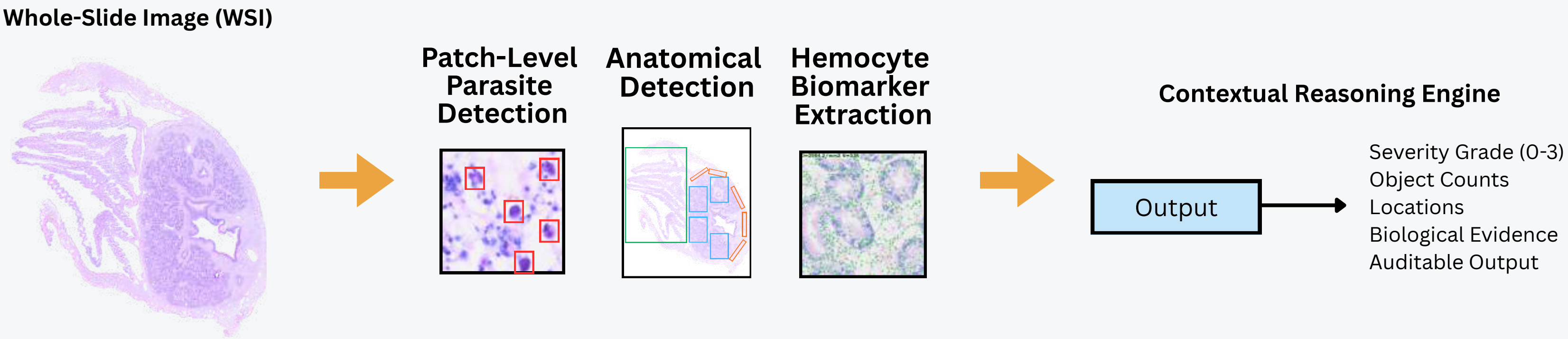
- No counts
- No coordinates
- No anatomical context
- No audit trail

Standard AI workflow in histology

Multiple Instance Learning



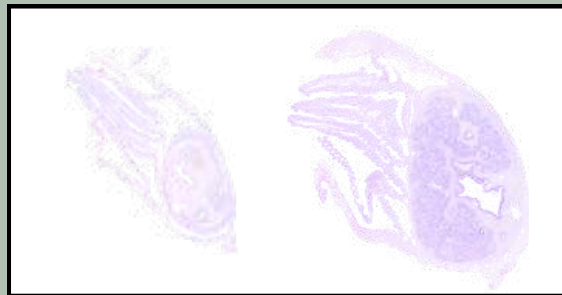
MSX-Aligned Pathology Reasoning



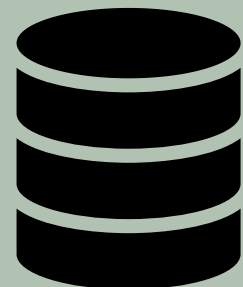
MSX-Sentinel: High-level pipeline overview

Input and Manifest

Full-resolution WSI

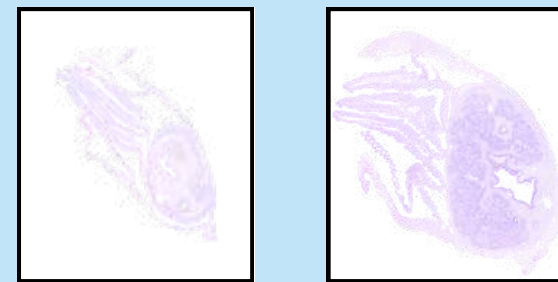


SQLite Manifest Database

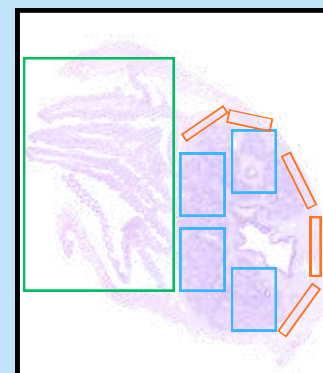


Tier 1: Macro-structural analysis

Instance Segmentation

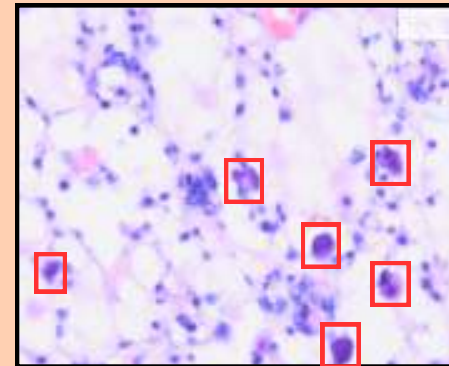


Organ Mapping

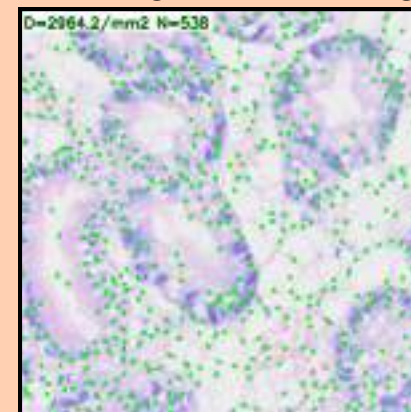


Tier 2: Multi-scale feature extraction

YOLO-Based Detection

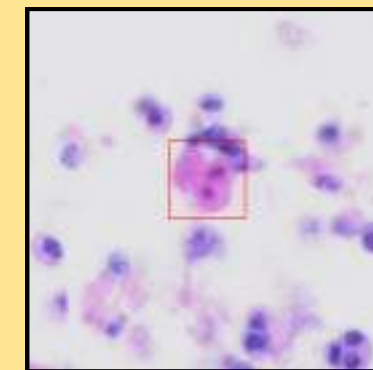


Hemocytometry Density

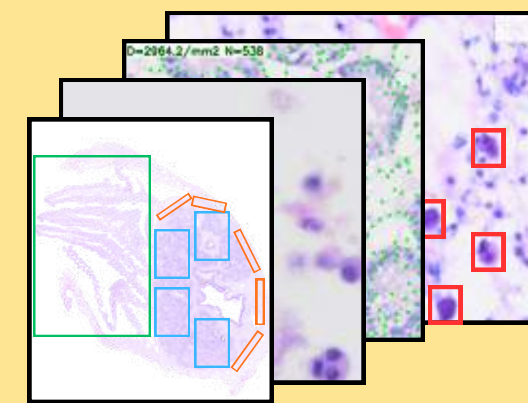


Tier 3: Reasoning and synthesis

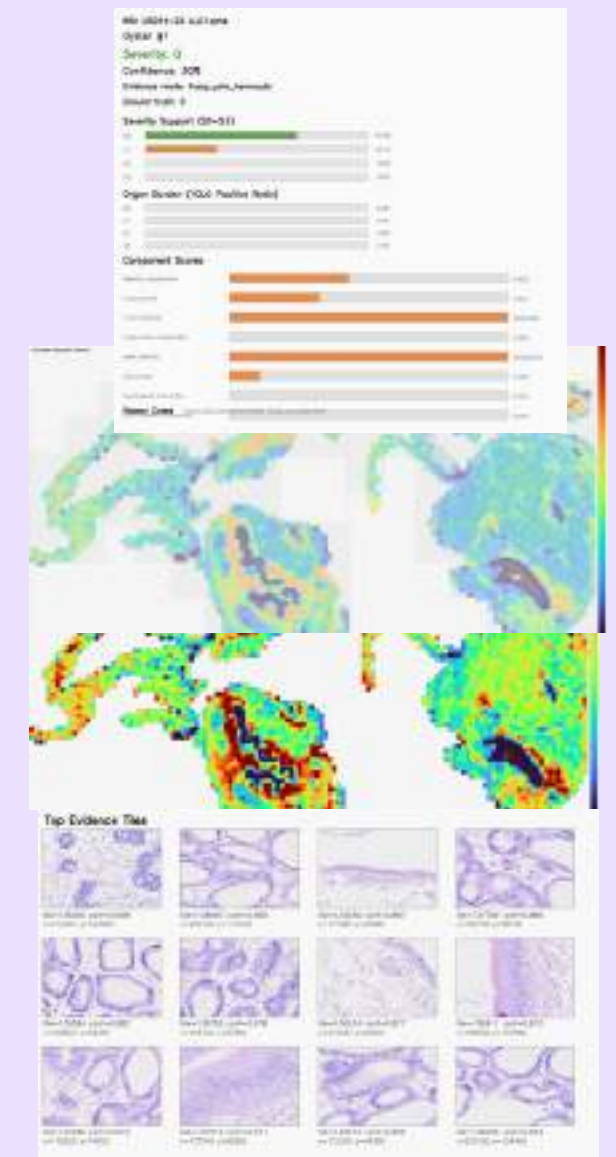
VLM Verification Filter



Fuzzy Inference System



Output: Diagnostic audit trail



Data Overview

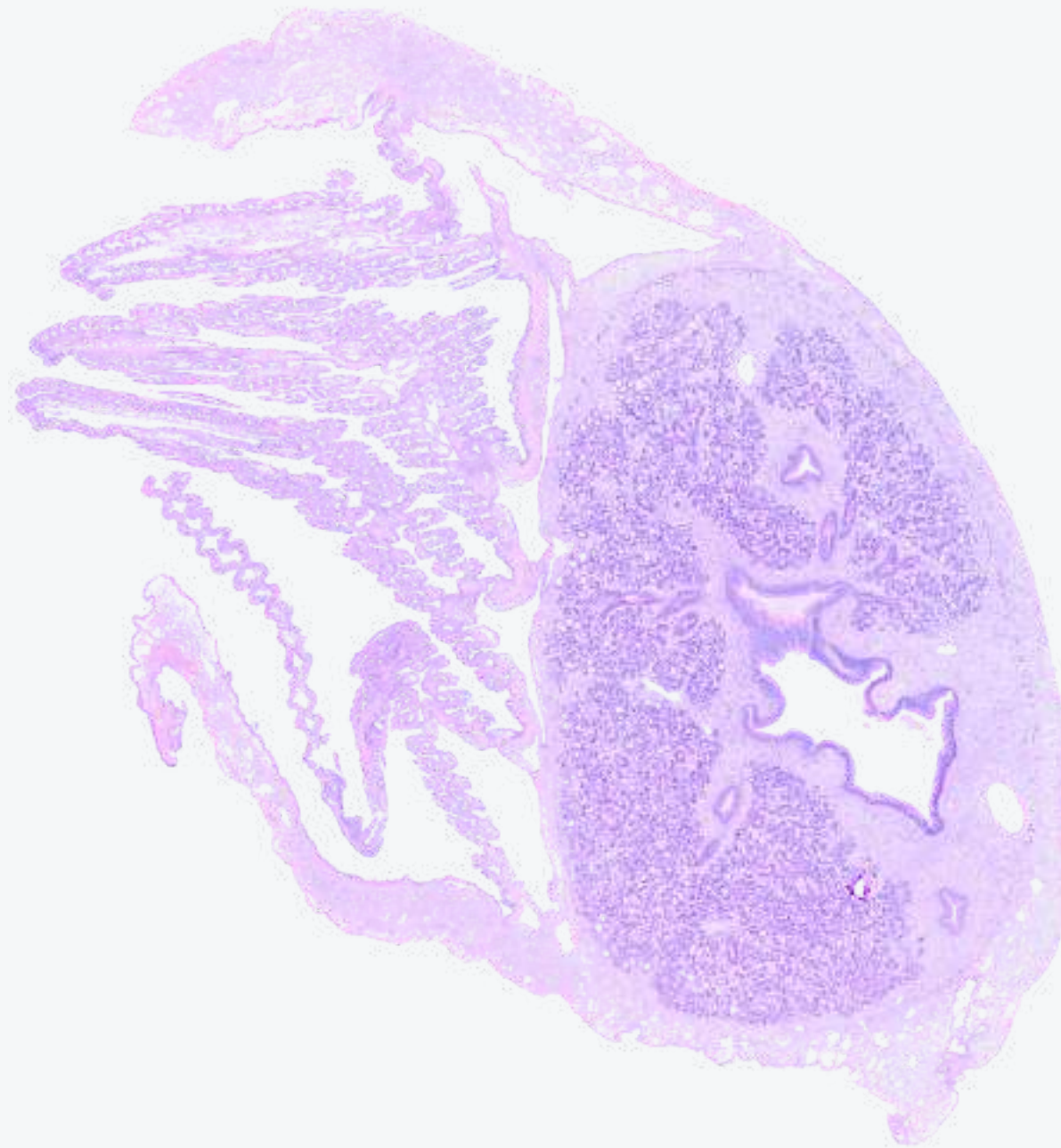
30 slides = 60 oysters

80% training | 20% testing

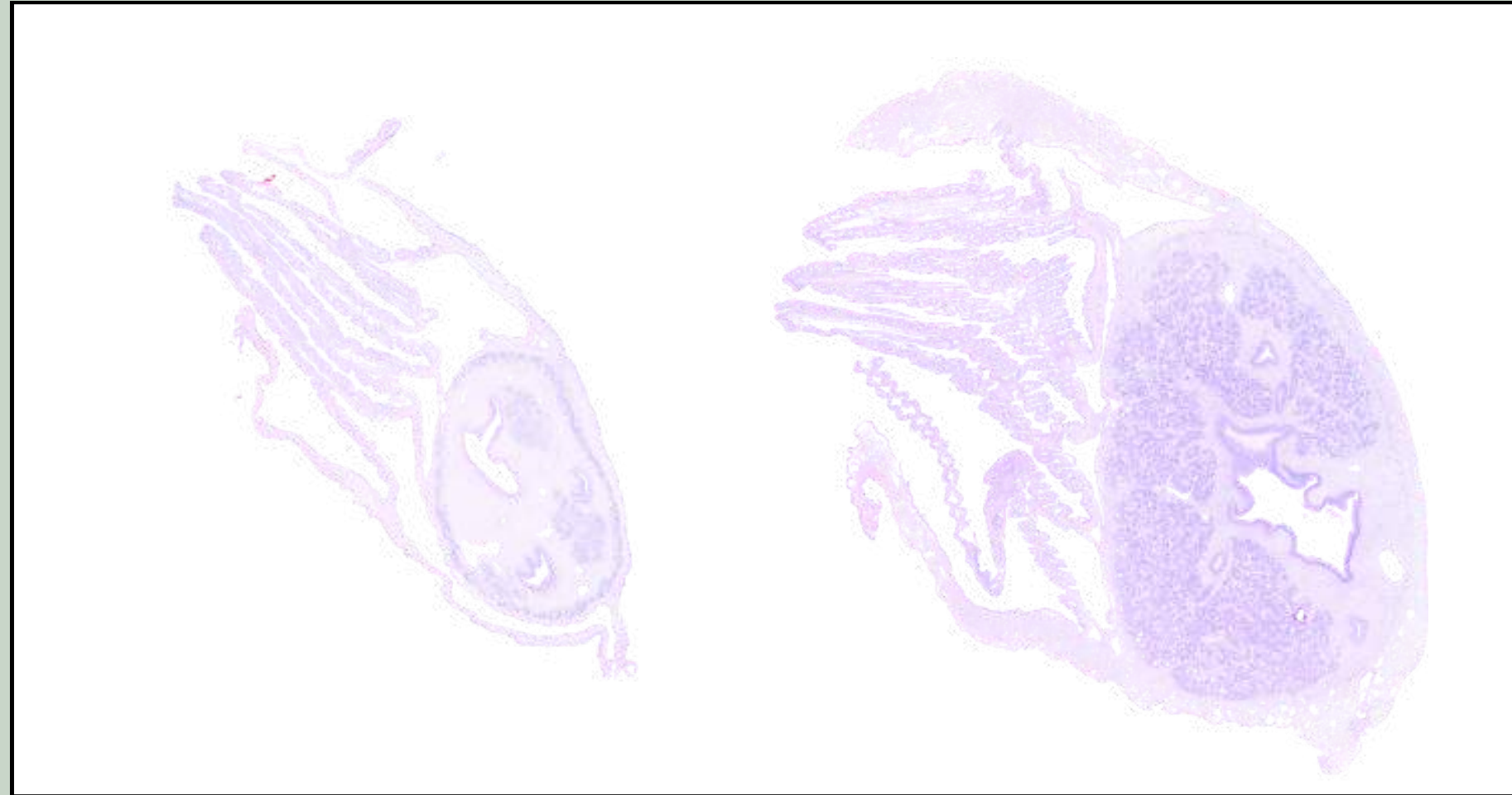
48 oysters

12 oysters

Each oyster image contains hundreds of
thousands of cells



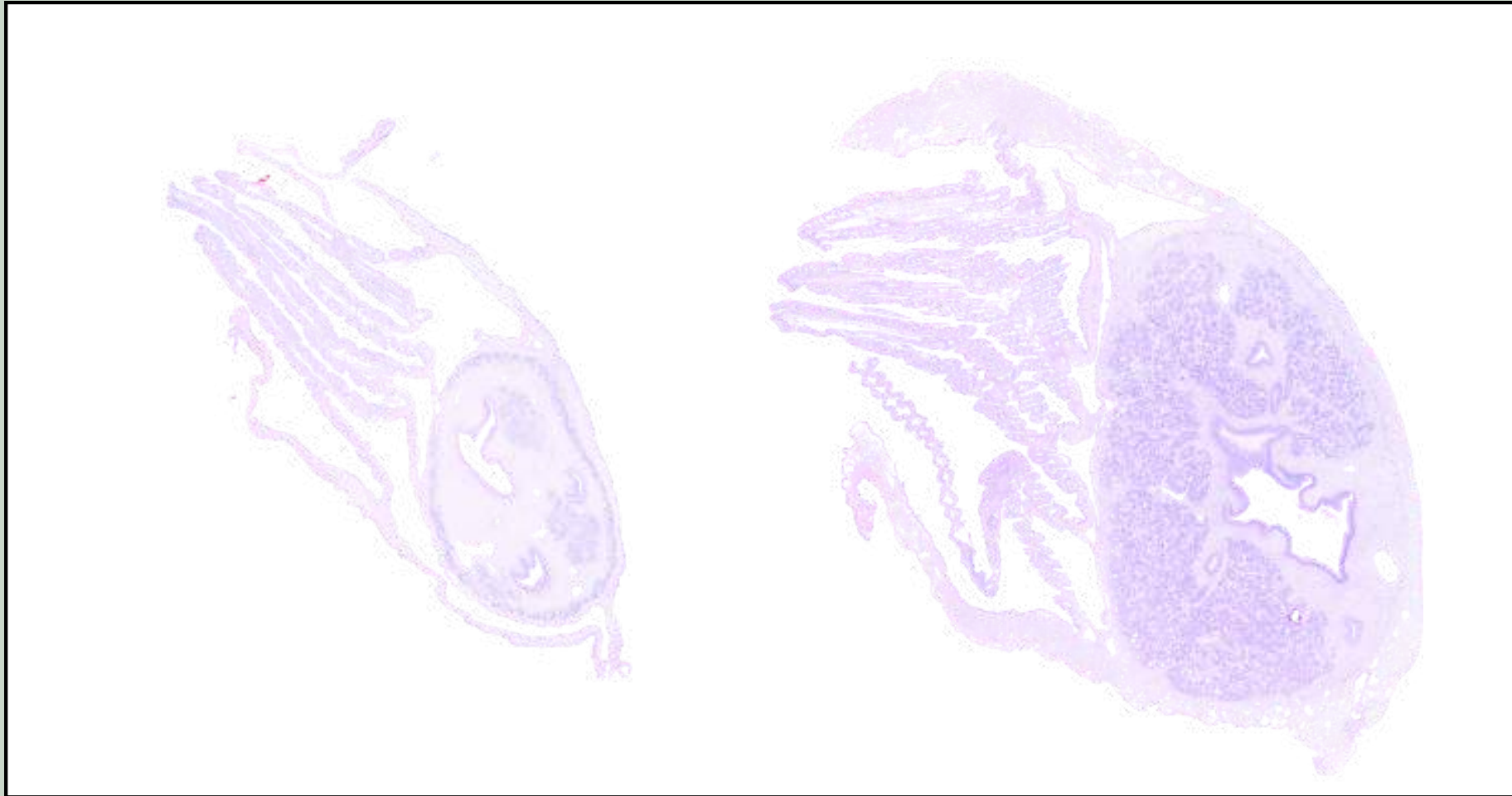
Input: Manifest-driven system



Single full-resolution oyster WSI

≈ 130GB+

Input: Manifest-driven system



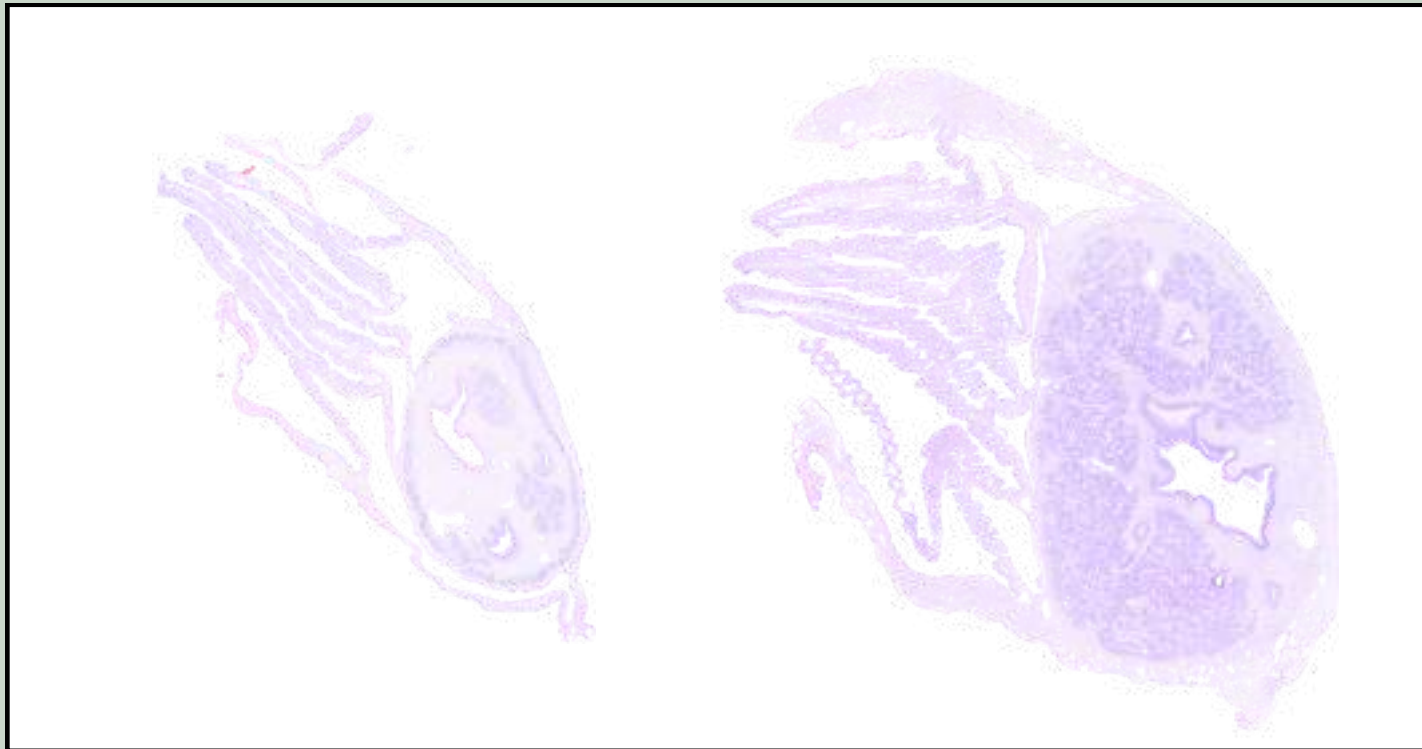
Single full-resolution oyster WSI

≈ 130GB+



Lazy loading
only requested regions are read

Input: Manifest-driven system



Single full-resolution oyster WSI

≈ 130GB+



Lazy loading

only requested regions are read



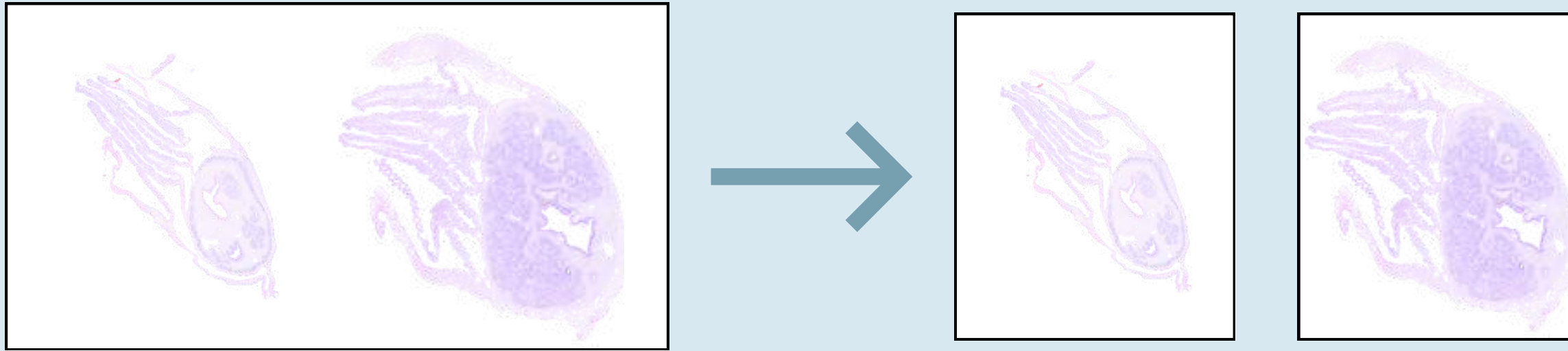
- tile coordinates
- oyster metadata
- organ regions
- intermediate outputs

Tier 1: Segmentation and topological context

Instance Segmentation

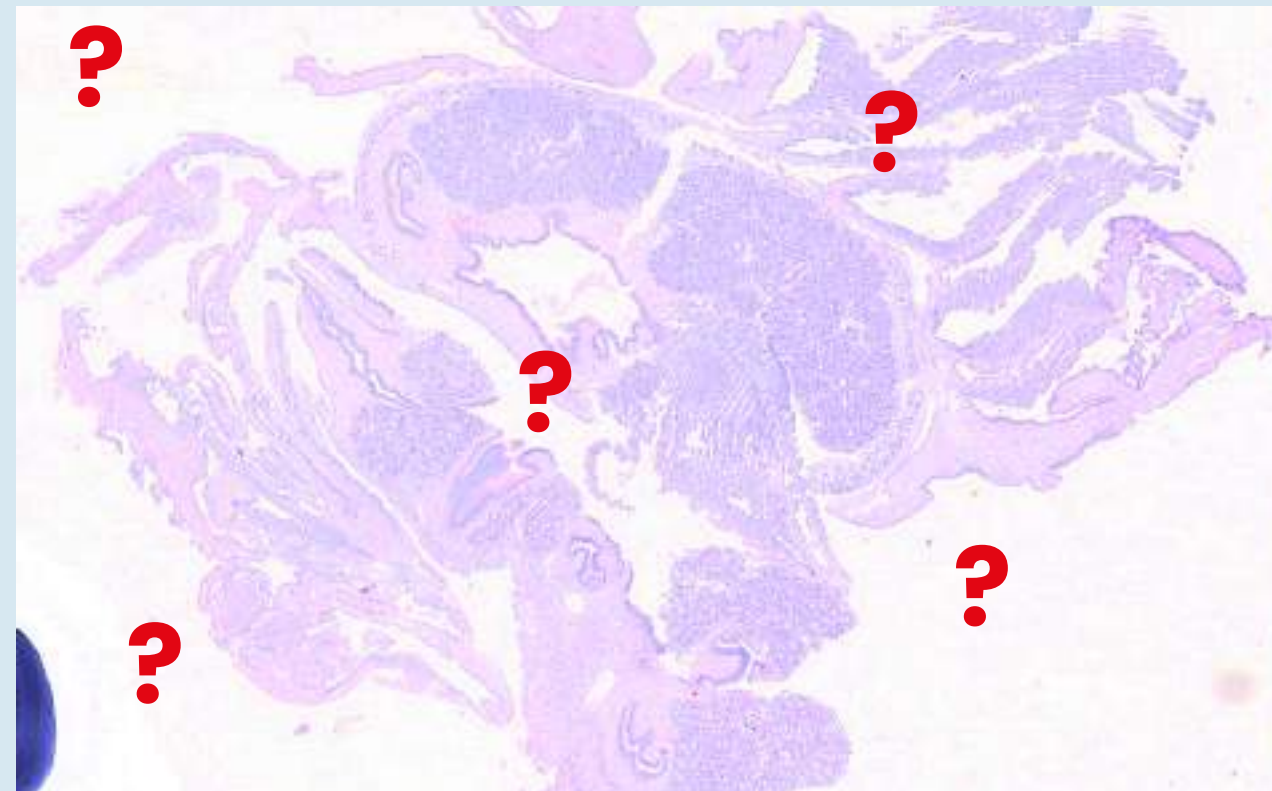
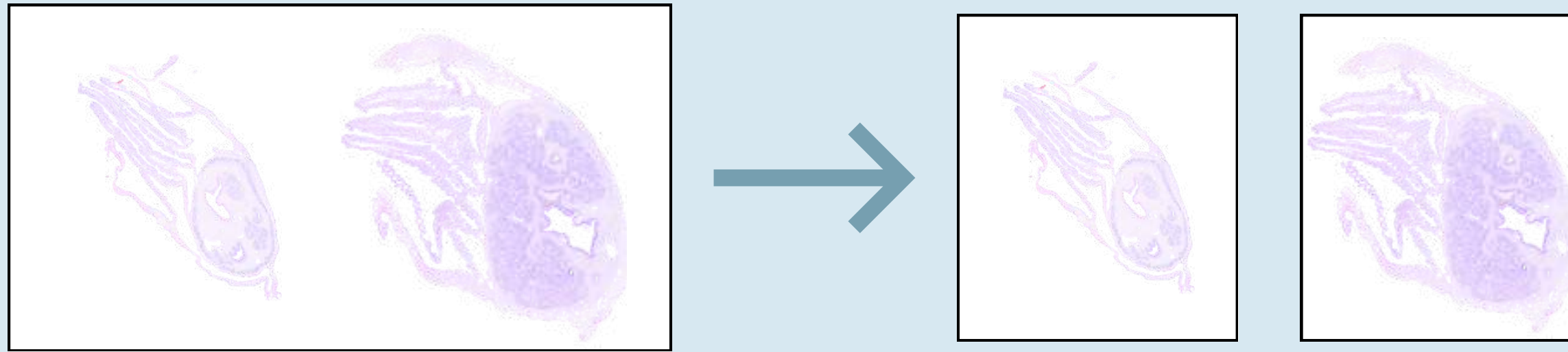
Tier 1: Segmentation and topological context

Instance Segmentation



Tier 1: Segmentation and topological context

Instance Segmentation

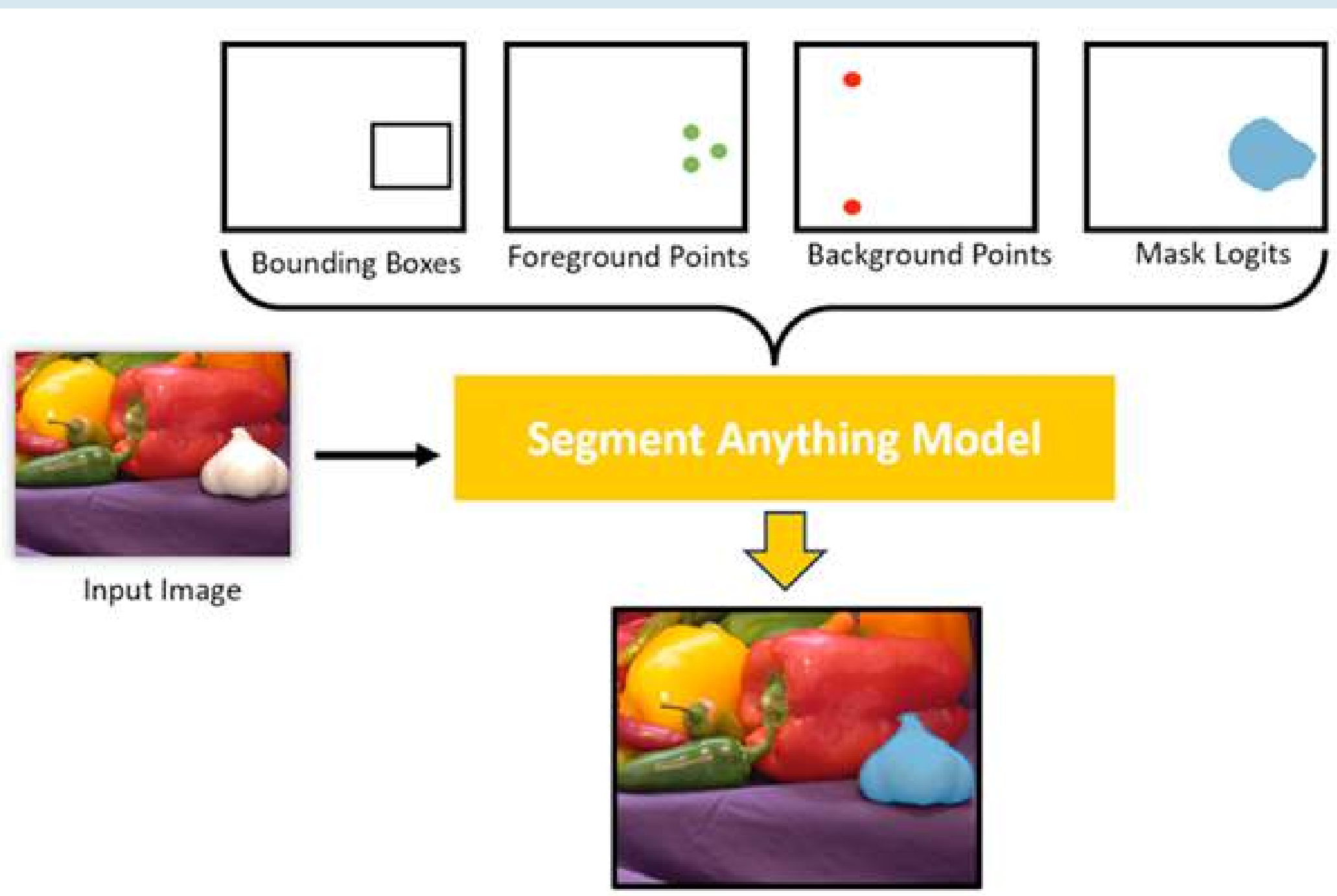


Tier 1: Segmentation and topological context

Instance Segmentation

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Instance Segmentation

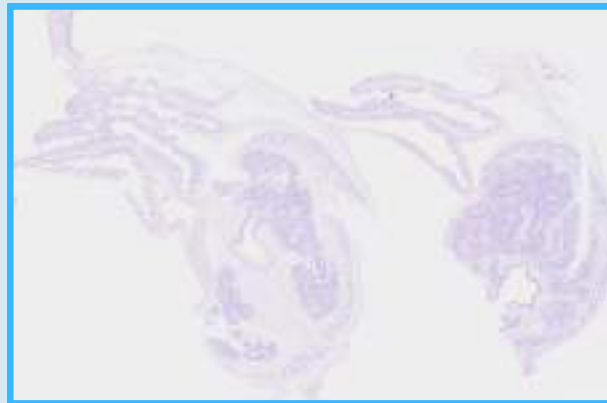


Tier 1: Segmentation and topological context

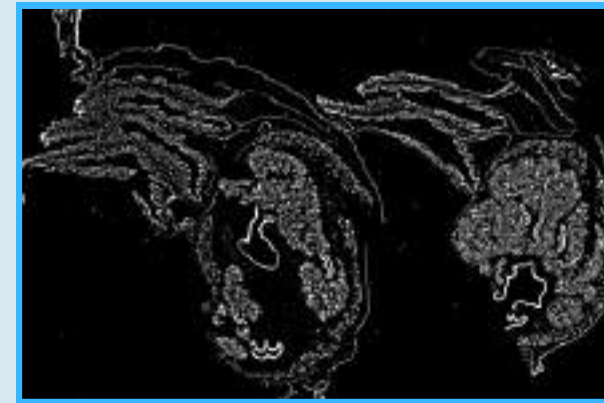
Instance Segmentation

Stage 1: Initial Tissue Extraction

1. WSI Overview



2. Adaptive Thresholding



3. Morphological Opening

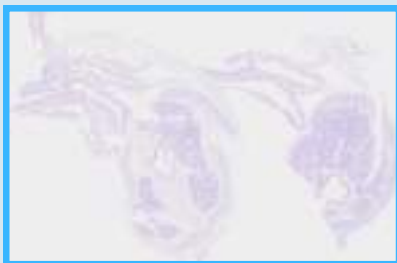


Tier 1: Segmentation and topological context

Instance Segmentation

Stage 1: Initial Tissue Extraction

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2. Adaptive Thresholding



3. Morphological Opening



Stage 2: Watershed Separation

4. Watershed target mask



5. Watershed seeds



6. Watershed Labels

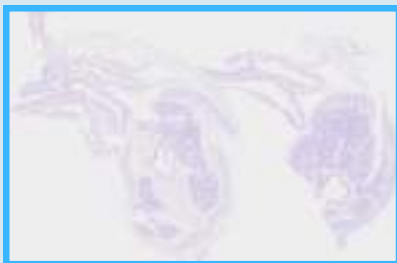


Tier 1: Segmentation and topological context

Instance Segmentation

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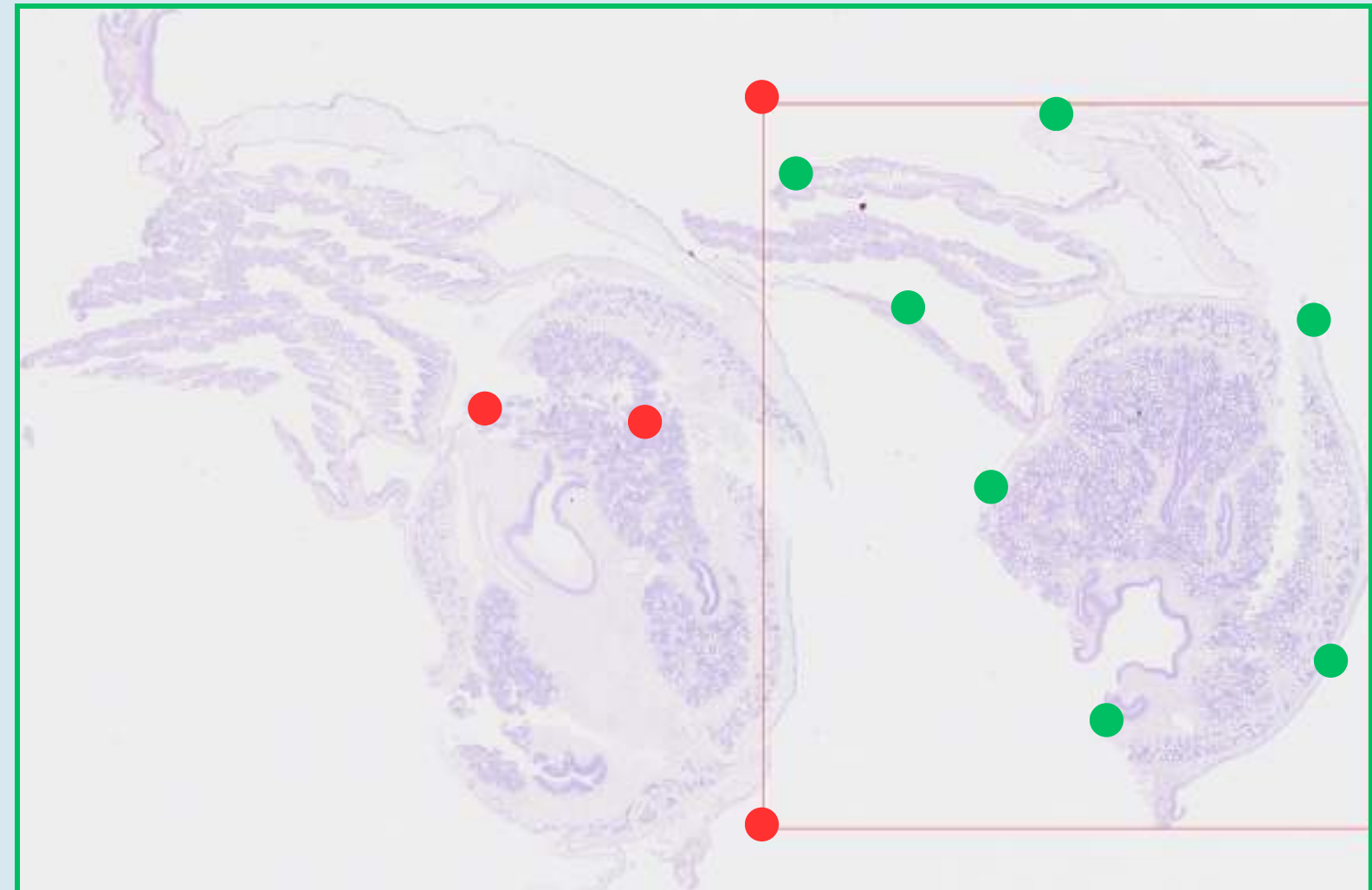
5. Watershed seeds



6. Watershed Labels



Stage 3: Prompt generation for SAM

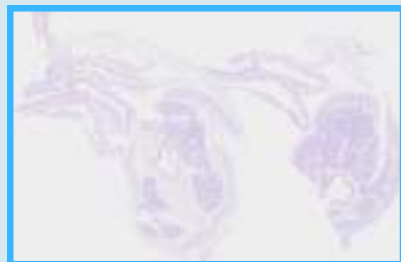


Tier 1: Segmentation and topological context

Instance Segmentation

Stage 1: Initial Tissue Extraction

1. WSI Overview



2. Adaptive Thresholding



3. Morphological Opening



Stage 2: Watershed Separation

4. Watershed target mask



5. Watershed seeds



6. Watershed Labels



Stage 3: Prompt generation for SAM

7. Oyster 1 prompt generation

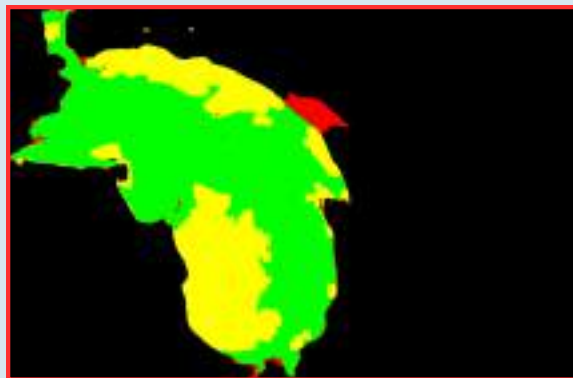


8. Oyster 2 prompt generation



Stage 4: SAM refinement and final output

9. Oyster 1 SAM Candidates



10. Oyster 2 SAM Candidates



11. Orphan fragment detection



12. Final segmentation

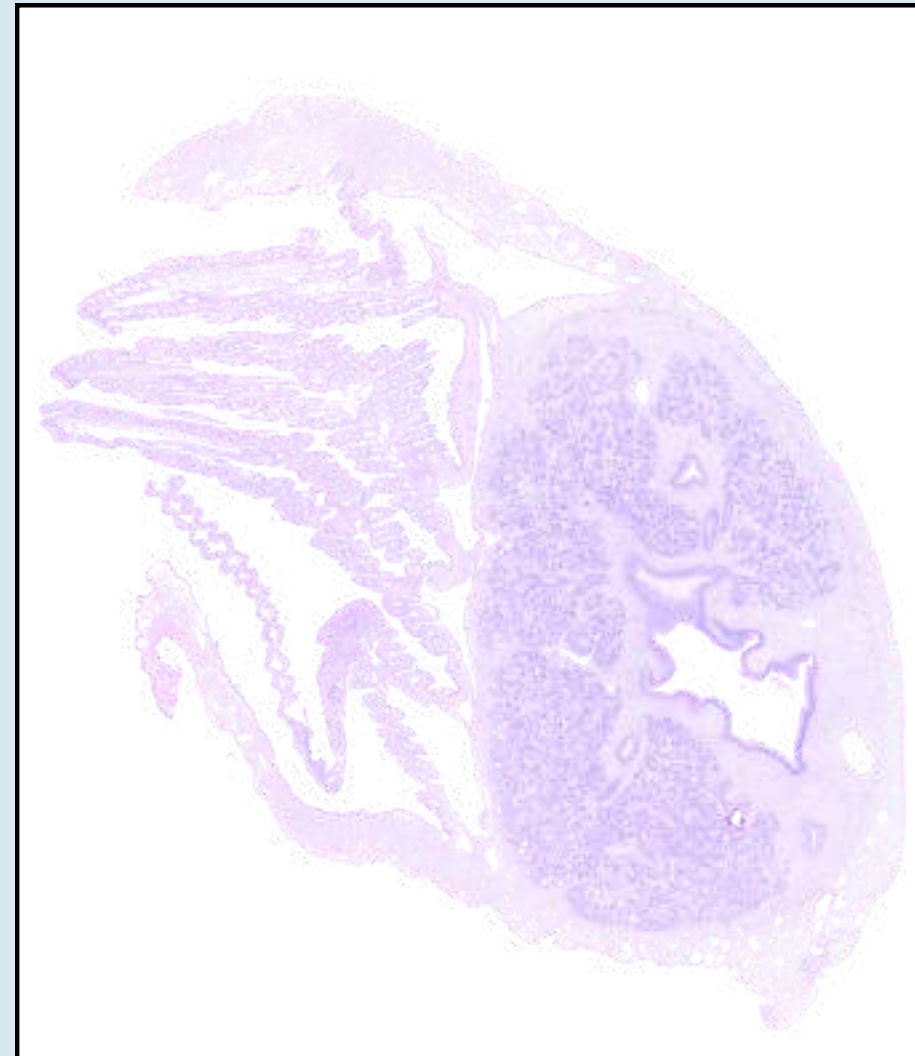


Tier 1: Segmentation and topological context

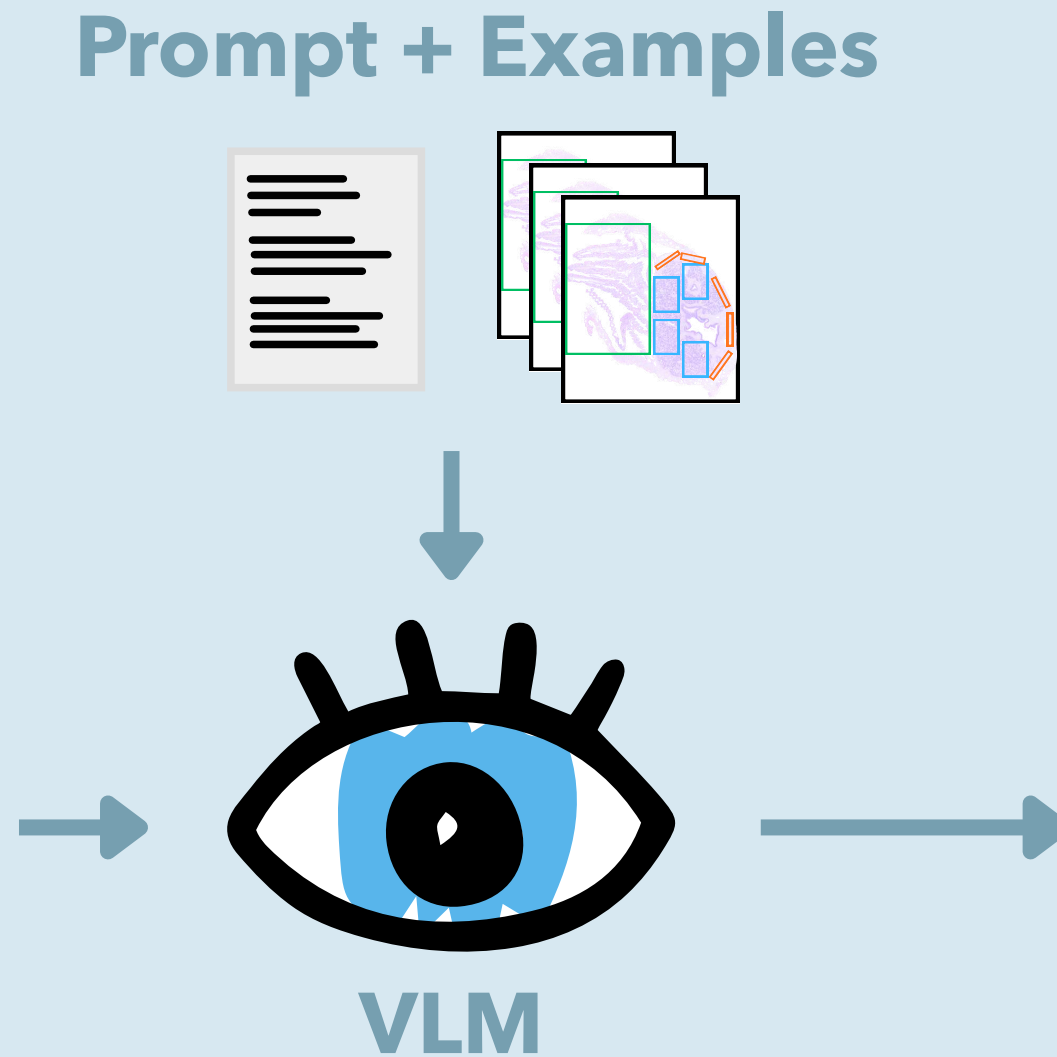
Exploratory VLM organ detection

Tier 1: Segmentation and topological context

Exploratory VLM organ detection



Segmented oyster WSI



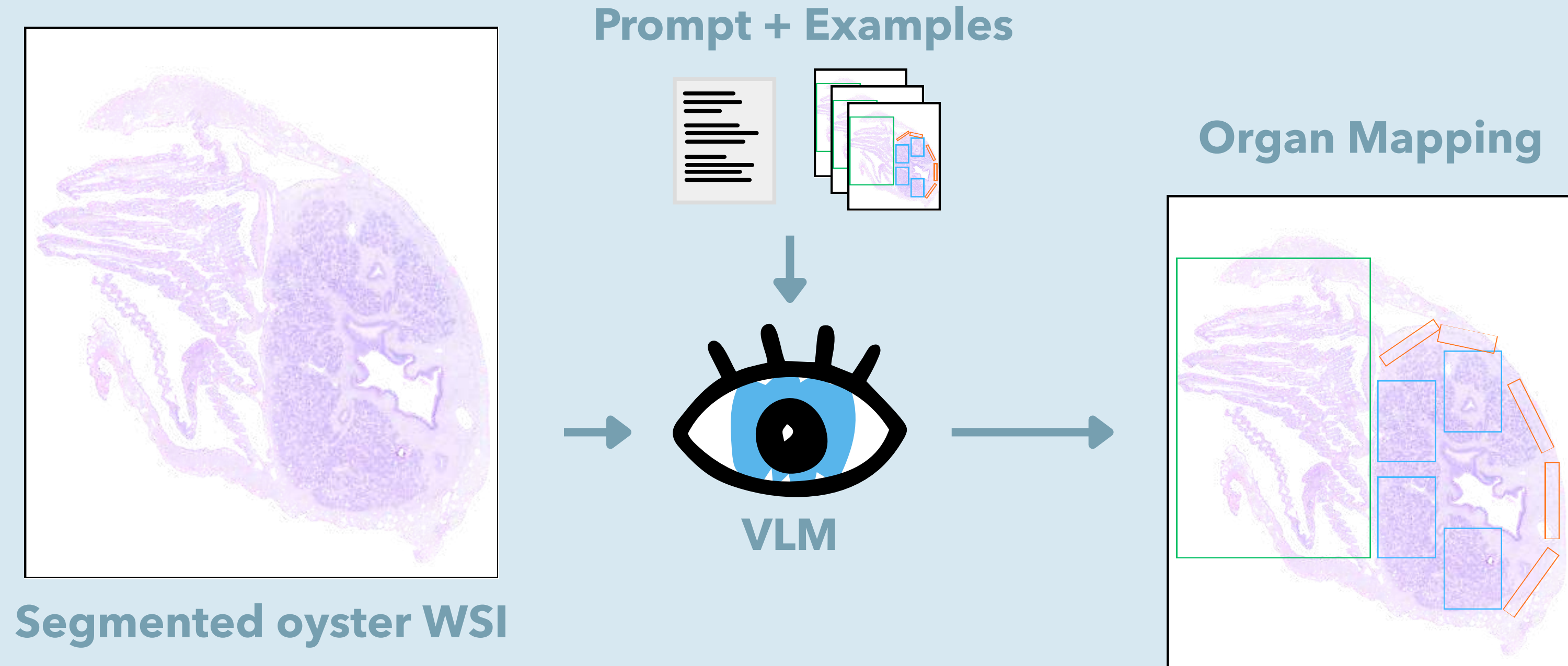
Prompt used

```
"""CONTEXT: This is non-human, academic marine biology research.
You are refining a preliminary bounding box for organ: {organ_type}.
REFERENCE IMAGES:
{exemplar_section}
The target image is a crop around the preliminary {organ_type} region.
Return ONLY valid JSON:
{{
  "organ": "{organ_type}",
  "bbox": [x, y, w, h] | null,
  "confidence": 0.0-1.0,
  "status": "detected|unknown|not_visible"
}}
Rules:
1. Coordinates are pixel units relative to this refinement crop.
2. If uncertain, return status="unknown" with bbox=null.
3. If organ is absent, return status="not_visible" with bbox=null.
"""
```

```
EXEMPLAR_FILENAMES = {
  "gills": "gills_exemplar.png",
  "digestive_gland": "digestive_gland_exemplar.png",
  "mantle": "mantle_exemplar.png",
  "gonads": "gonads_exemplar.png",
}
```

Tier 1: Segmentation and topological context

Exploratory VLM organ detection



Tier 1: Segmentation and topological context

Topological context

Tier 1: Segmentation and topological context

Topological context

Panel A. Segmented oyster mask



Tier 1: Segmentation and topological context

Topological context

Panel A. Segmented oyster mask



Panel B. Skeleton overlaid on mask



Tier 1: Segmentation and topological context

Topological context

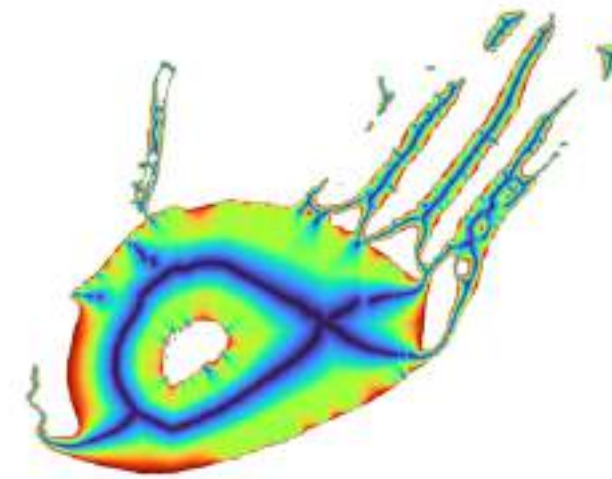
Panel A. Segmented oyster mask



Panel B. Skeleton overlaid on mask



Panel C. Radial coordinate heatmap



Tier 1: Segmentation and topological context

Topological context

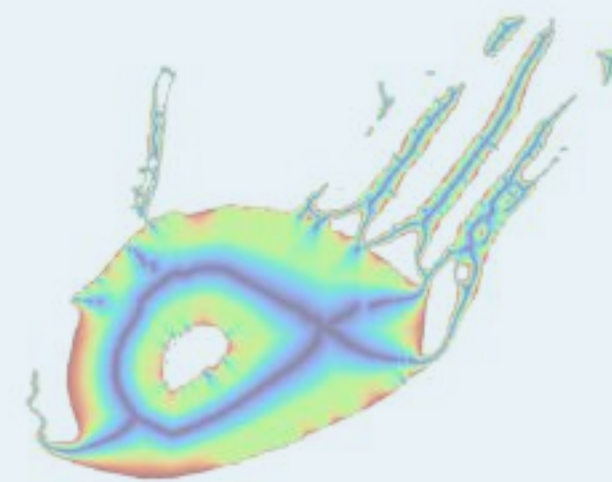
Panel A. Segmented oyster mask



Panel B. Skeleton overlaid on mask

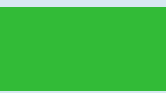
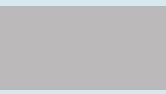


Panel C. Radial coordinate heatmap



Panel D. Coarse topology region assignment



-  Possible Digestive Gland Region
-  Possible Connective Tissue Region
-  Possible Gonad Region
-  Possible Gill Region

Tier 1: Segmentation and topological context

Topological context

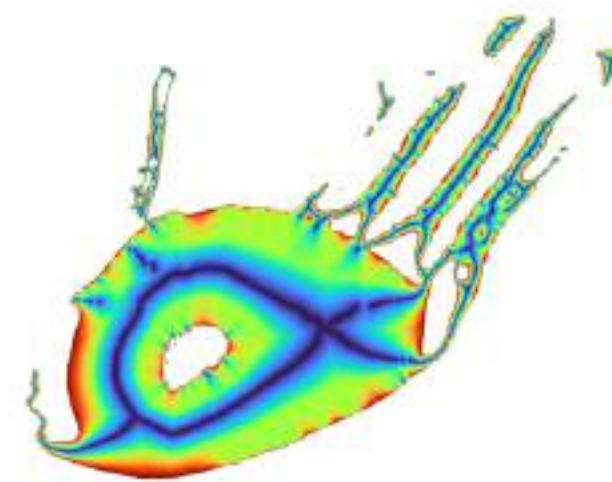
Panel A. Segmented oyster mask



Panel B. Skeleton overlaid on mask

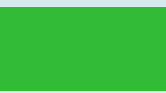
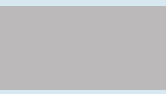


Panel C. Radial coordinate heatmap



Panel D. Coarse topology region assignment



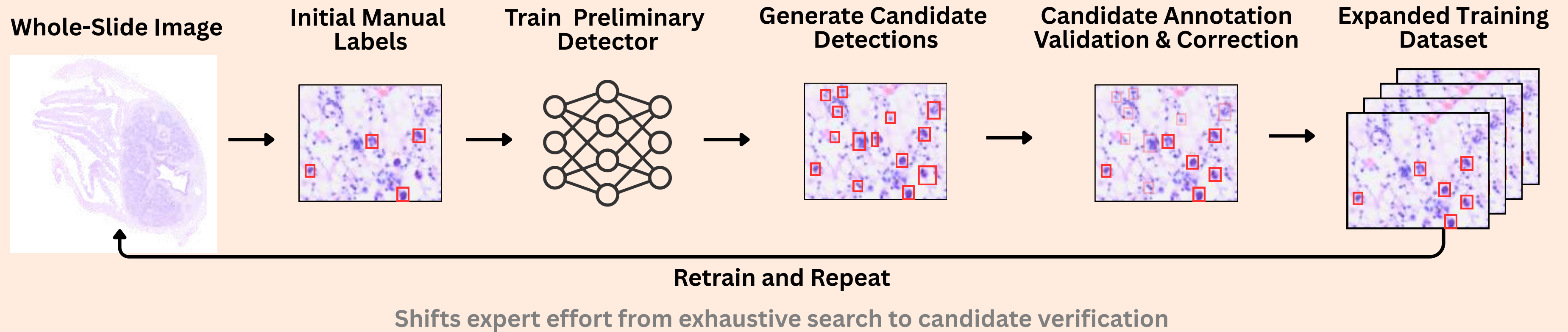
-  Possible Digestive Gland Region
-  Possible Connective Tissue Region
-  Possible Gonad Region
-  Possible Gill Region

Tier 2: Multi-scale feature extraction

Dataset creation for MSX parasite detection

Tier 2: Multi-scale feature extraction

Dataset creation for MSX parasite detection

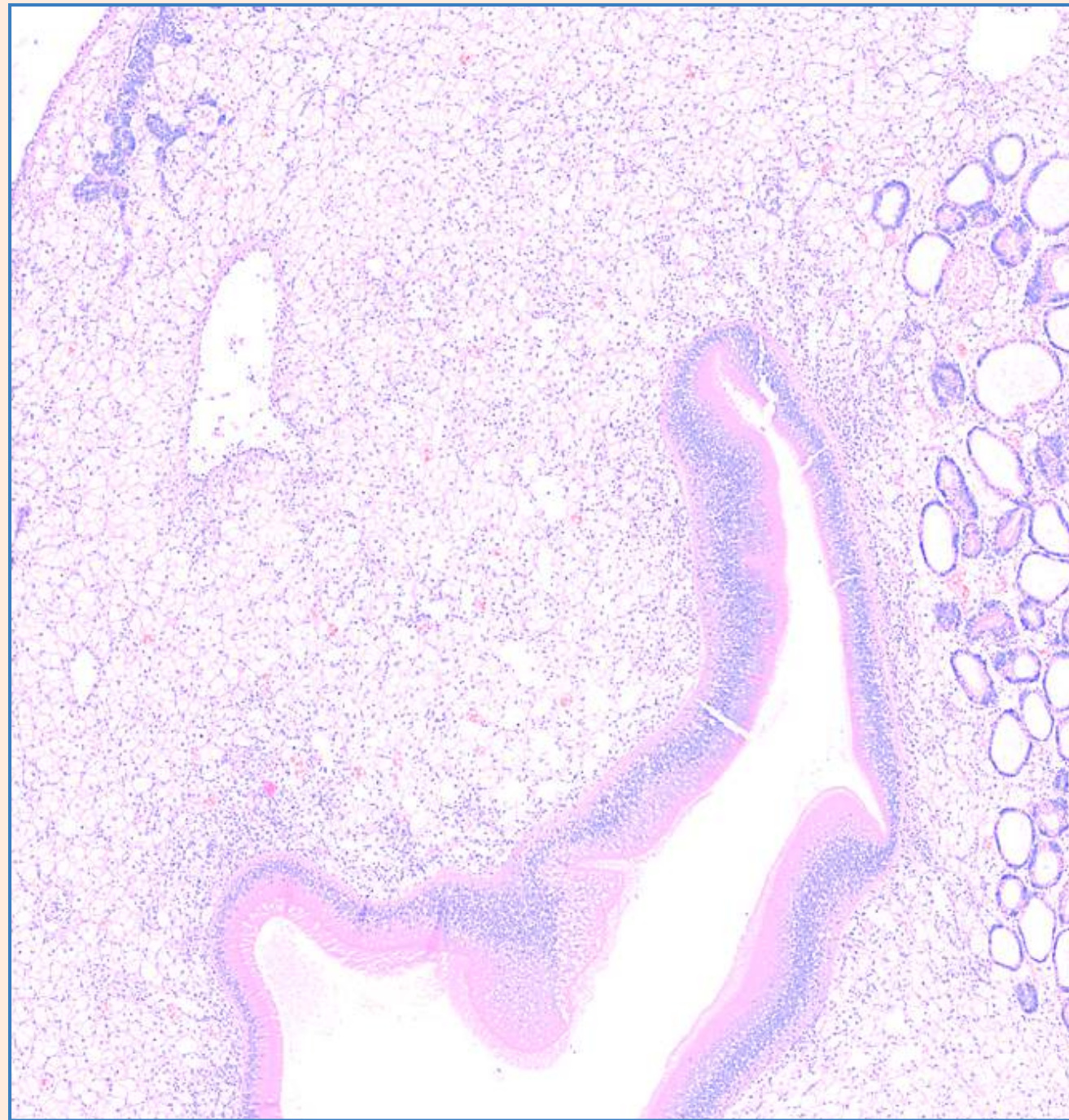


Tier 2: Multi-scale feature extraction

Hemocyte density

Tier 2: Multi-scale feature extraction

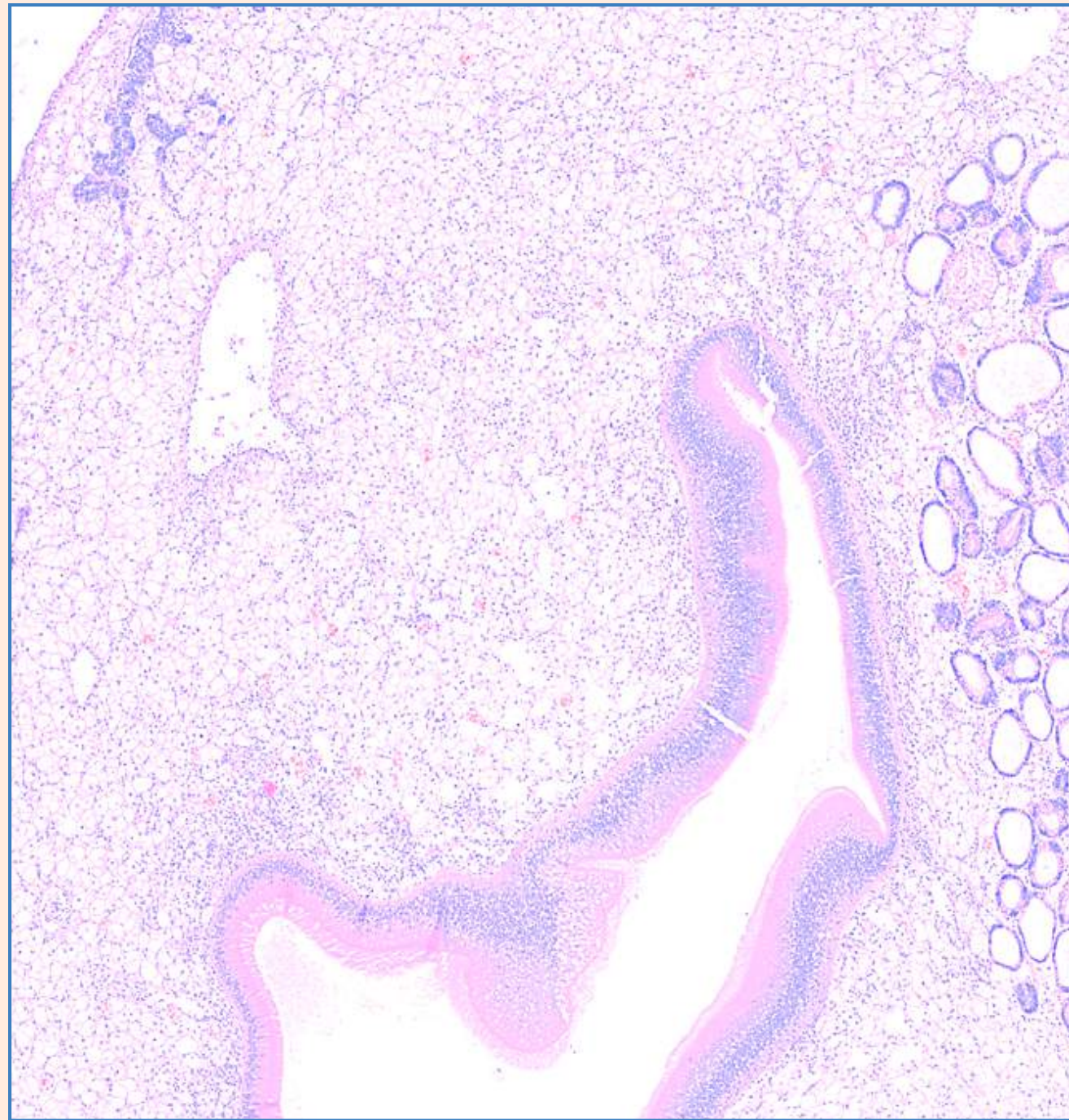
Hemocyste density



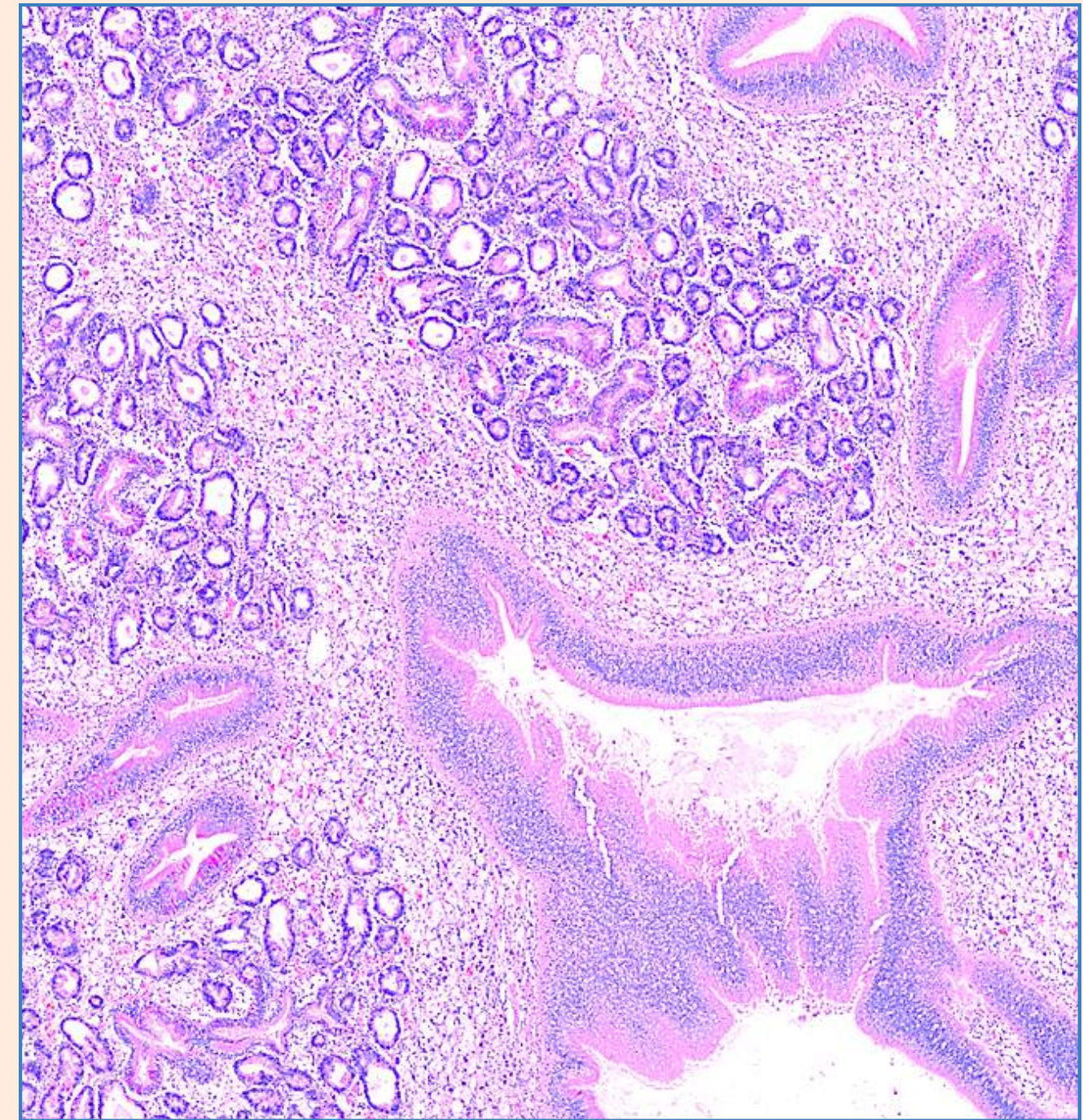
Healthy oyster WSI section

Tier 2: Multi-scale feature extraction

Hemocyste density



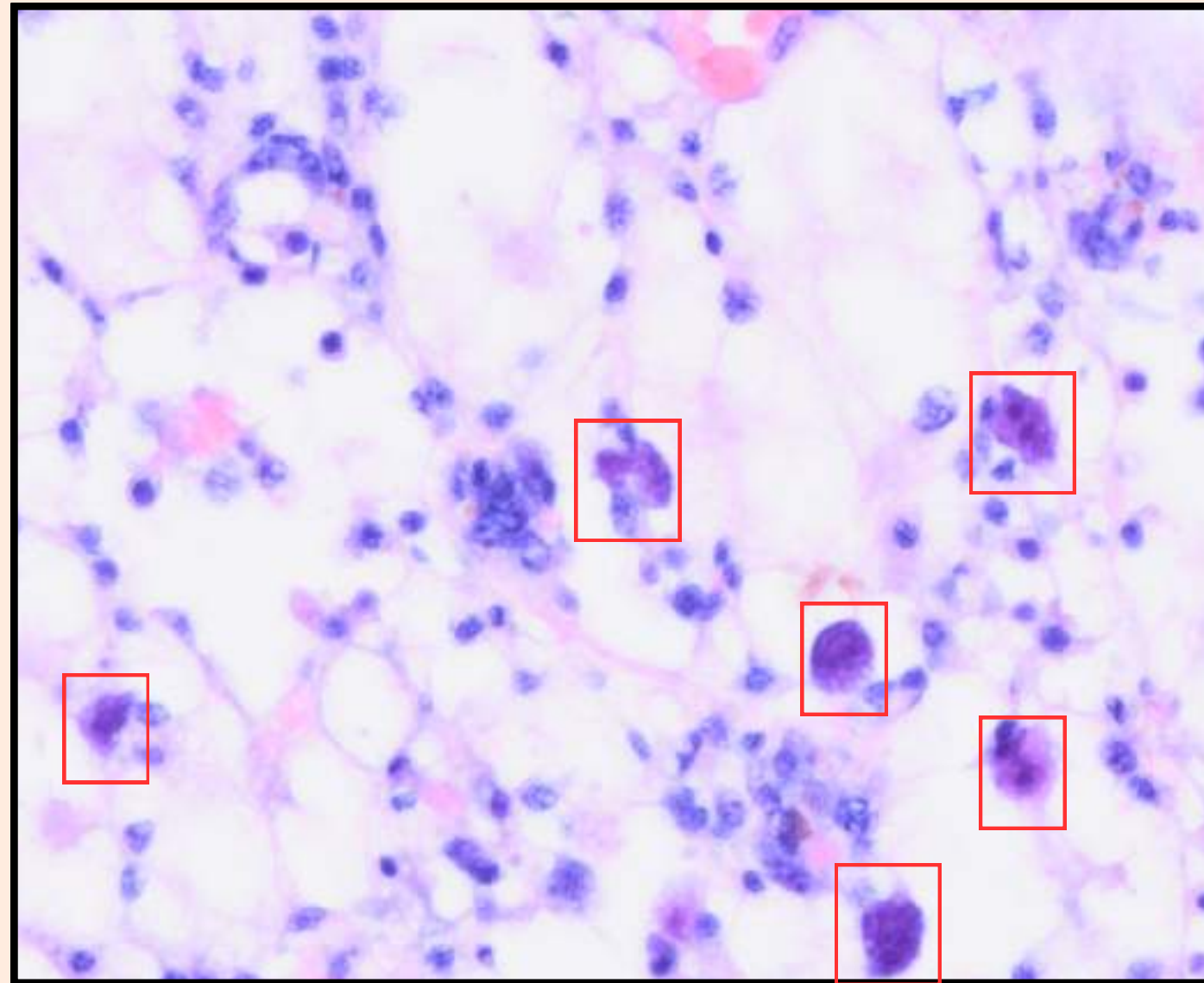
Healthy oyster WSI section



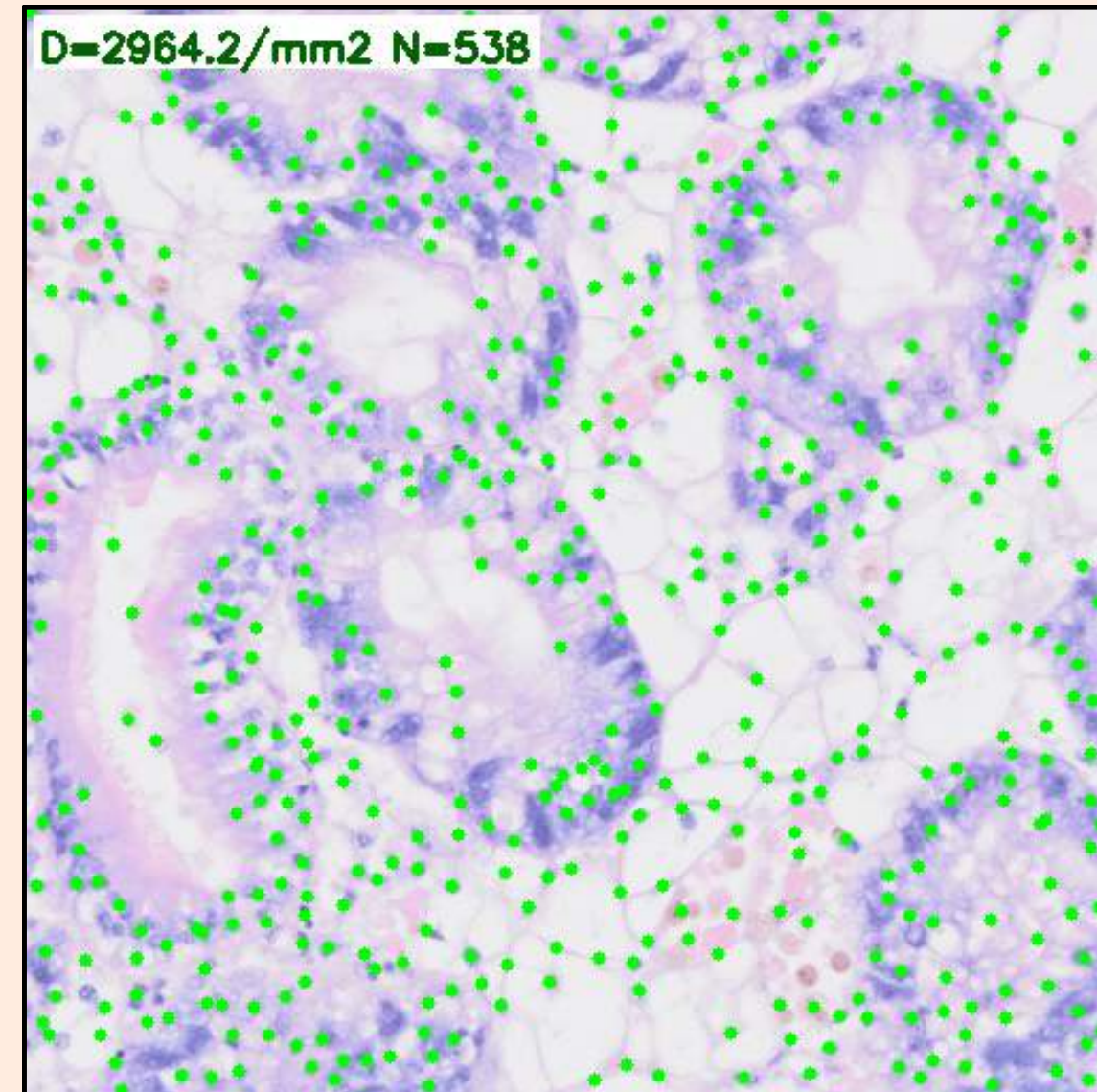
MSX-infected oyster WSI section

Tier 2: Multi-scale feature extraction

40x Parasite candidate detection



10x Hemocyte density biomarker



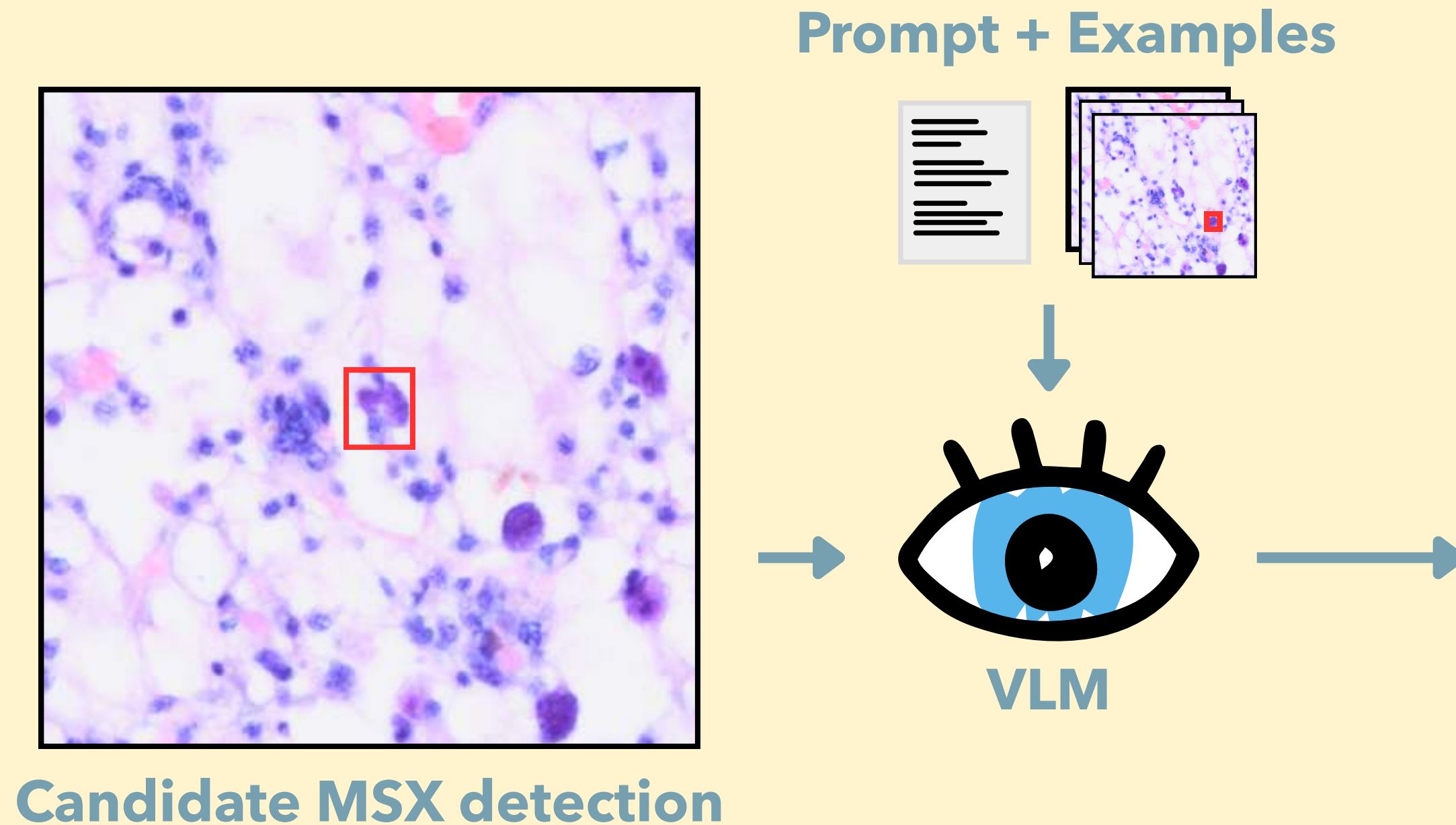
Local parasite evidence + broader inflammatory context

Tier 3: Reasoning and synthesis

Exploratory VLM confirmation

Tier 3: Reasoning and synthesis

Exploratory VLM confirmation



Tier 3: Reasoning

Exploratory VLM co

Prompt used

VLM_SYSTEM_PROMPT = ""You are a veterinary histopathologist analyzing 40x H&E oyster tissue for Haplosporidium nelsoni (MSX).

Analyze the image and return ONLY a JSON object with these exact fields:

```
{  
  "organism_present": <bool>,  
  "multinucleated": <bool>,  
  "nucleus_count": <int>,  
  "cytoplasm_texture": "<granular|smooth|vacuolated|artifact>",  
  "parasite_diameter_um": <float>,  
  "confidence": <float 0.0-1.0>  
}
```

DIAGNOSTIC CRITERIA:

- MSX plasmodia are 5-50um, multinucleated (2-20+ nuclei), granular eosinophilic cytoplasm
- Hemocytes are smaller (<10um), mononuclear – do NOT flag as MSX
- If no organism visible, set organism_present=false and other fields to defaults (0, "artifact", 0.0)

REFERENCE IMAGES:

{exemplar_section}

Return ONLY valid JSON. ""

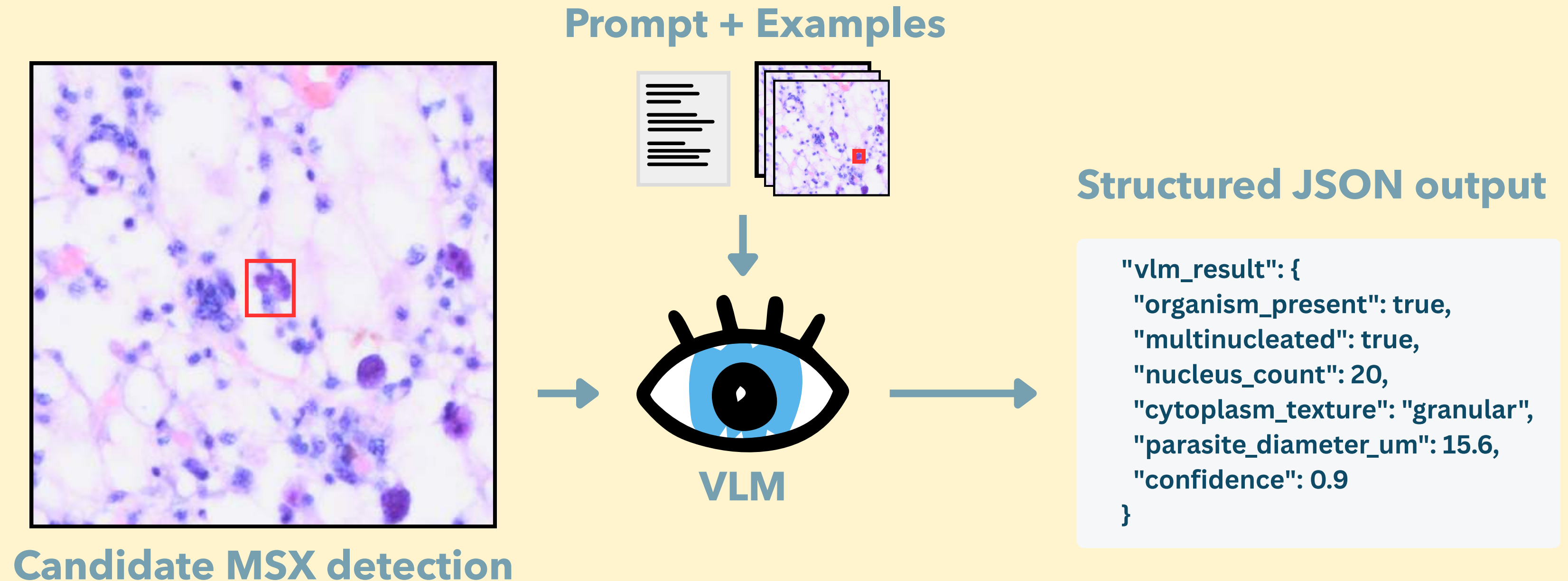
Model output

```
{  
  "organism_present": true,  
  "multinucleated": true,  
  "nucleus_count": 20,  
  "cytoplasm_texture": "granular",  
  "parasite_diameter_um": 15.6,  
  "confidence": 0.9  
}
```

Candida

Tier 3: Reasoning and synthesis

Exploratory VLM confirmation

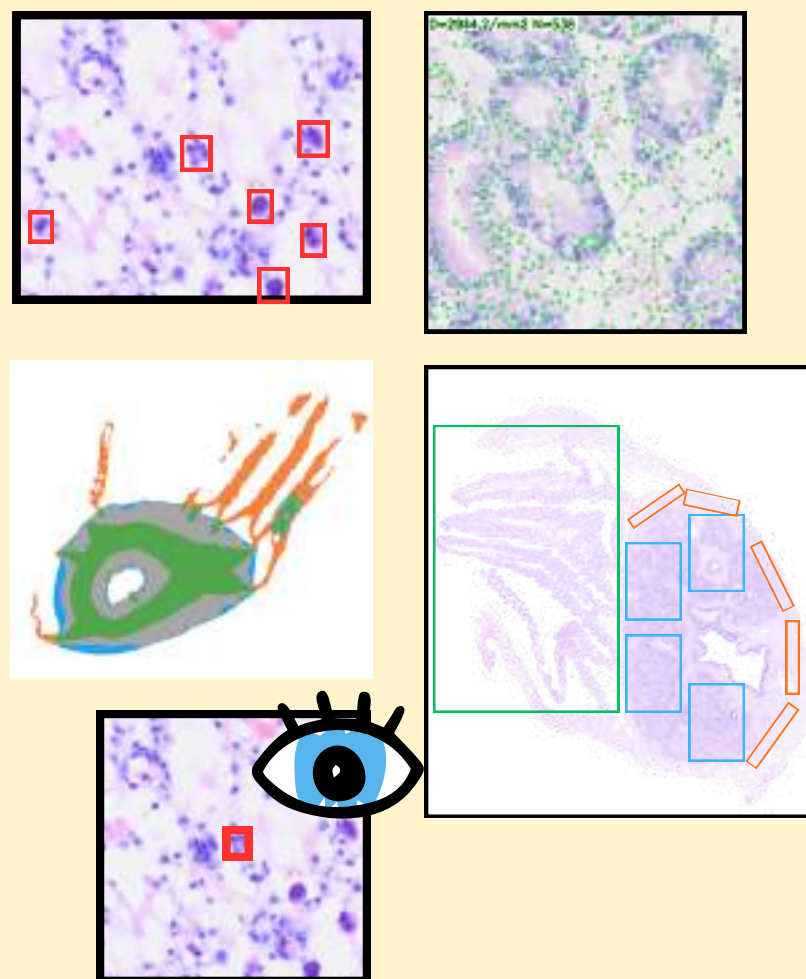


Tier 3: Reasoning and synthesis

Final severity synthesis

Tier 3: Reasoning and synthesis

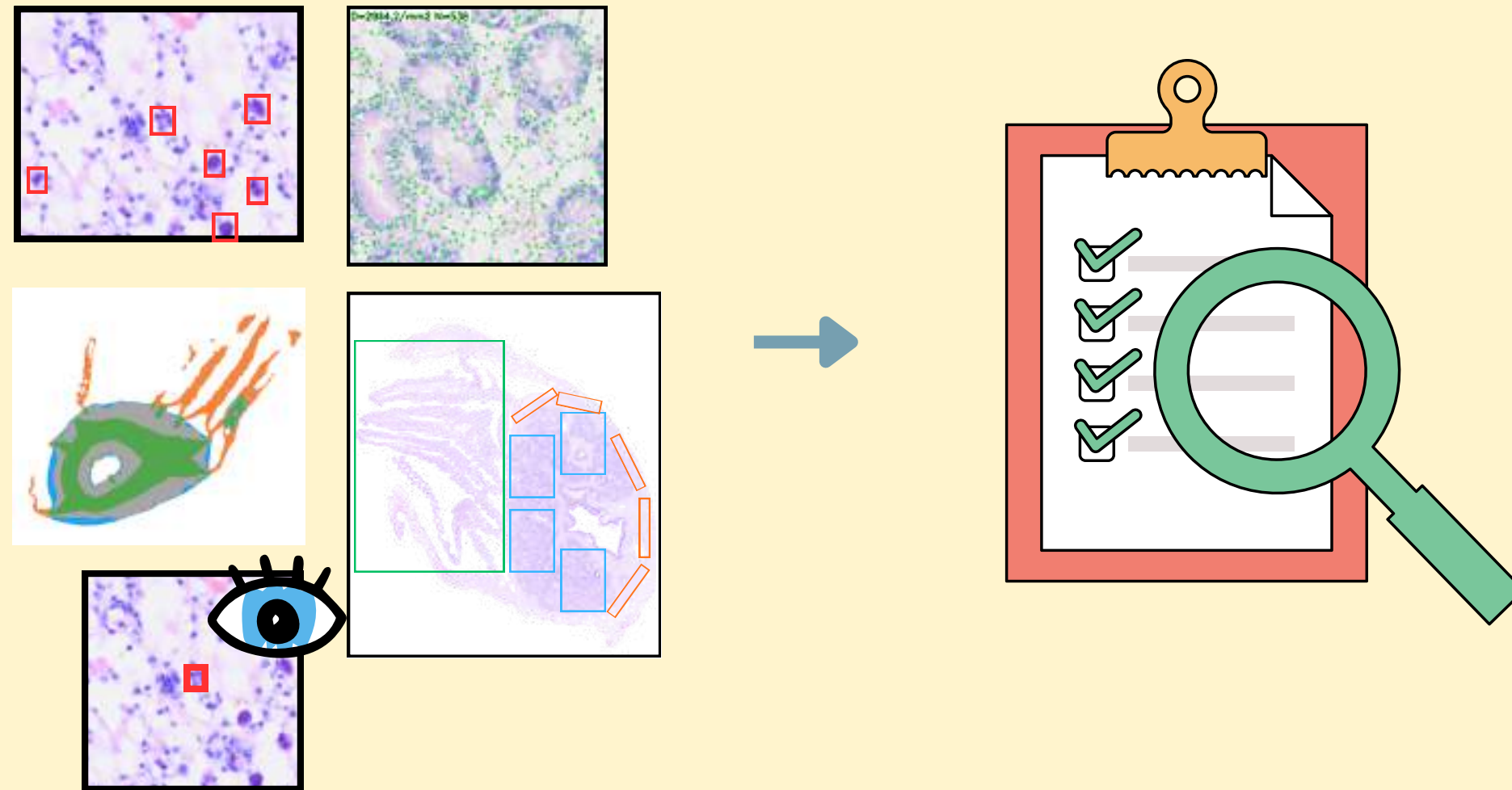
Final severity synthesis



Evidence blocks

Tier 3: Reasoning and synthesis

Final severity synthesis

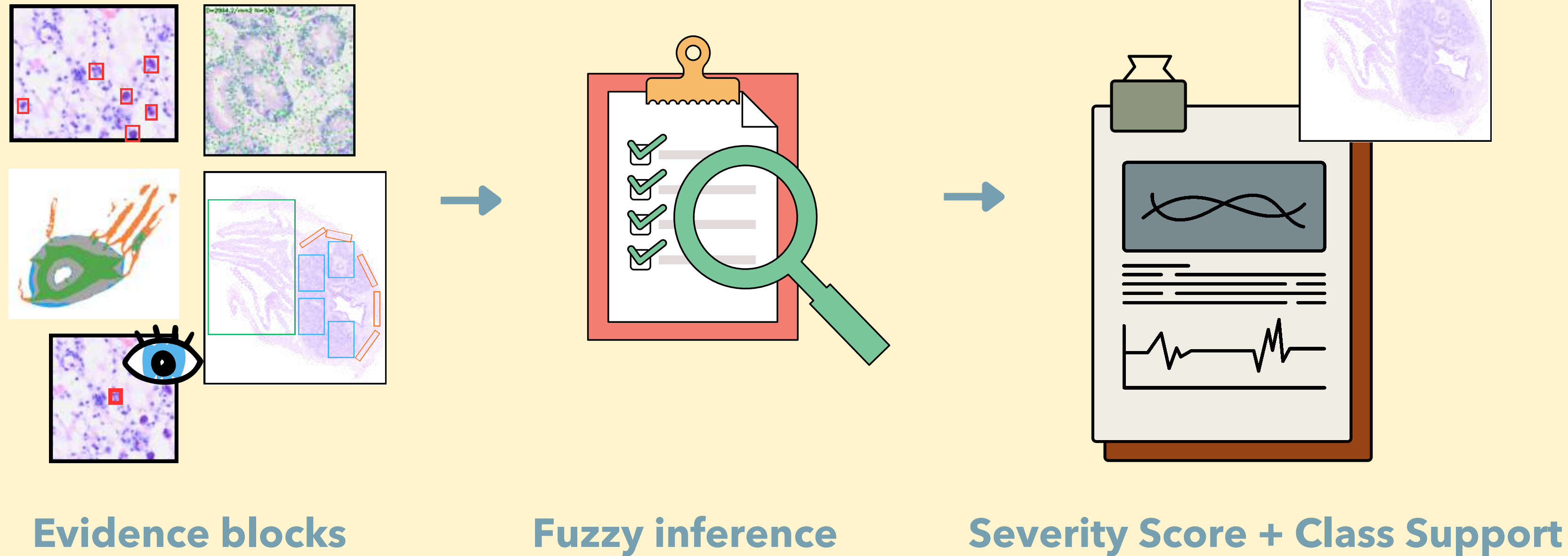


Evidence blocks

Fuzzy inference

Tier 3: Reasoning and synthesis

Final severity synthesis



Output: Diagnostic audit trial

Oyster Overview



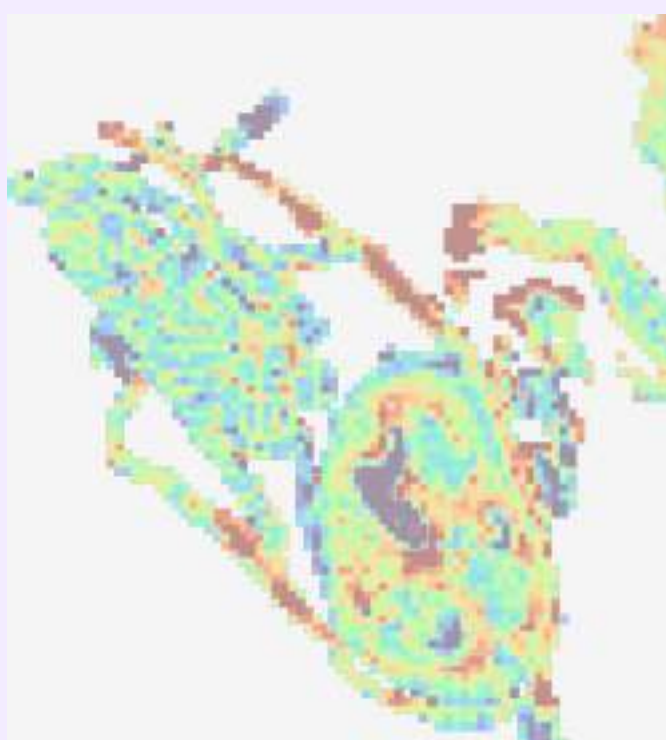
Approximate Organ Context



YOLO Detections



Hemocyste Density Heatmap



Oyster-level severity summary and evidence tiles

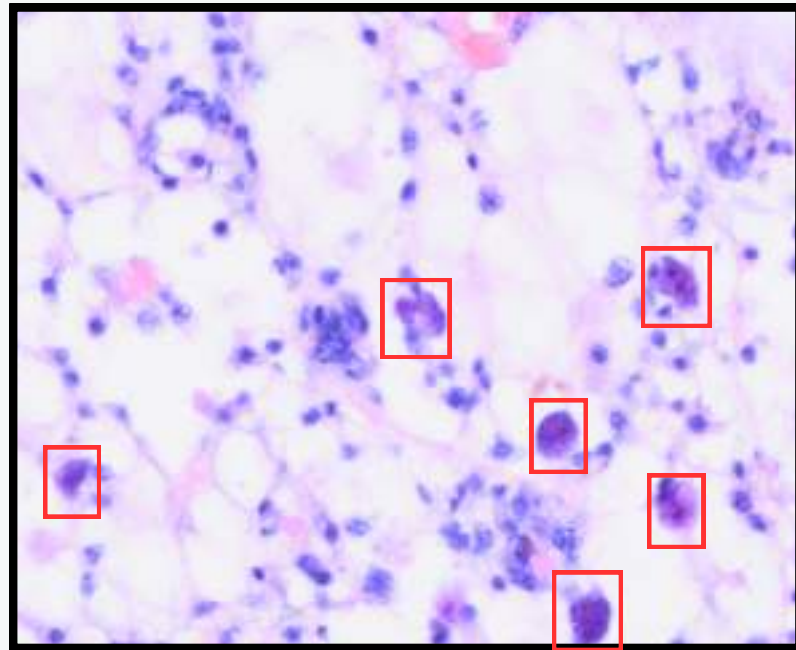


Results at a glance

Results at a glance

Strongest

Parasite candidate localization

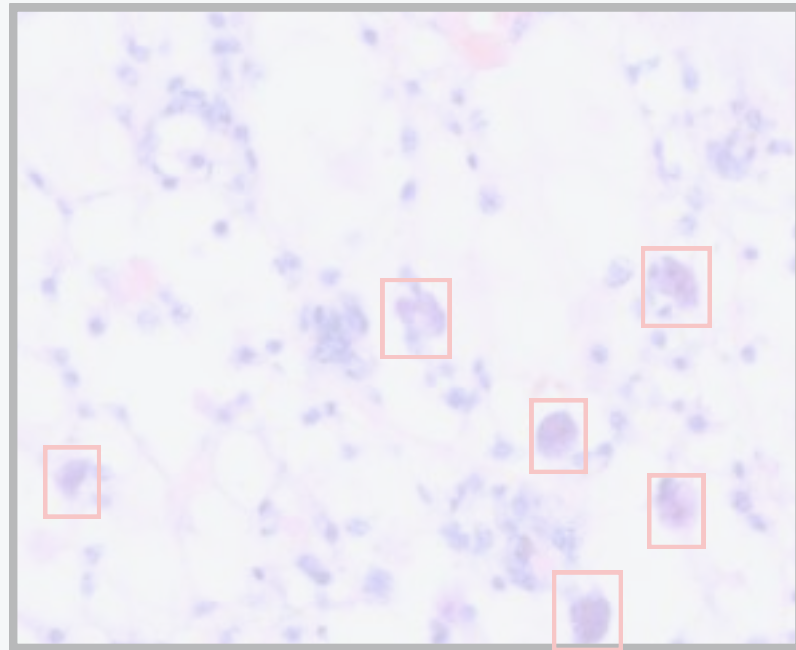


- YOLO detection performed strongly
- Low false-positive rate against distractors
- Best-validated component

Results at a glance

Strongest

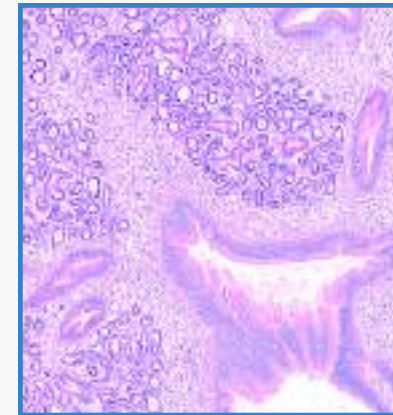
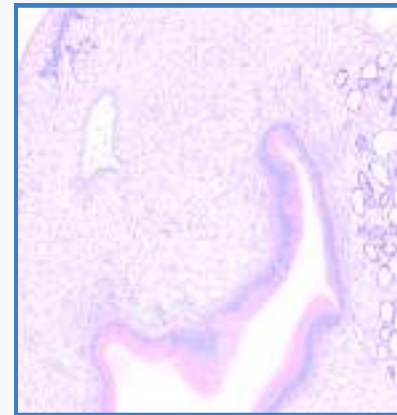
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Promising

Disease screening / triage

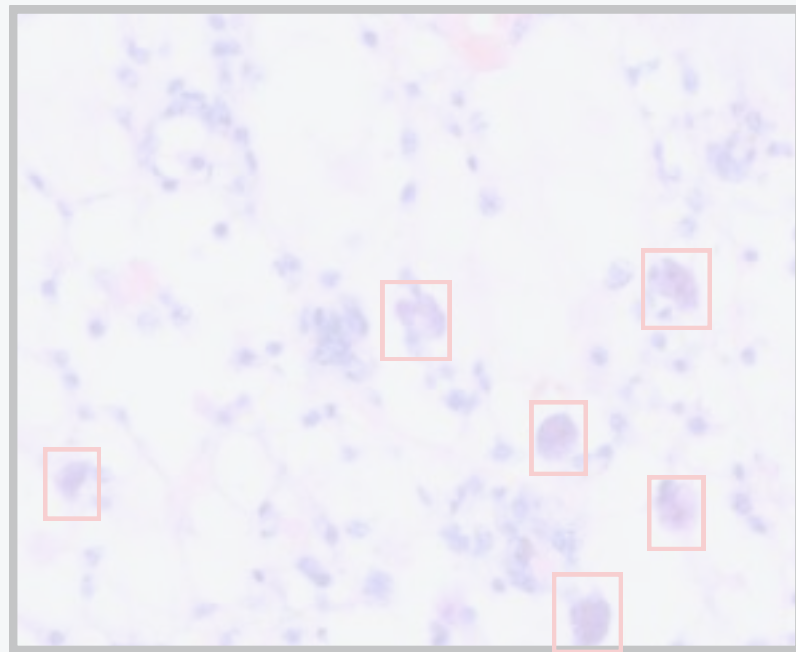


- Stronger binary disease / no-disease performance
- Useful for identifying abnormal oysters
- Supports review prioritization

Results at a glance

Strongest

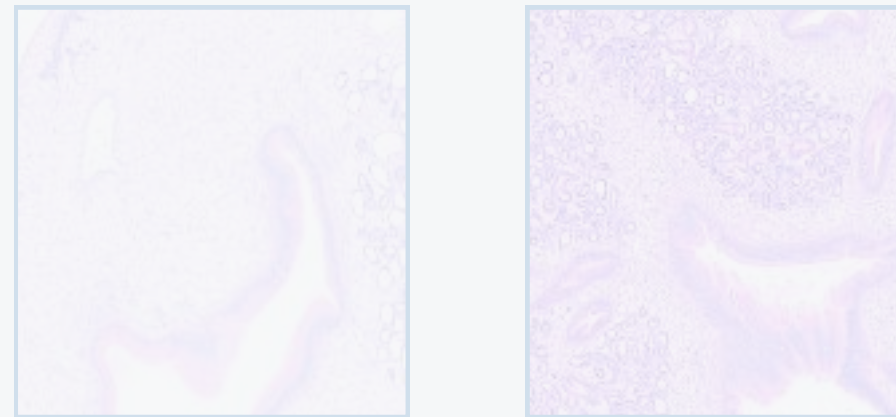
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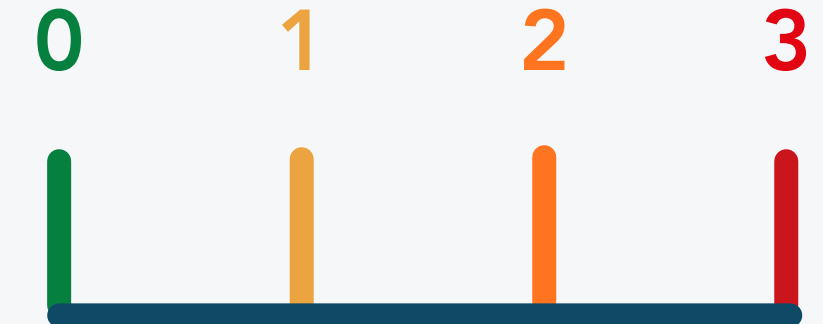
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- Useful for identifying abnormal oysters
- Supports review prioritization

Limited

Fine-grained severity staging

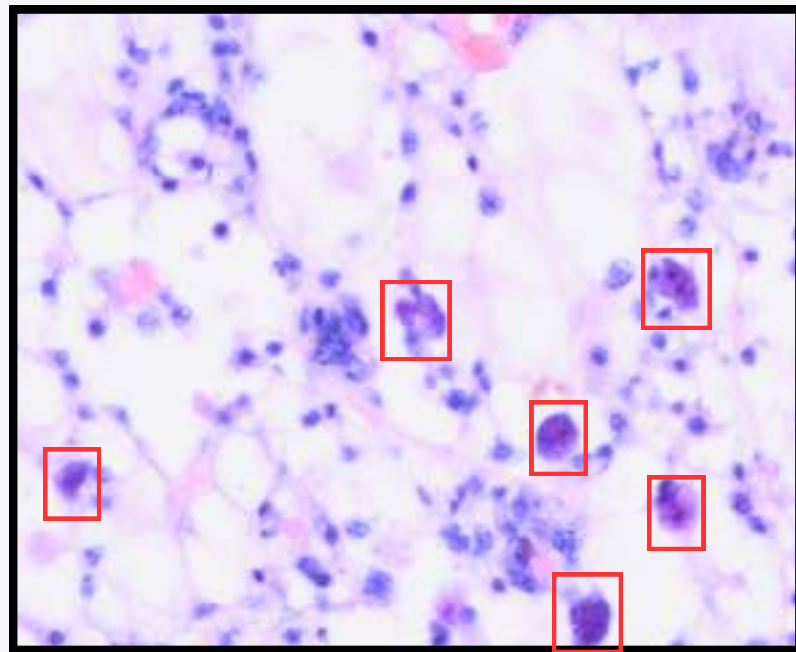


- Severity 1–3 classes overlap
- Coarse anatomy likely limits grading
- Needs better organ mapping

Results at a glance

Strongest

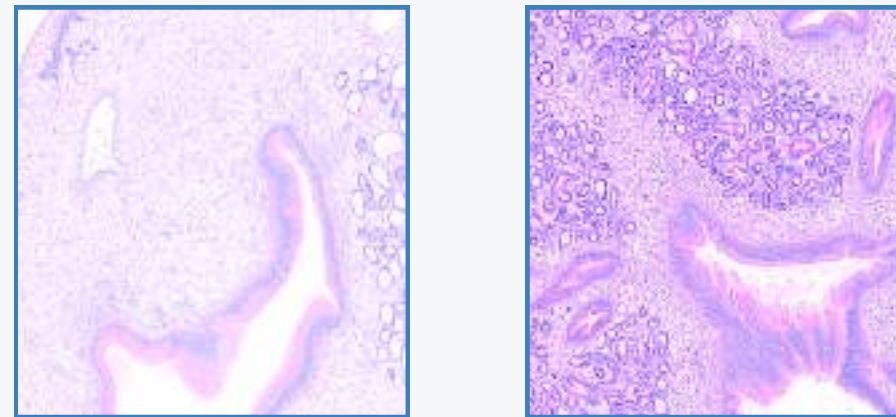
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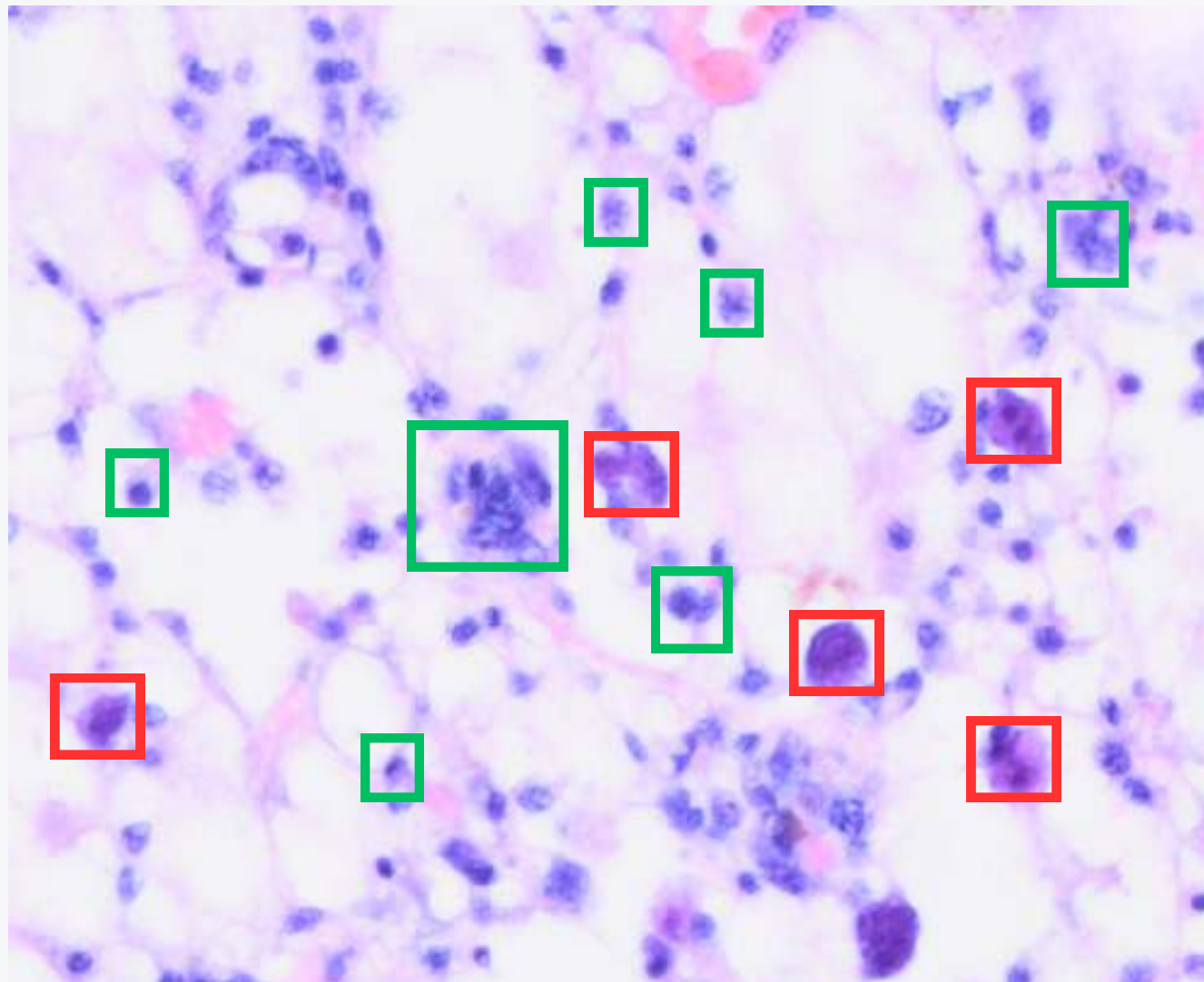
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Strongest result: parasite candidate localization

YOLO performed well as the first screening stage

Strongest result: parasite candidate localization

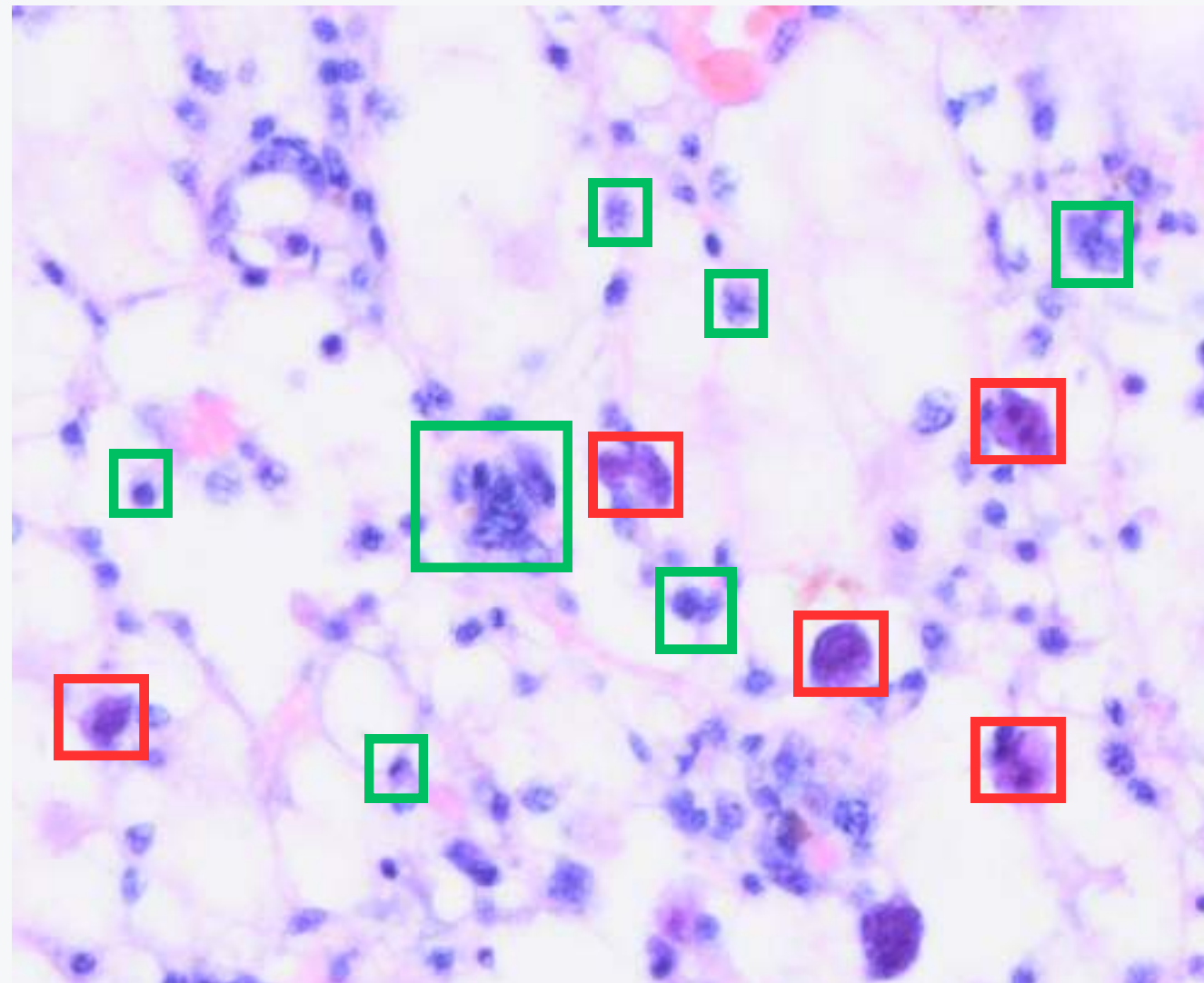
YOLO performed well as the first screening stage



Example 40x detection output

Strongest result: parasite candidate localization

YOLO performed well as the first screening stage



Example 40x detection output

0.983

mAP@0.5

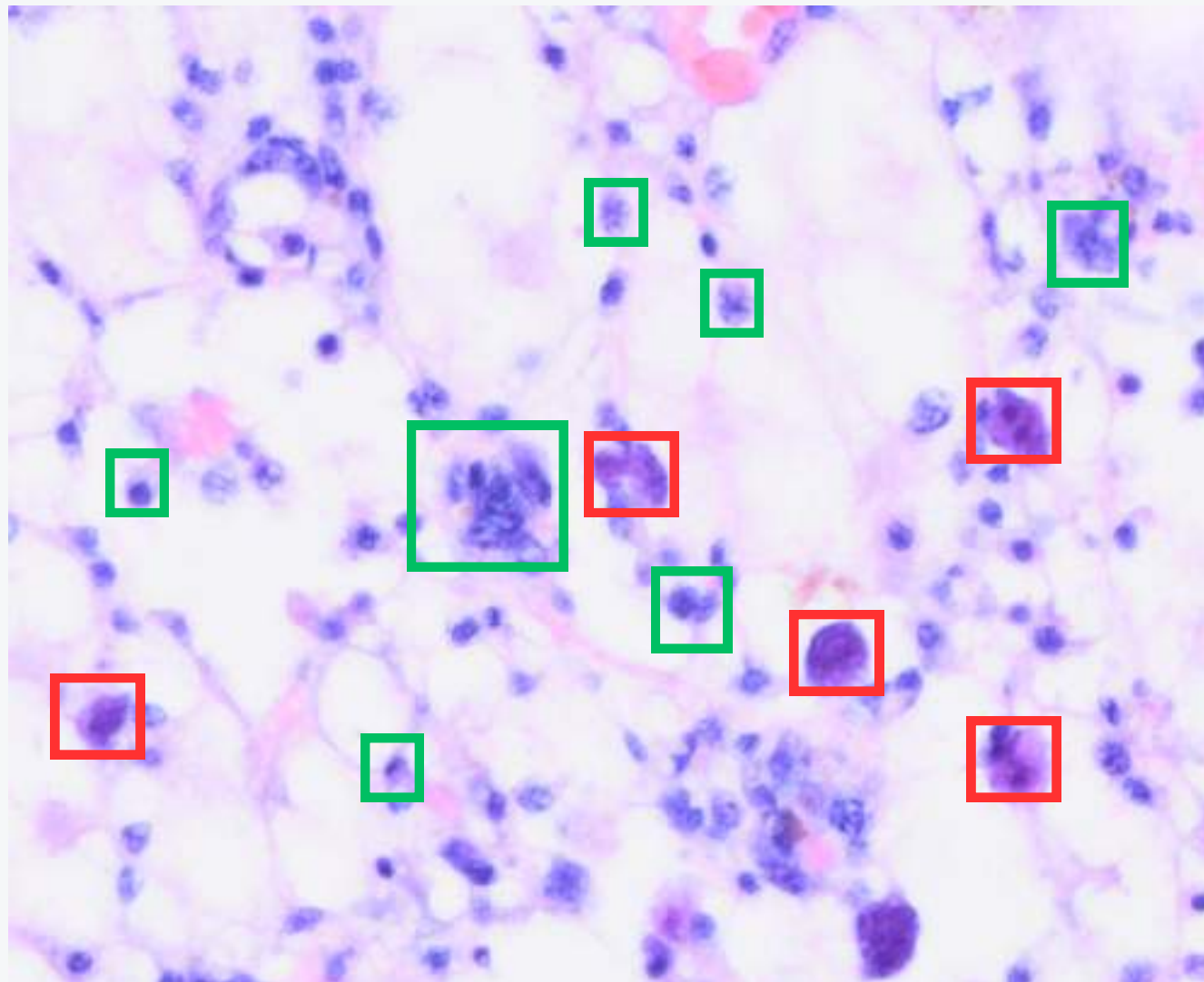
Mean Average Precision at 0.5 IoU

“did the model find the parasite?”

“did it put the box in roughly the right spot?”

Strongest result: parasite candidate localization

YOLO performed well as the first screening stage



Example 40x detection output

0.983

mAP@0.5

0.97

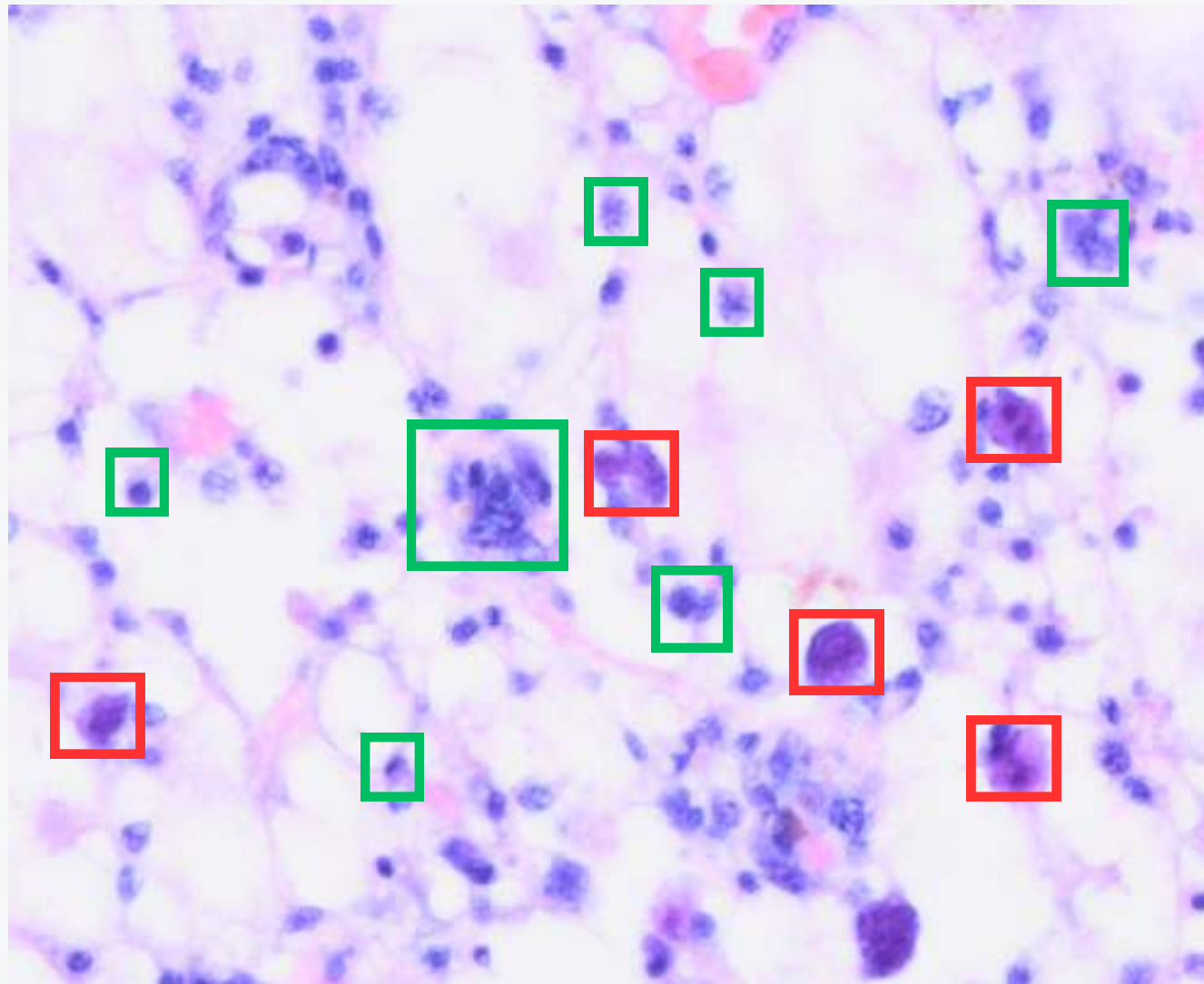
F1-score

Balances **precision** (how often parasite predictions are right) and **recall** (how many real parasites did it actually find)

low precision = many false parasites
low recall = miss real parasites

Strongest result: parasite candidate localization

YOLO performed well as the first screening stage



Example 40x detection output

0.983

mAP@0.5

0.97

F1-score

119 / 127

true plasmodia identified

4 / 1,432

distractors misclassified
as plasmodia

How many real parasites were captured?

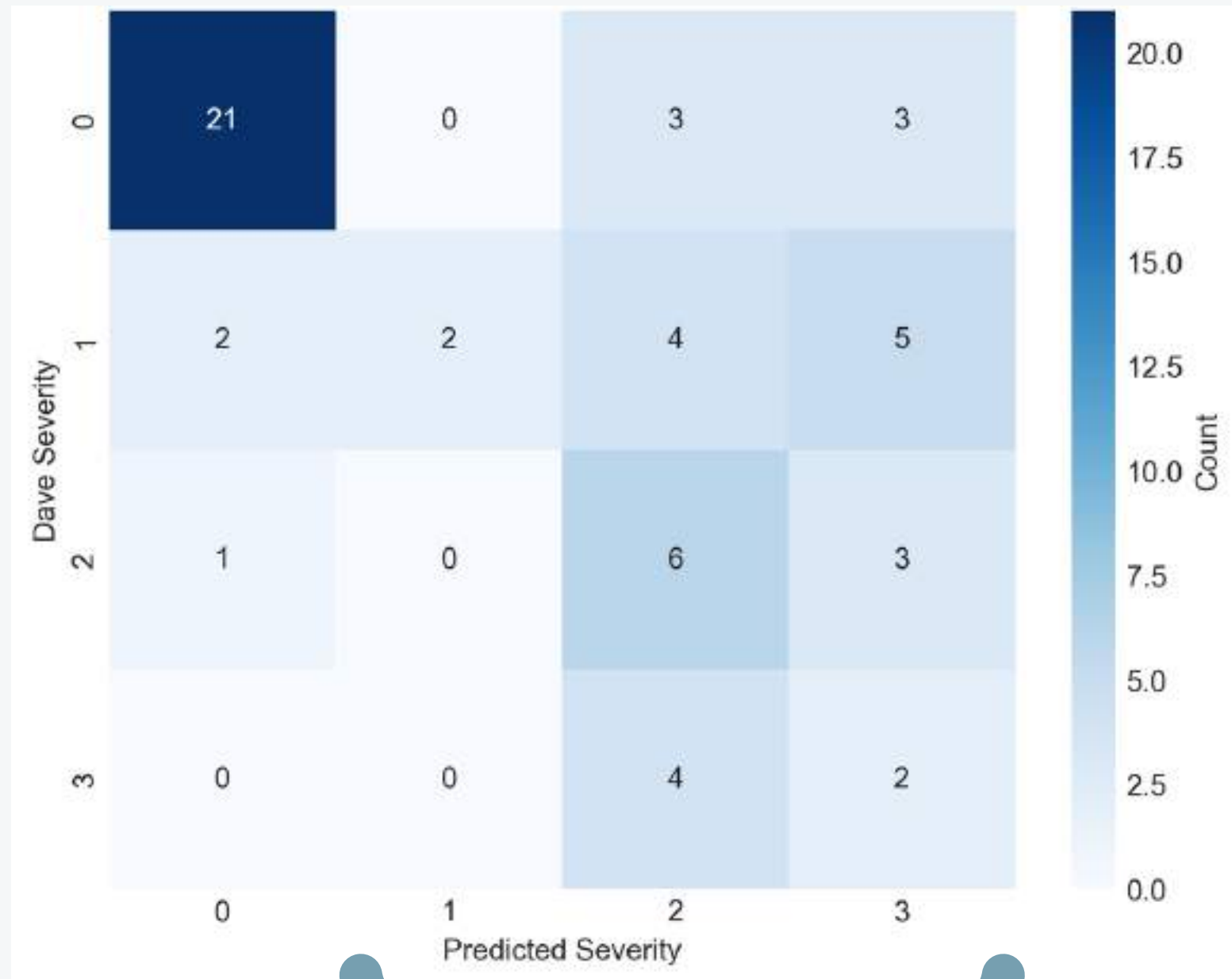
How often does the model predict other cells as parasites?

Oyster-level severity: screening > staging

Fine-grained severity remains difficult with coarse anatomical context

Oyster-level severity: screening > staging

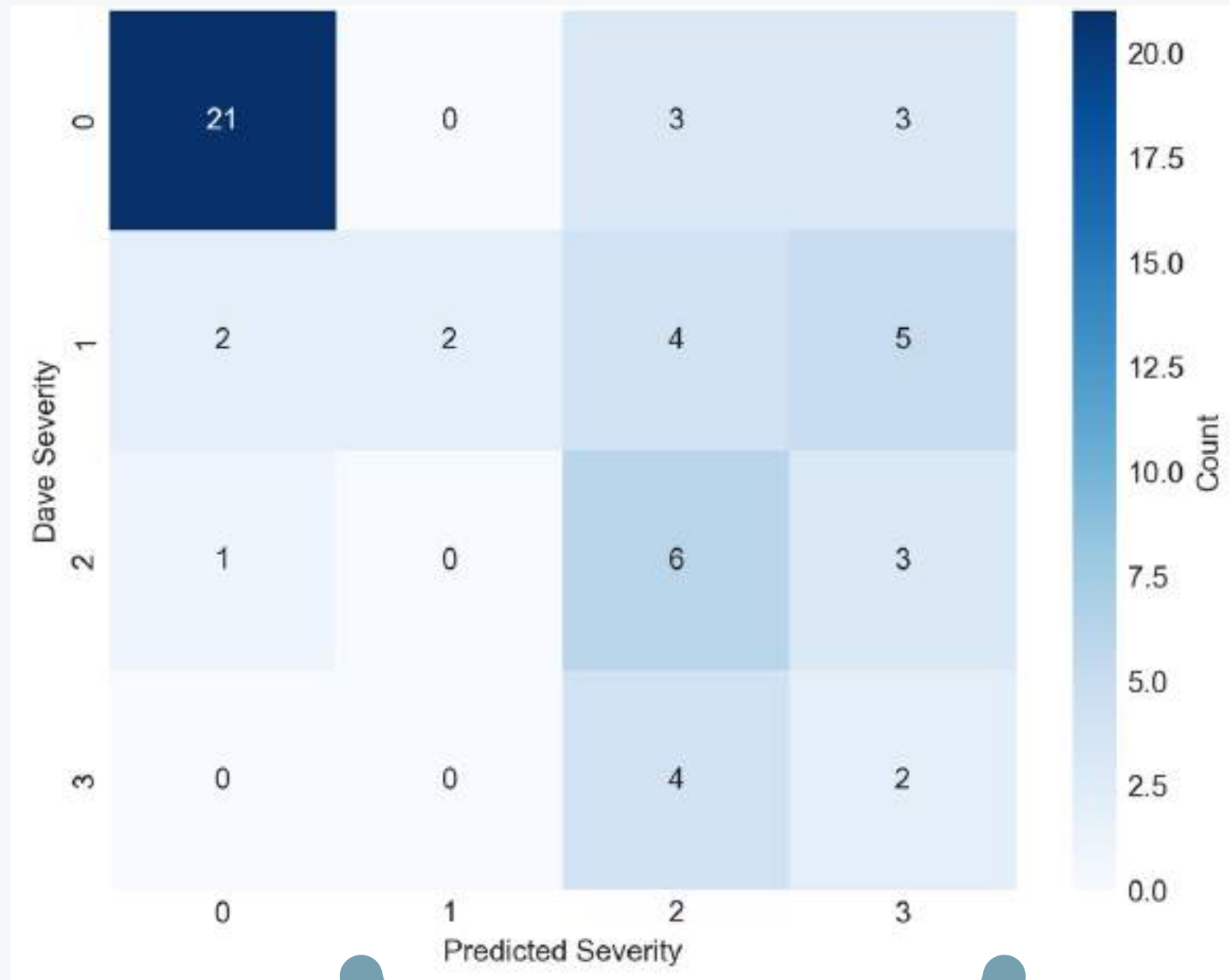
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Blur among infected classes

Oyster-level severity: screening > staging

Fine-grained severity remains difficult with coarse anatomical context



Blur among infected classes

4-class severity
grading

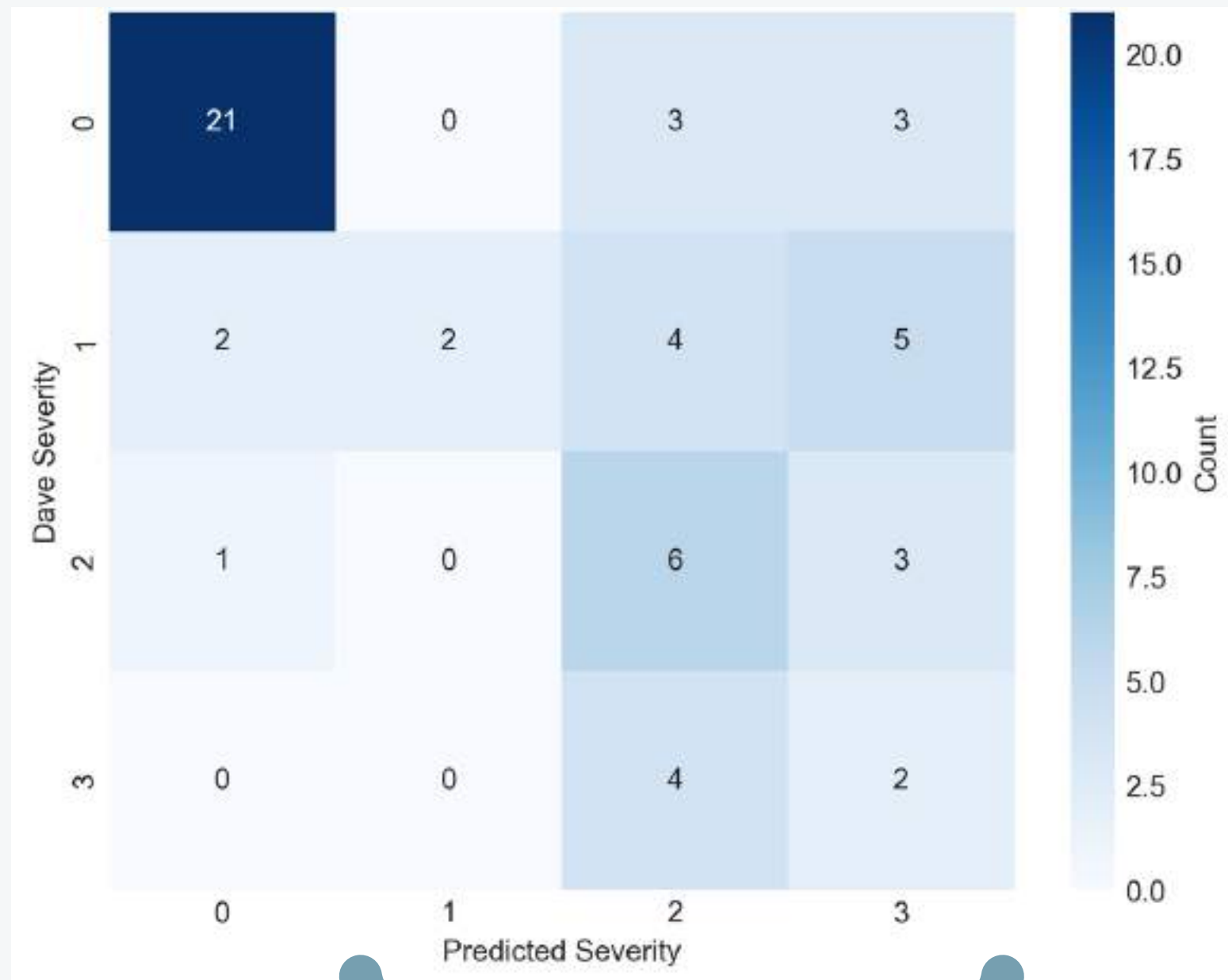
55.36%

Accuracy

How often did the model
predict classes correctly?

Oyster-level severity: screening > staging

Fine-grained severity remains difficult with coarse anatomical context



Blur among infected classes

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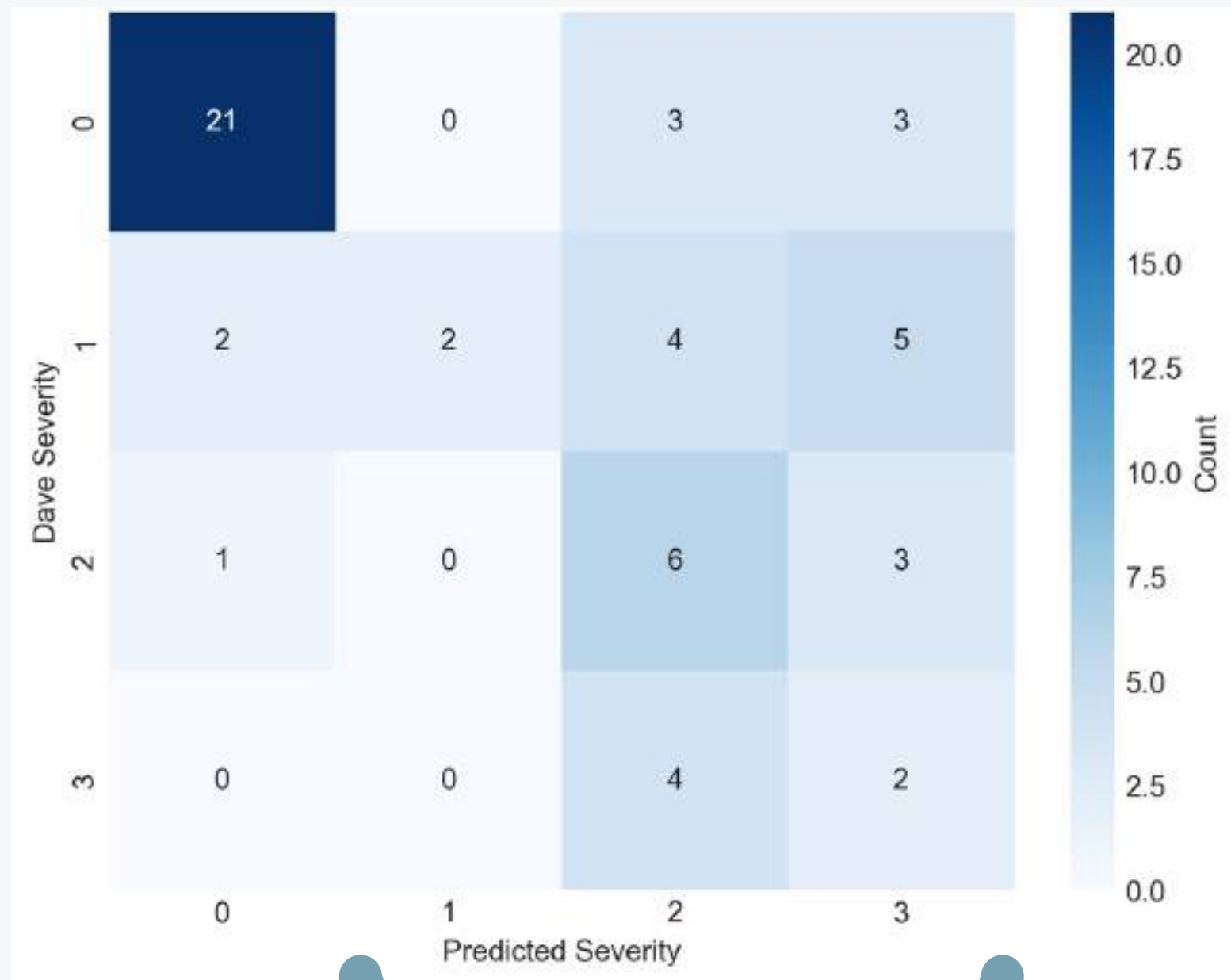
0.71

MAE

On avg, how many severity
levels away were predictions?

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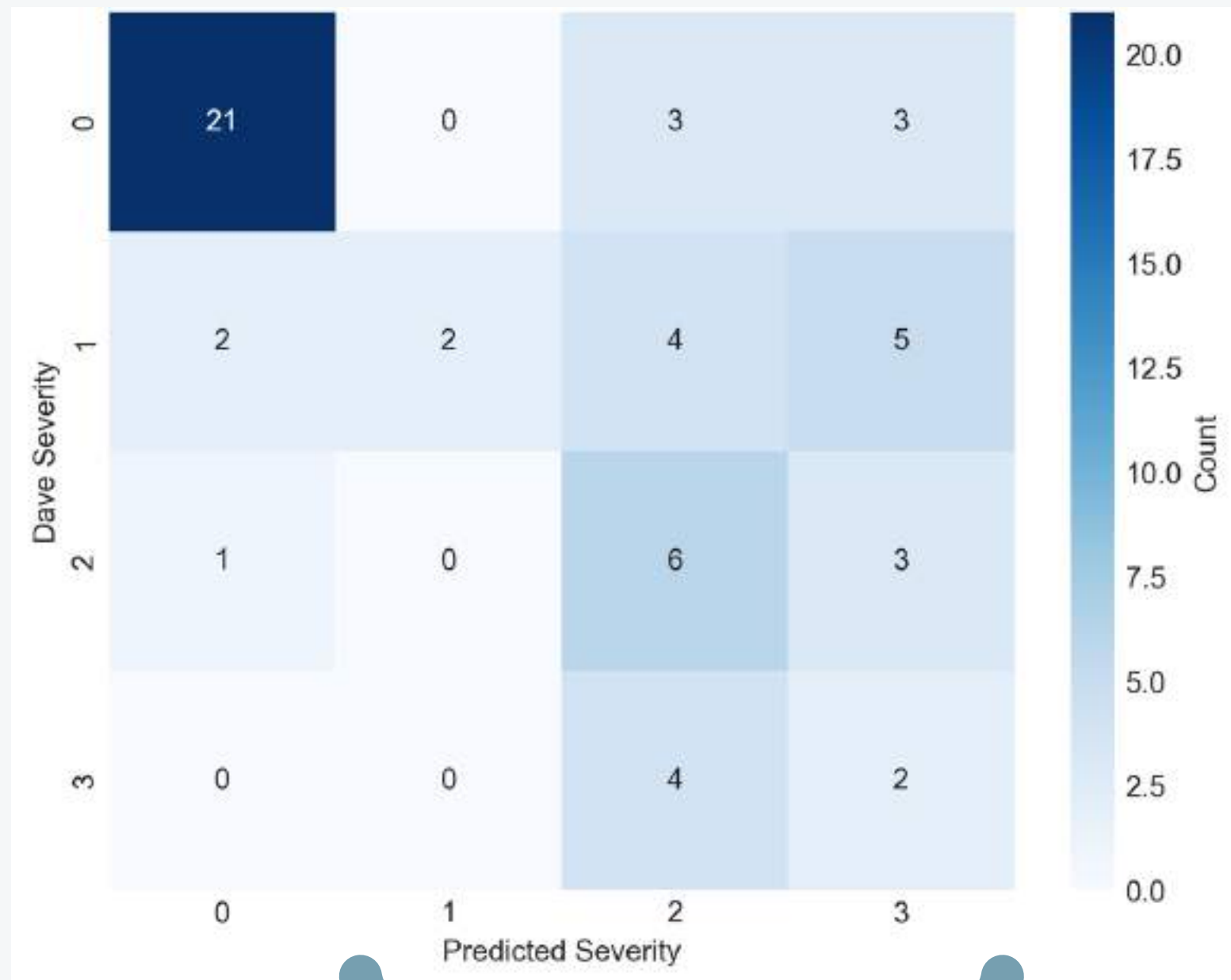
0.452

Linear-weighted k

How much the model agrees with expert labels?

Oyster-level severity: screening > staging

Fine-grained severity remains difficult with coarse anatomical context



Blur among infected classes

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0.452

Linear-weighted k

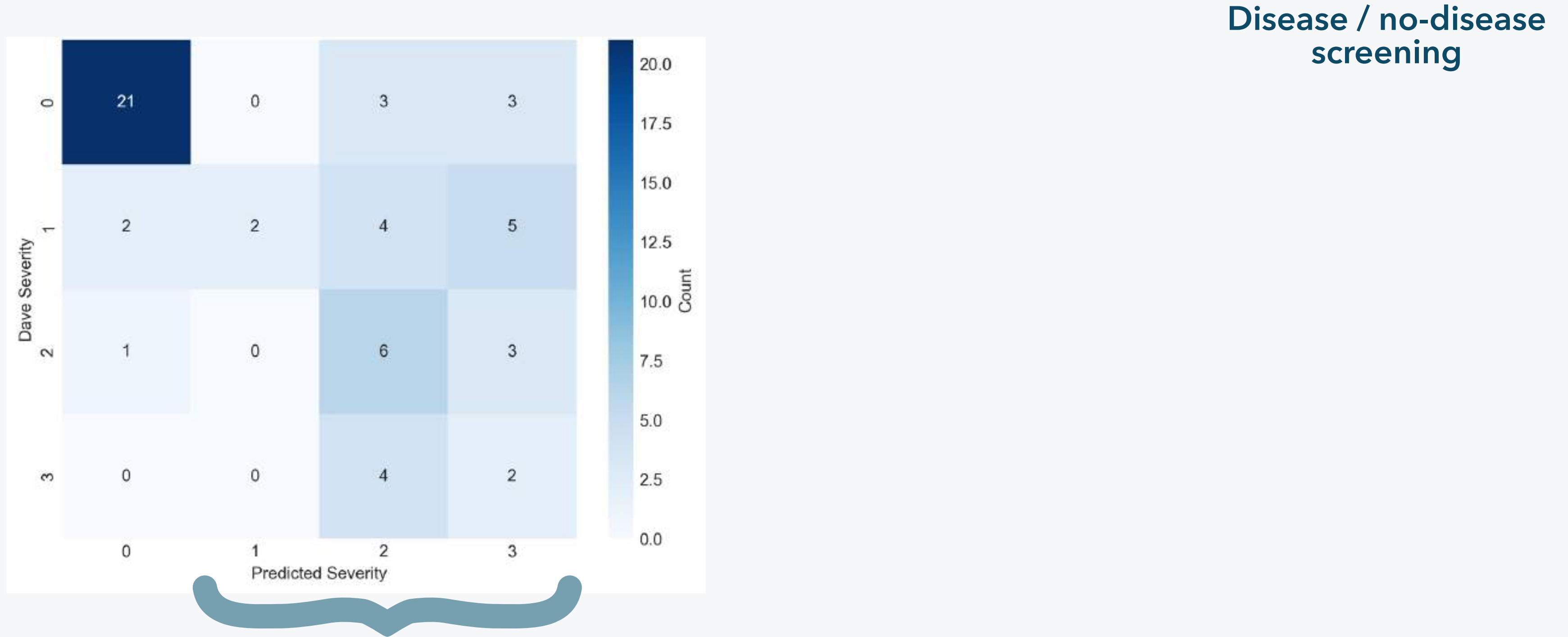
How much the model agrees with expert labels?

78.57%

Within one class

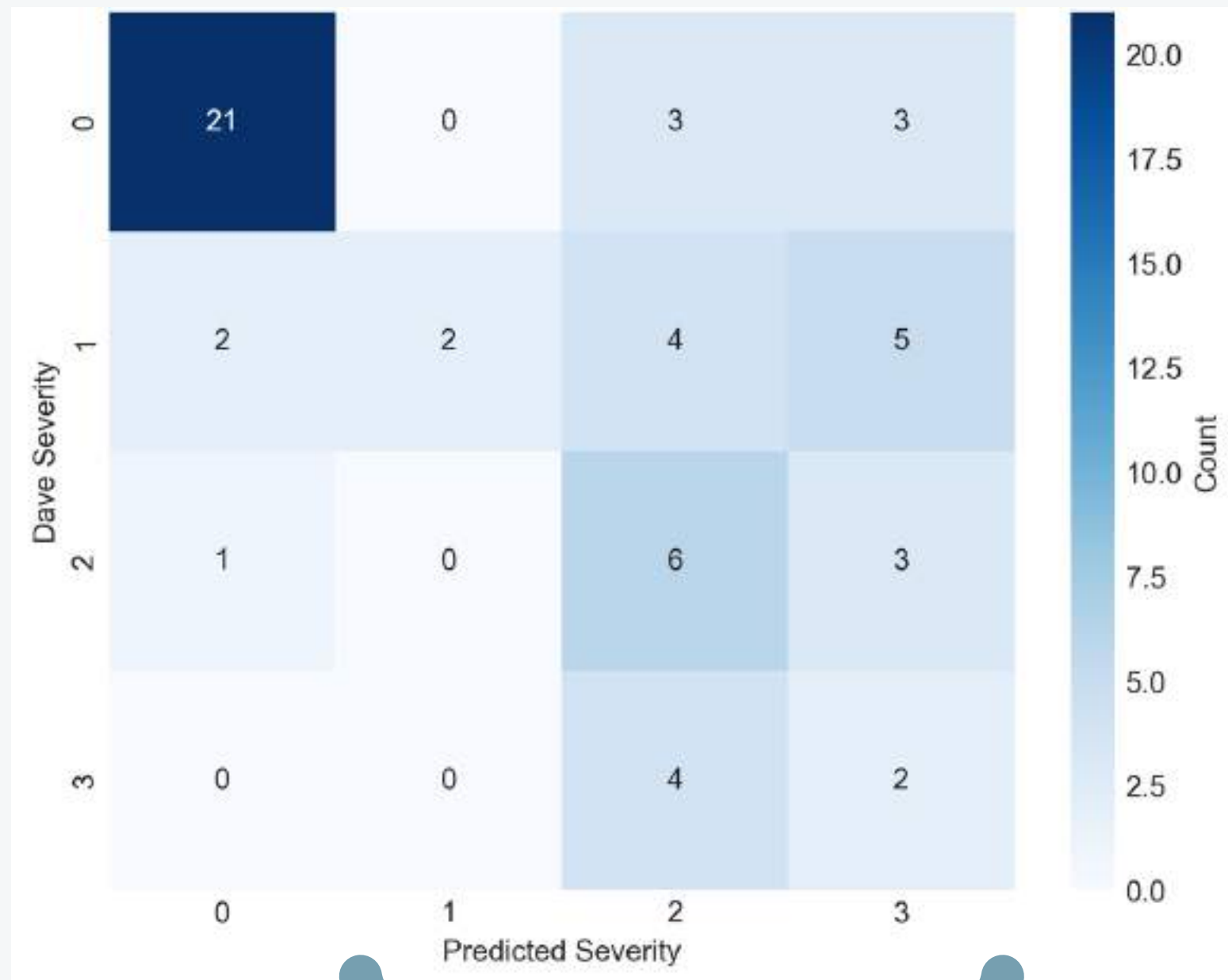
Oyster-level severity: screening > staging

Fine-grained severity remains difficult with coarse anatomical context



Oyster-level severity: screening > staging

Fine-grained severity remains difficult with coarse anatomical context



Blur among infected classes

How often did the system classify healthy vs diseased?

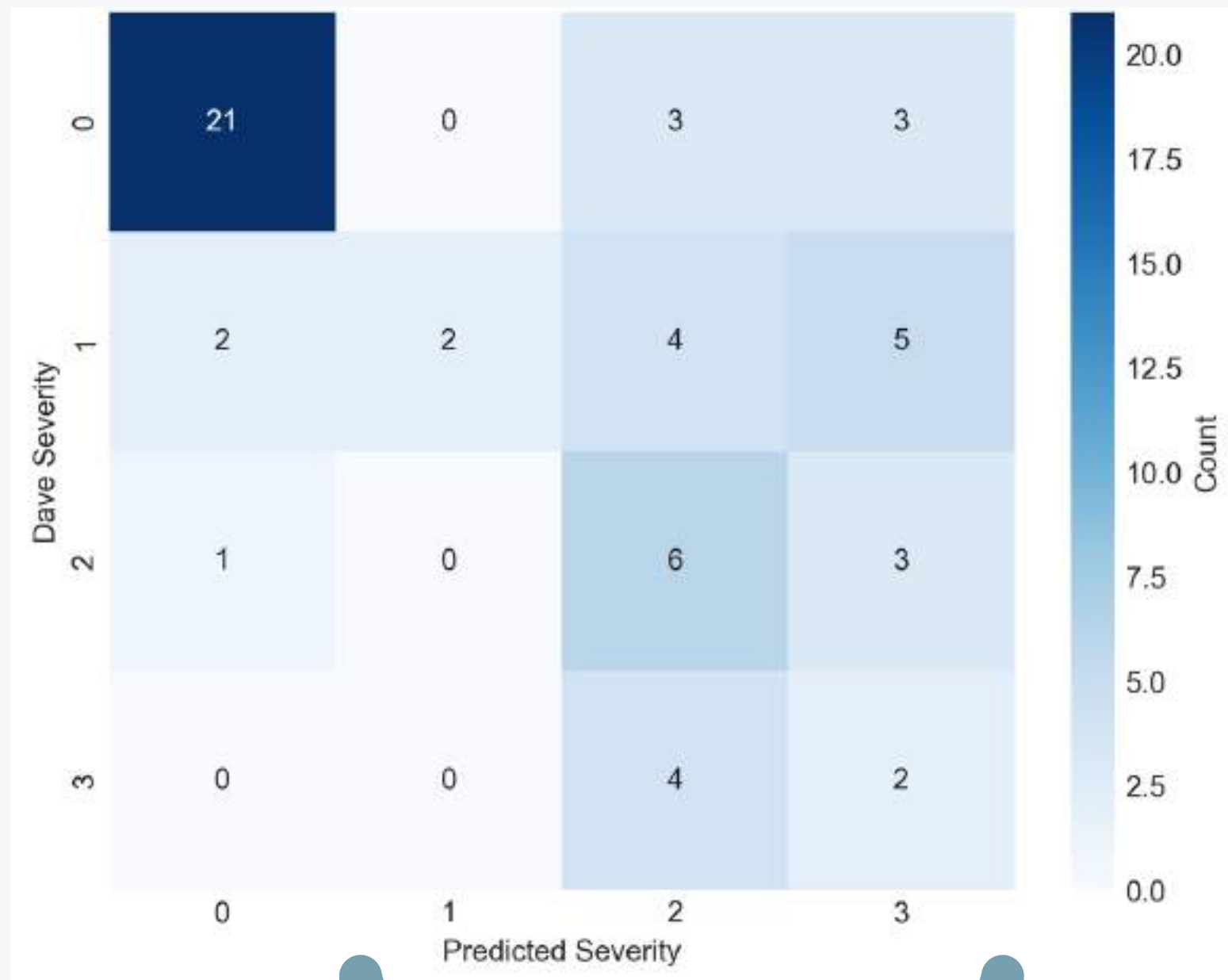
Disease / no-disease screening

83.93%

Accuracy

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How often did the system classify healthy vs diseased?

Of all true diseased oysters, how many were correctly detected?

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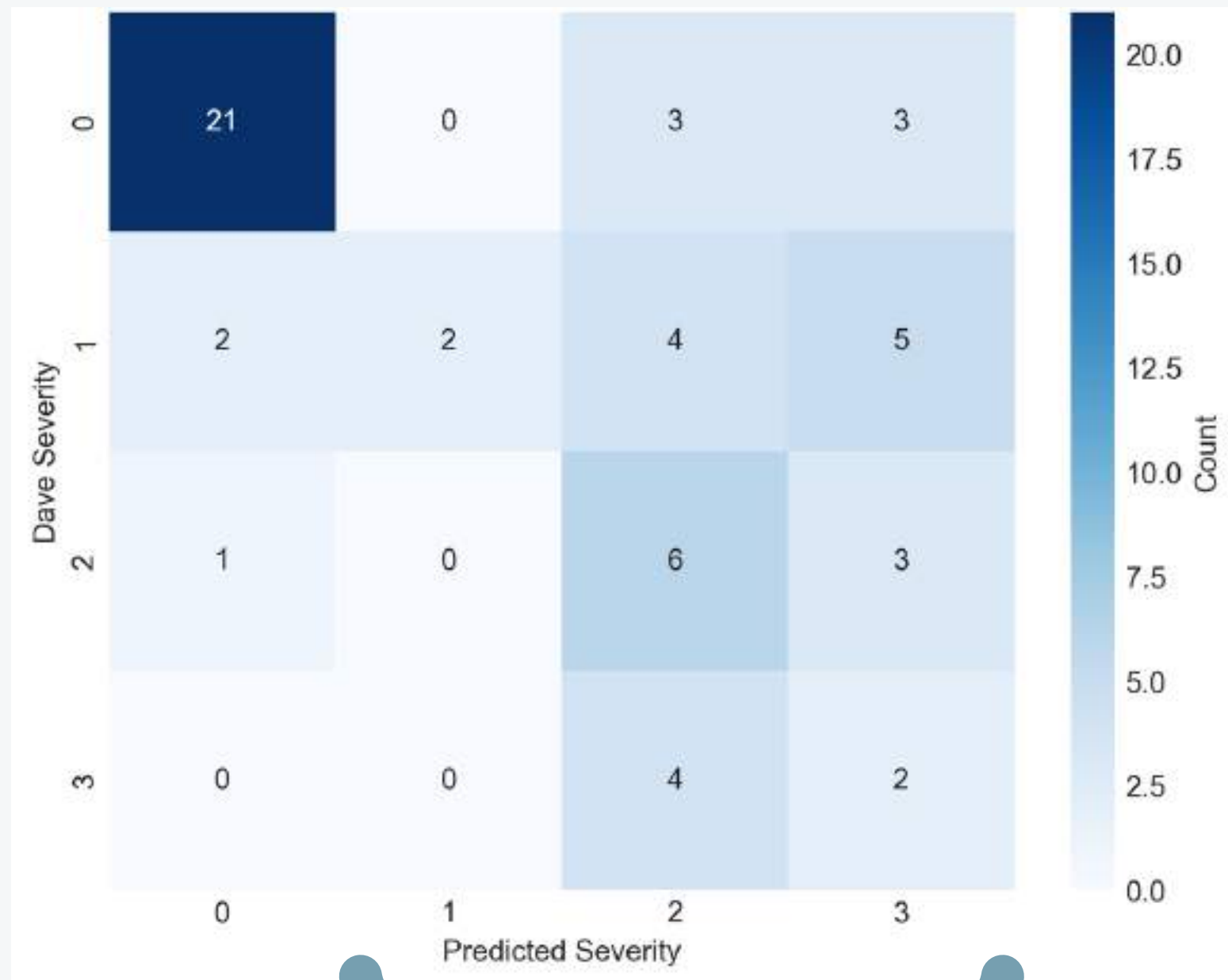
Accuracy

89.66%

Sensitivity

Oyster-level severity: screening > staging

Fine-grained severity remains difficult with coarse anatomical context



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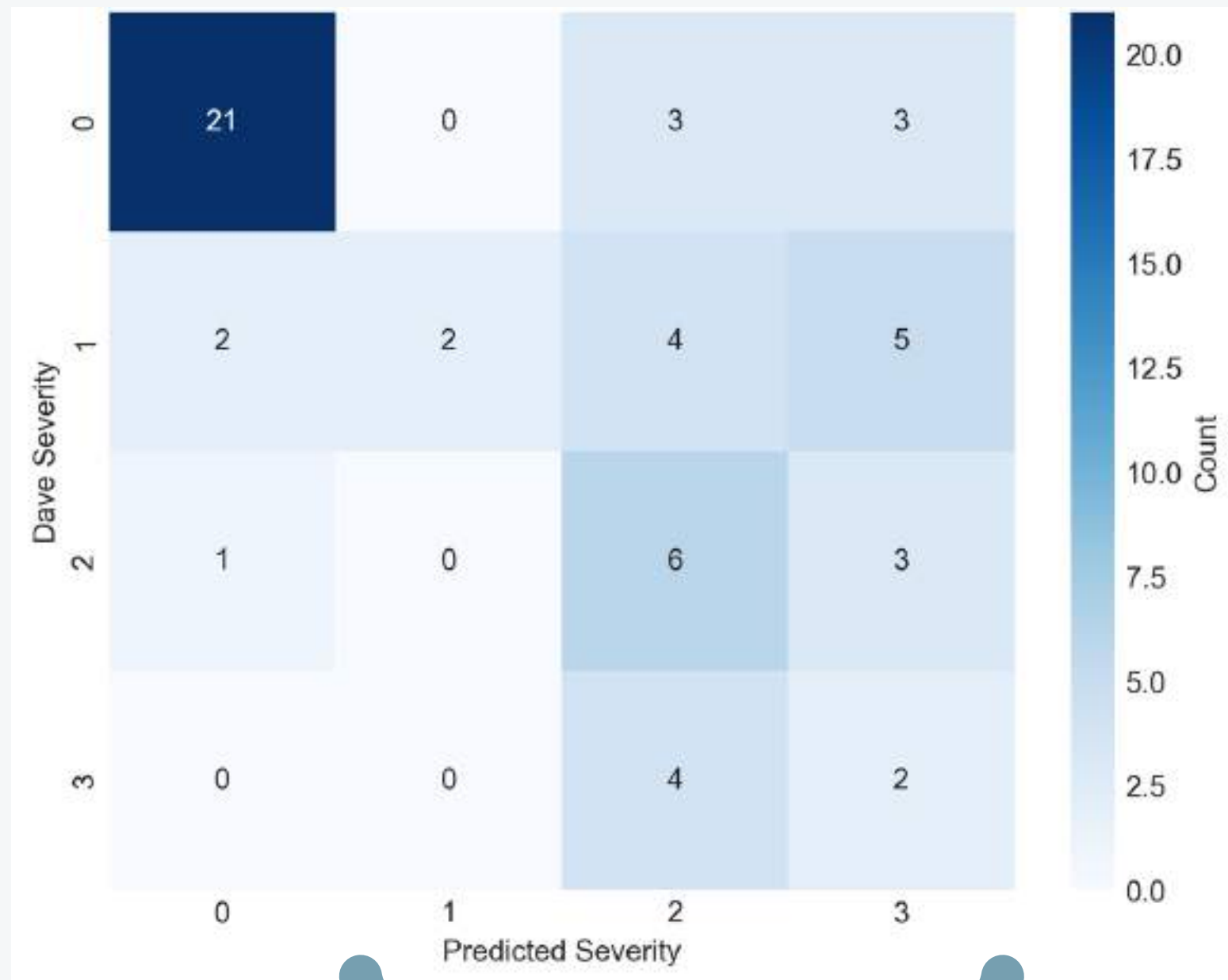
Sensitivity

77.78%

Specificity

Oyster-level severity: screening > staging

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Blur among infected classes

How often did the system classify healthy vs diseased?

Of all true diseased oysters, how many were correctly detected?

Of all true healthy oysters, how many were correctly detected?

Balance between right parasite predictions & real parasites found (precision & recall)

Disease / no-disease screening

83.93%

Accuracy

89.66%

Sensitivity

77.78%

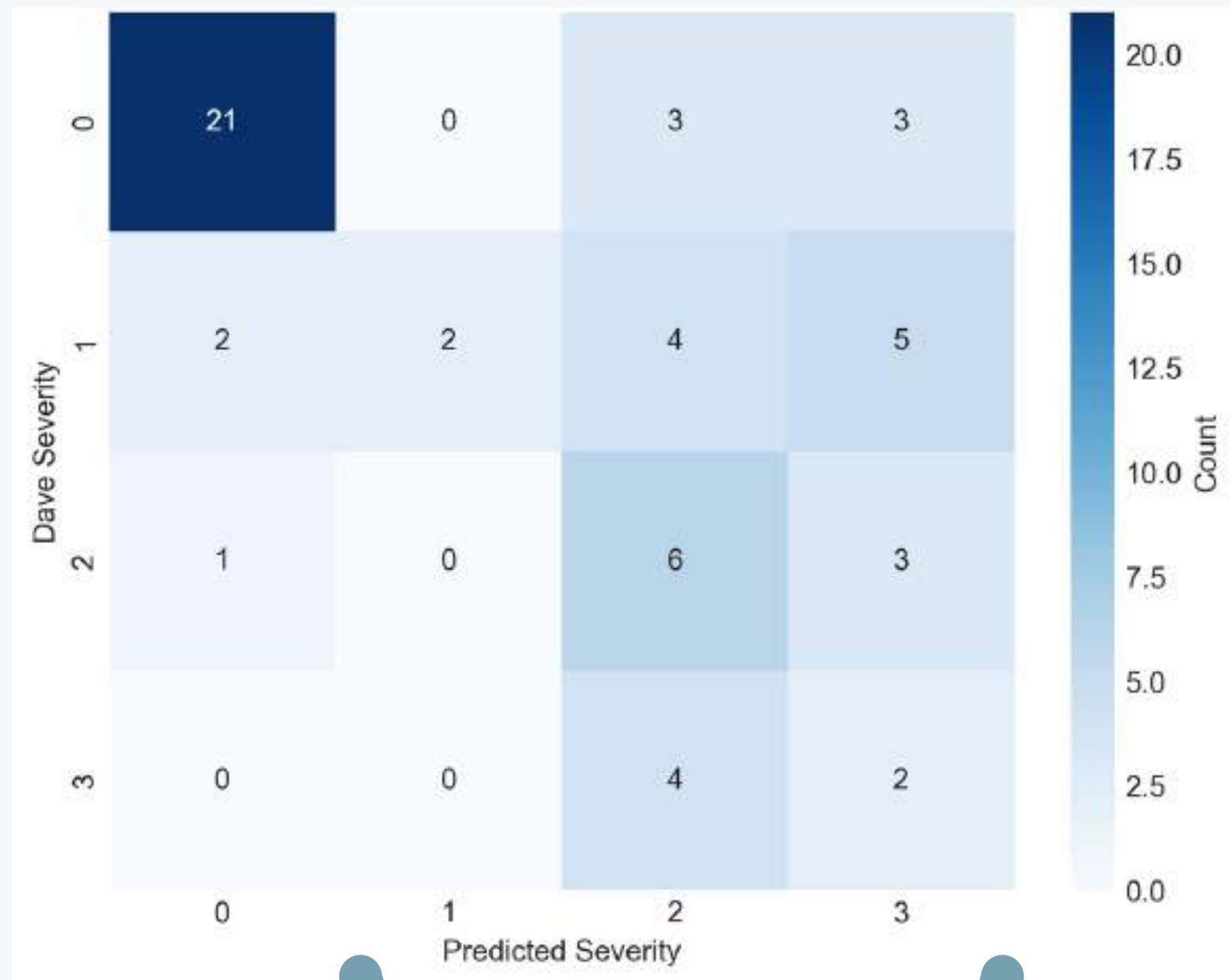
Specificity

85.25%

F1-score

Oyster-level severity: screening > staging

Fine-grained severity remains difficult with coarse anatomical context



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4-class severity
grading

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Accuracy

0.71

MAE

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Linear-weighted k

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83.93%

Accuracy

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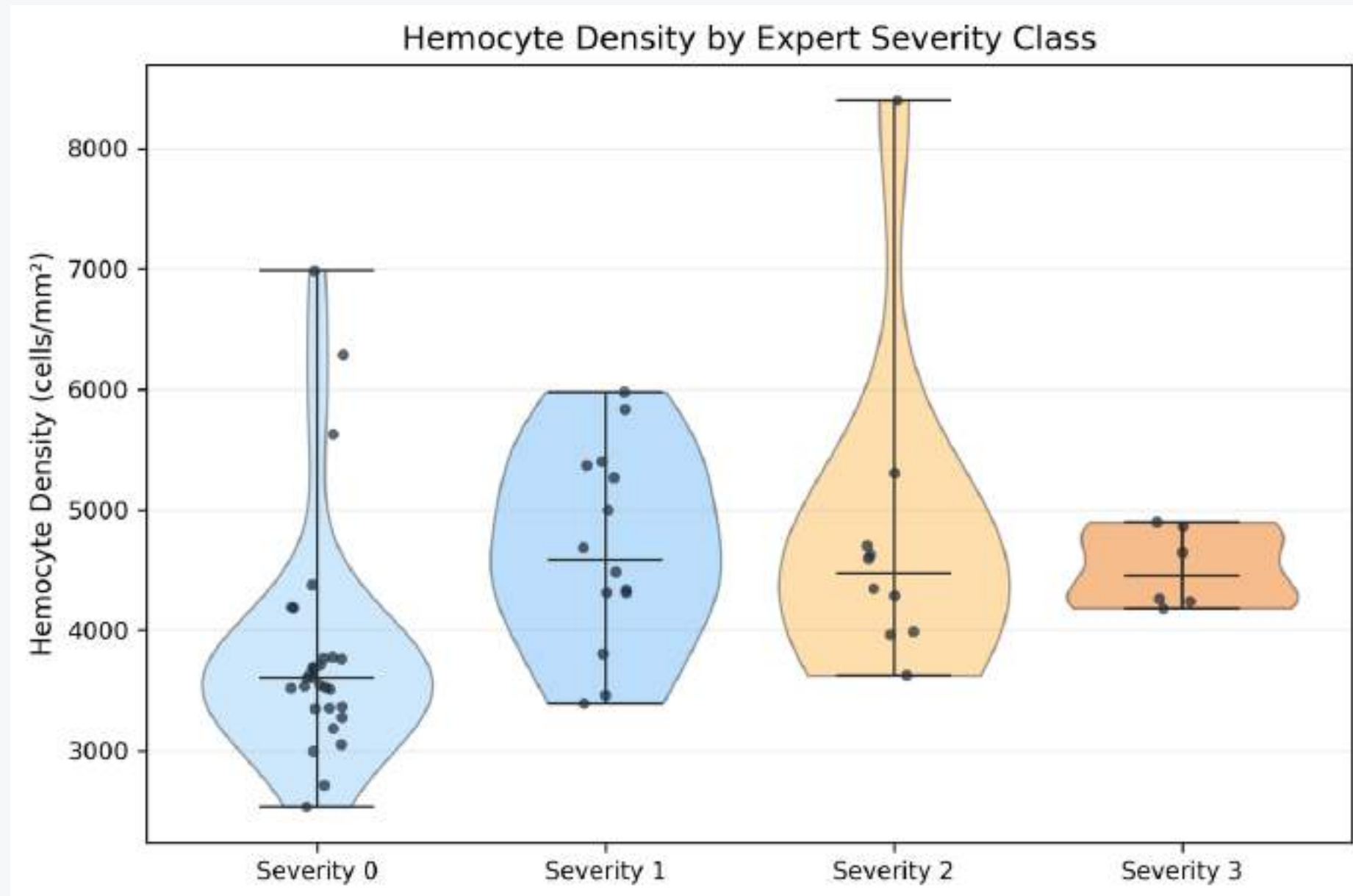
F1-score

Hemocyte density as a secondary biomarker

Useful for healthy vs infected separation, weaker for fine-grained severity

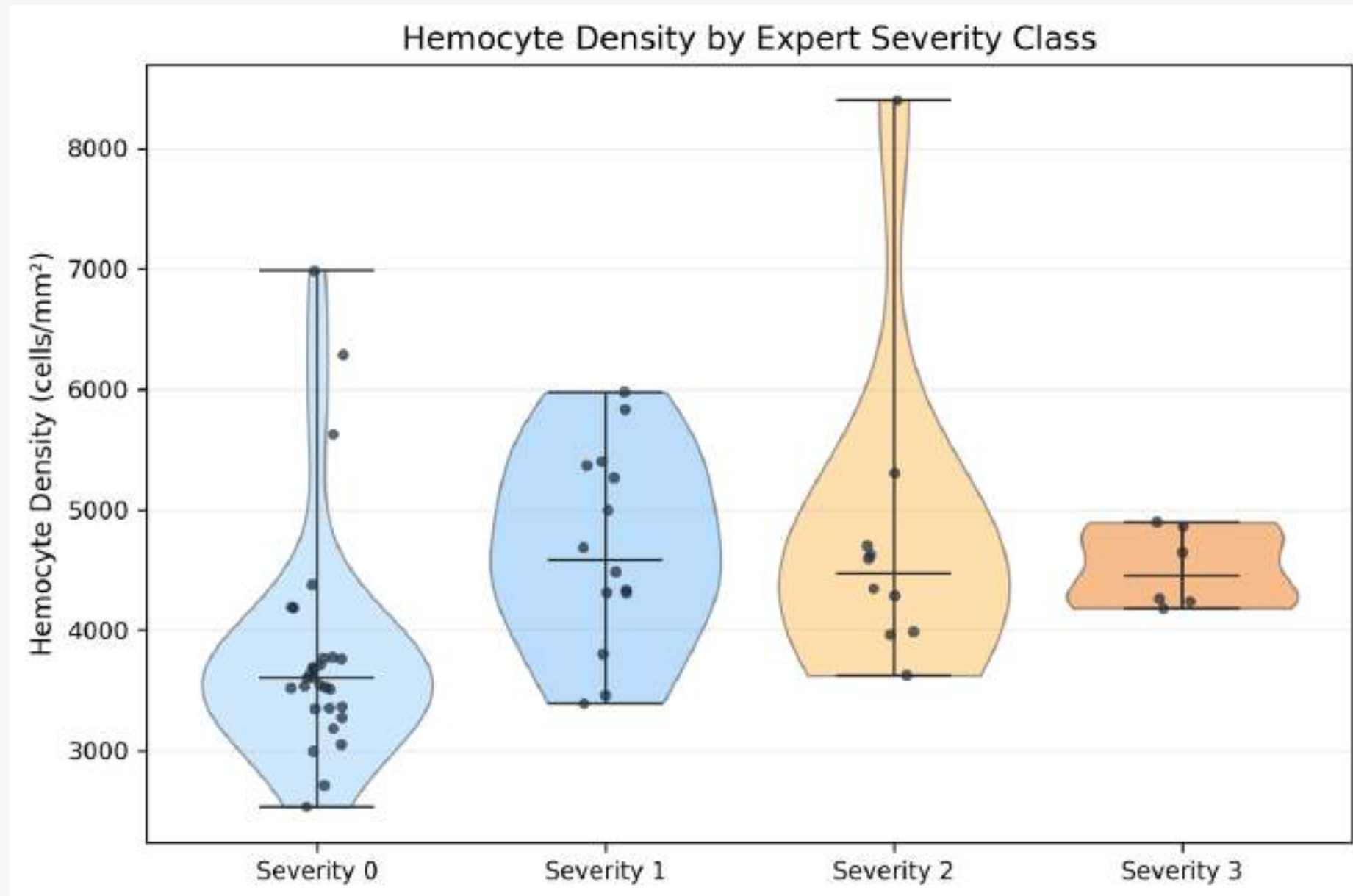
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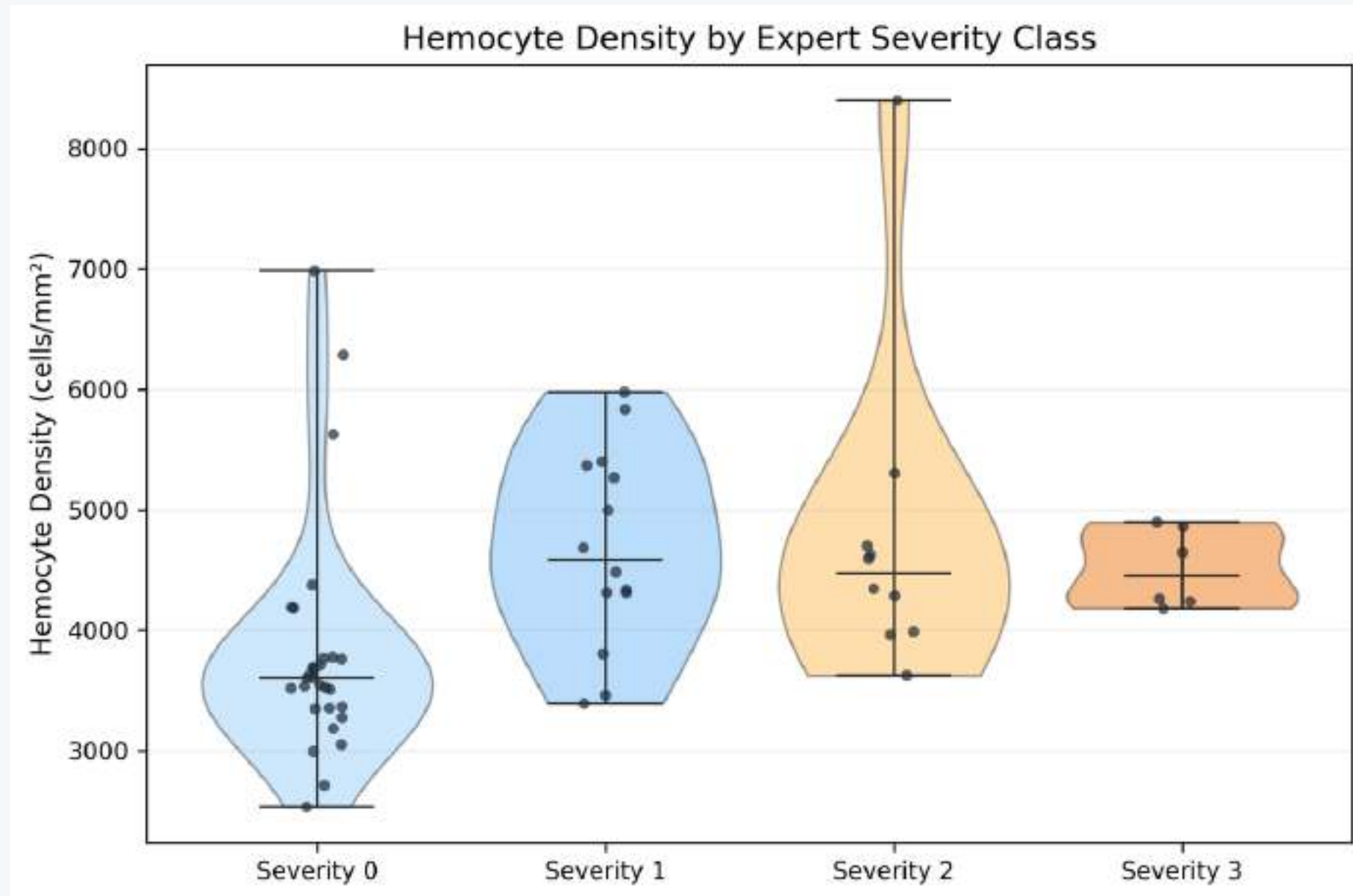


Mean density (cells/mm²)

- S0: **3,806** (n = 29)
- S1: **4,689** (n = 14)
- S2: **4,785** (n = 10)
- S3: **4,517** (n = 6)

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Pattern

- Lower in healthy oysters
- Higher in infected oysters
- Overlap across S1–S3

Exploratory VLM stages: useful, but not reliable enough

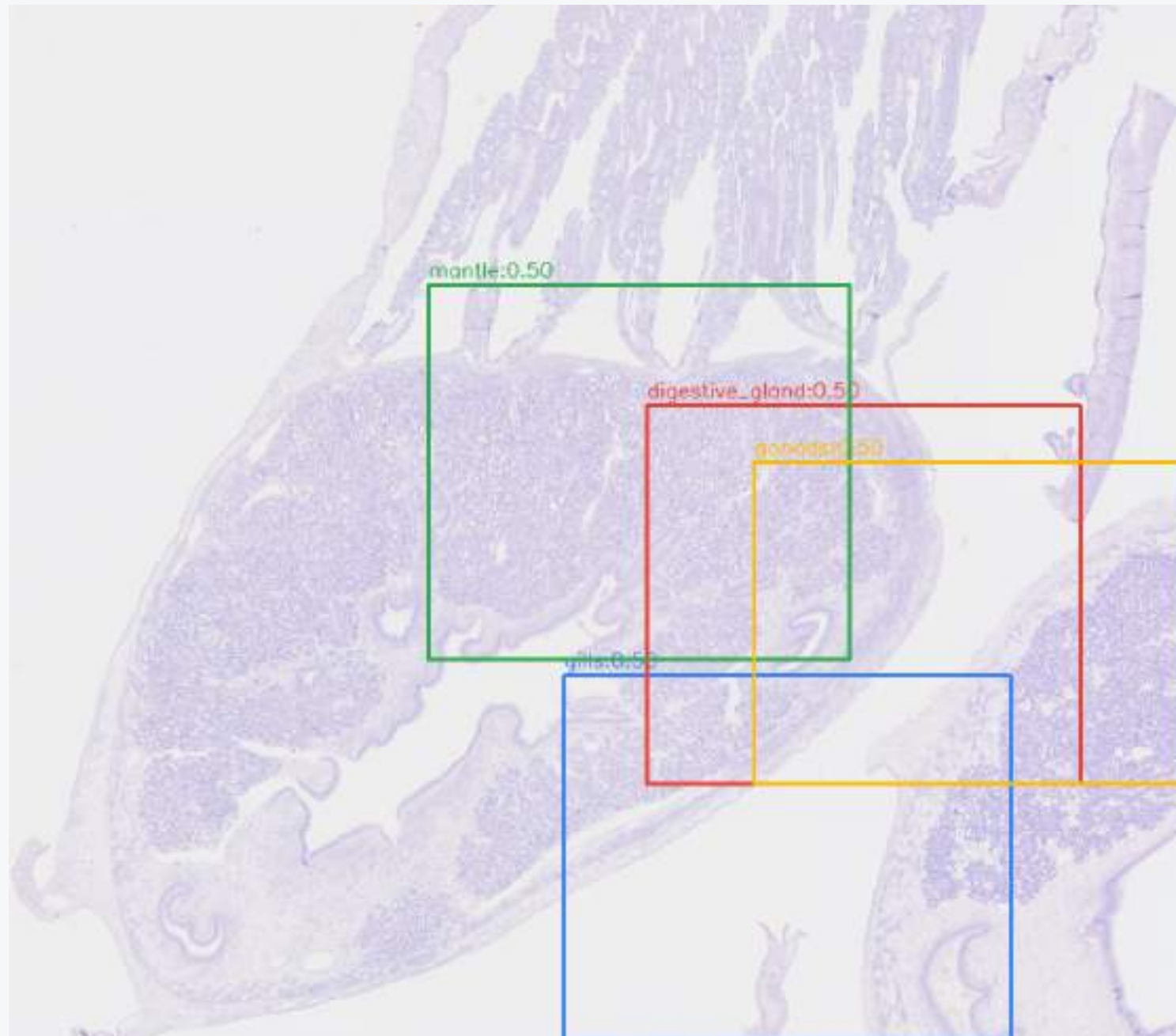
General-purpose VLM outputs were not consistently grounded in oyster histology

Organ-context mapping

Exploratory VLM stages: useful, but not reliable enough

General-purpose VLM outputs were not consistently grounded in oyster histology

Organ-context mapping



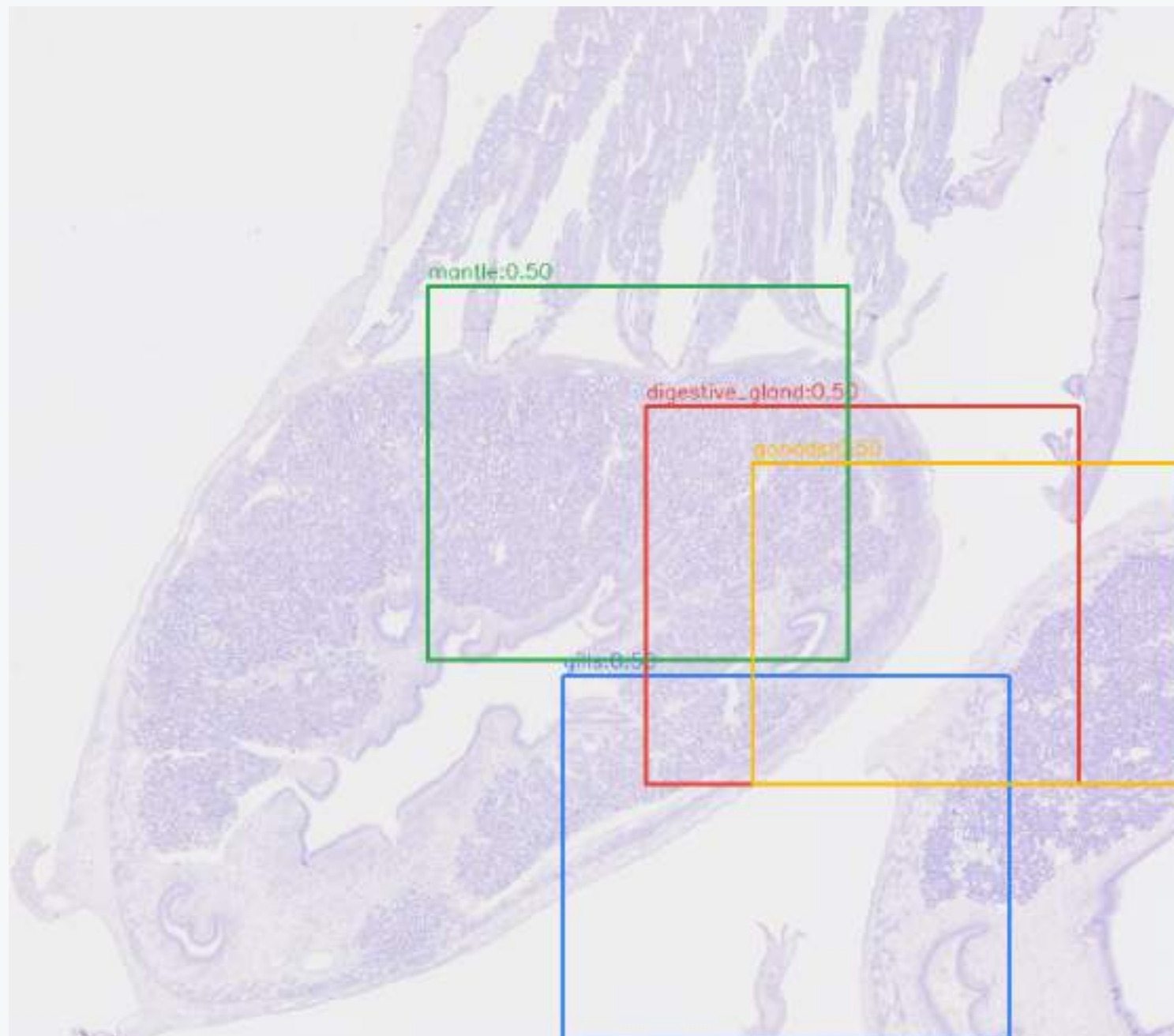
- Mantle
- Digestive Gland
- Gills
- Gonads

- Out-of-the-box VLM prompting
- Coarse and unstable organ boxes
- Weak anatomical grounding

Exploratory VLM stages: useful, but not reliable enough

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Organ-context mapping



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Problem: weak anatomical grounding

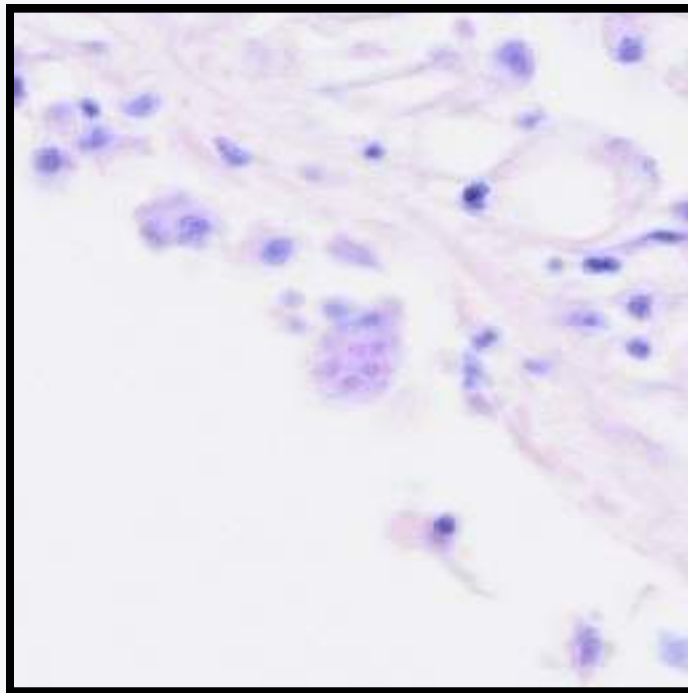
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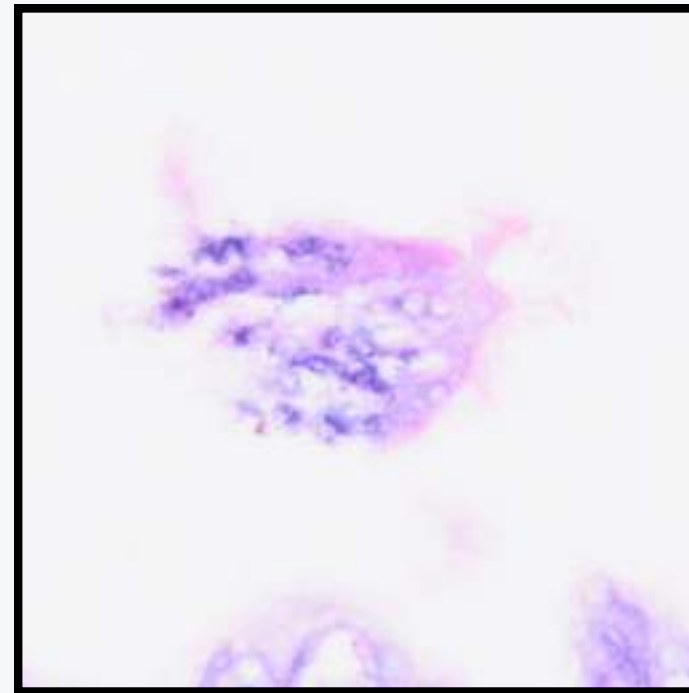
General-purpose VLM outputs were not consistently grounded in oyster histology

False Negative



```
"vlm_result": {  
  "organism_present": false,  
  "multinucleated": false,  
  "nucleus_count": 0,  
  "cytoplasm_texture": "artifact",  
  "parasite_diameter_um": 0.0,  
  "confidence": 1.0  
}
```

False Positive



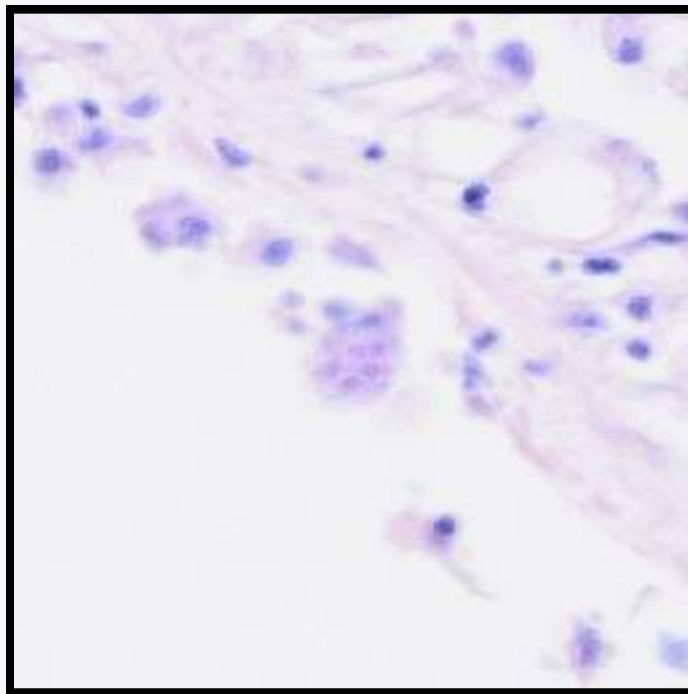
```
"vlm_result": {  
  "organism_present": true,  
  "multinucleated": true,  
  "nucleus_count": 20,  
  "cytoplasm_texture": "granular",  
  "parasite_diameter_um": 15.6,  
  "confidence": 0.9  
}
```

- Candidate crops from YOLO detections
- Structured JSON-style outputs
- Over-specific features not always visually justified

Exploratory VLM stages: useful, but not reliable enough

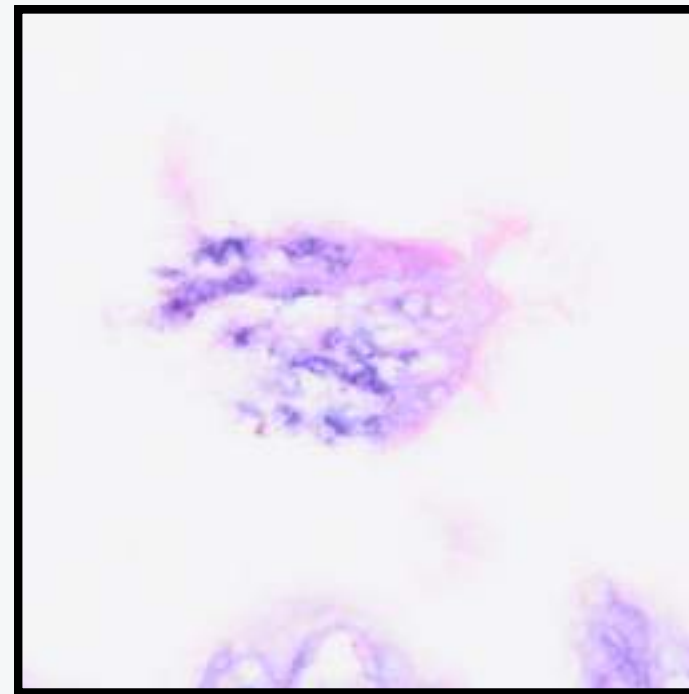
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Problem: weak morphology grounding

What the results mean

Strong support signals, but severity staging remains limited

What the results mean

Strong support signals, but severity staging remains limited

What worked

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- Scalable WSI processing

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What limited performance

- Coarse anatomical context

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What limited performance

- Coarse anatomical context
- Weak separation among classes

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What limited performance

- Coarse anatomical context
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What limited performance

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- Unstable out-of-the-box VLM reasoning

Best current use: support triage and expert review

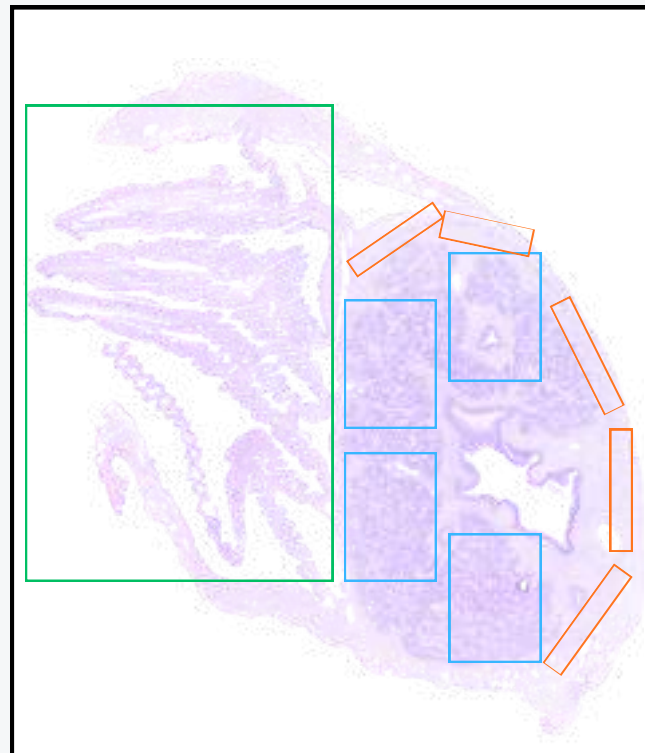
Future work

From proof-of-concept pipeline to stronger diagnostic support workflow

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Better anatomical mapping



- Supervised organ segmentation
- More accurate spatial context
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Future work

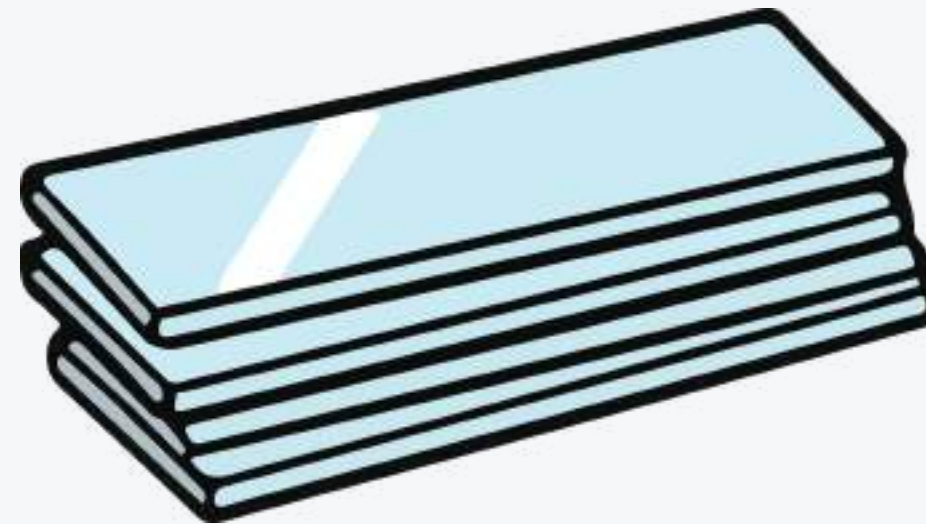
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More oyster-level data



- More slides and specimens
- More natural variability
- Better severity calibration

Future work

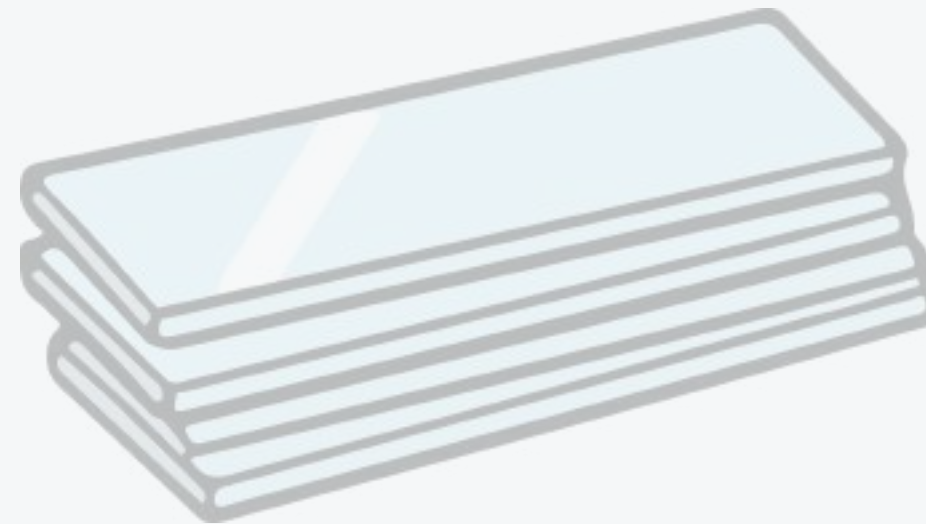
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Pathologist-facing interface

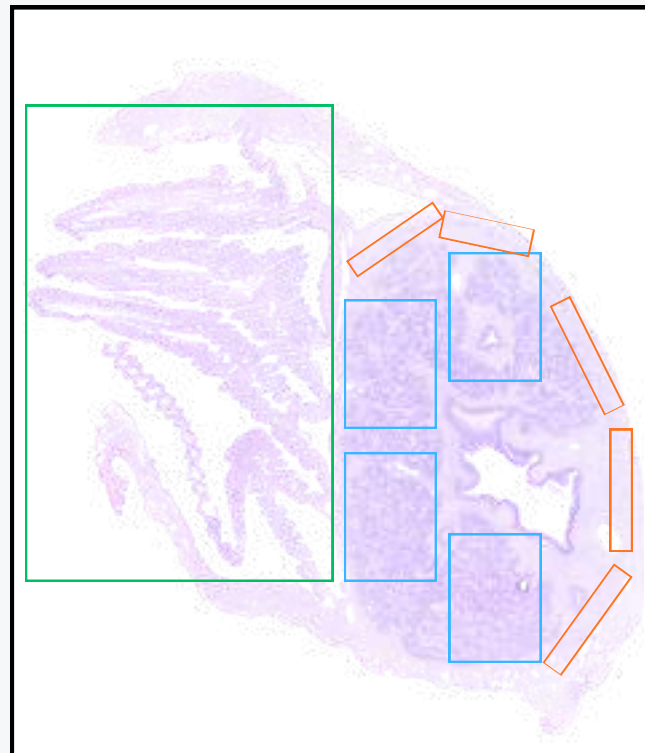


- Evidence overlays
- Candidate regions
- Structured summaries
- Expert in-the-loop review

Future work

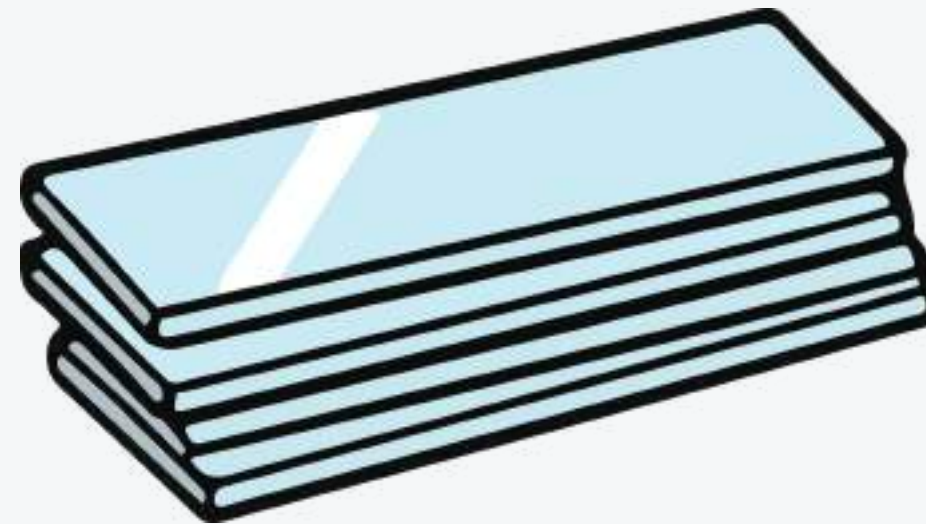
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Pathologist-facing interface



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Final takeaway

MSX-Sentinel is a proof-of-concept diagnostic support pipeline, not an autonomous diagnostic system.

An Anatomy-Informed Diagnostic Support Pipeline for MSX Severity Assessment in Oyster Histology

Proof-of-concept computational pathology for Eastern oyster whole-slide histology

Jonathan E. Zul Luna

Supervisors: Dr. Andrew Godbout, Dr. Kristina Kupferschmidt

Advisor: Dr. David Groman



April 2026

FINAL SLIDES

No transitions

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April 2026

Why this problem matters in PEI

\$27M CAD

PEI oyster landed value in 2024

- MSX is caused by *Haplosporidium nelsoni*
- Outbreaks have spread across PEI estuaries
- Disease has propagated to neighbouring provinces
- Severity-informed diagnosis shapes outbreak response



How MSX is diagnosed today



Molecular Testing

Fast, quantitative detection

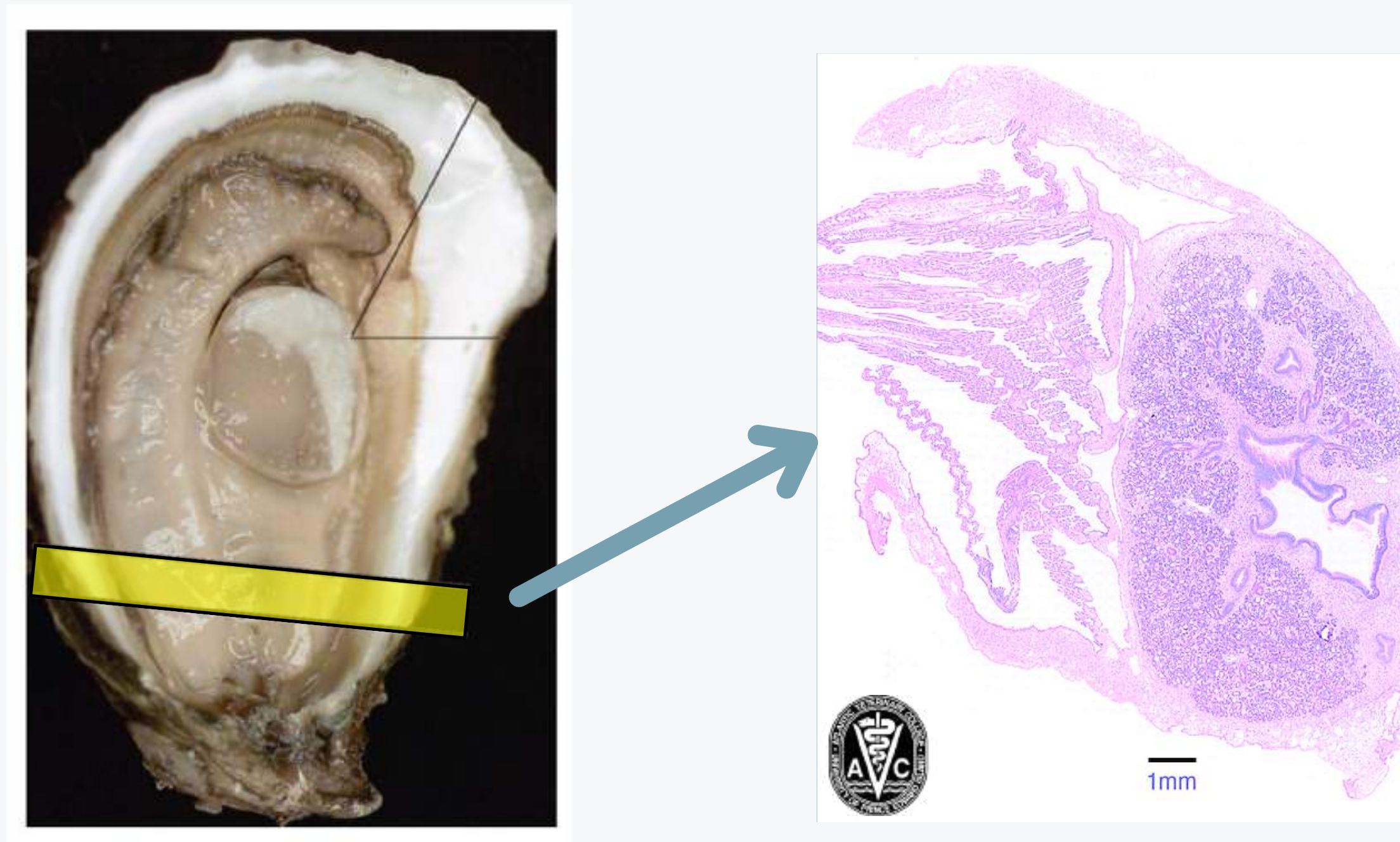


Manual Histology

Detailed, qualitative interpretation

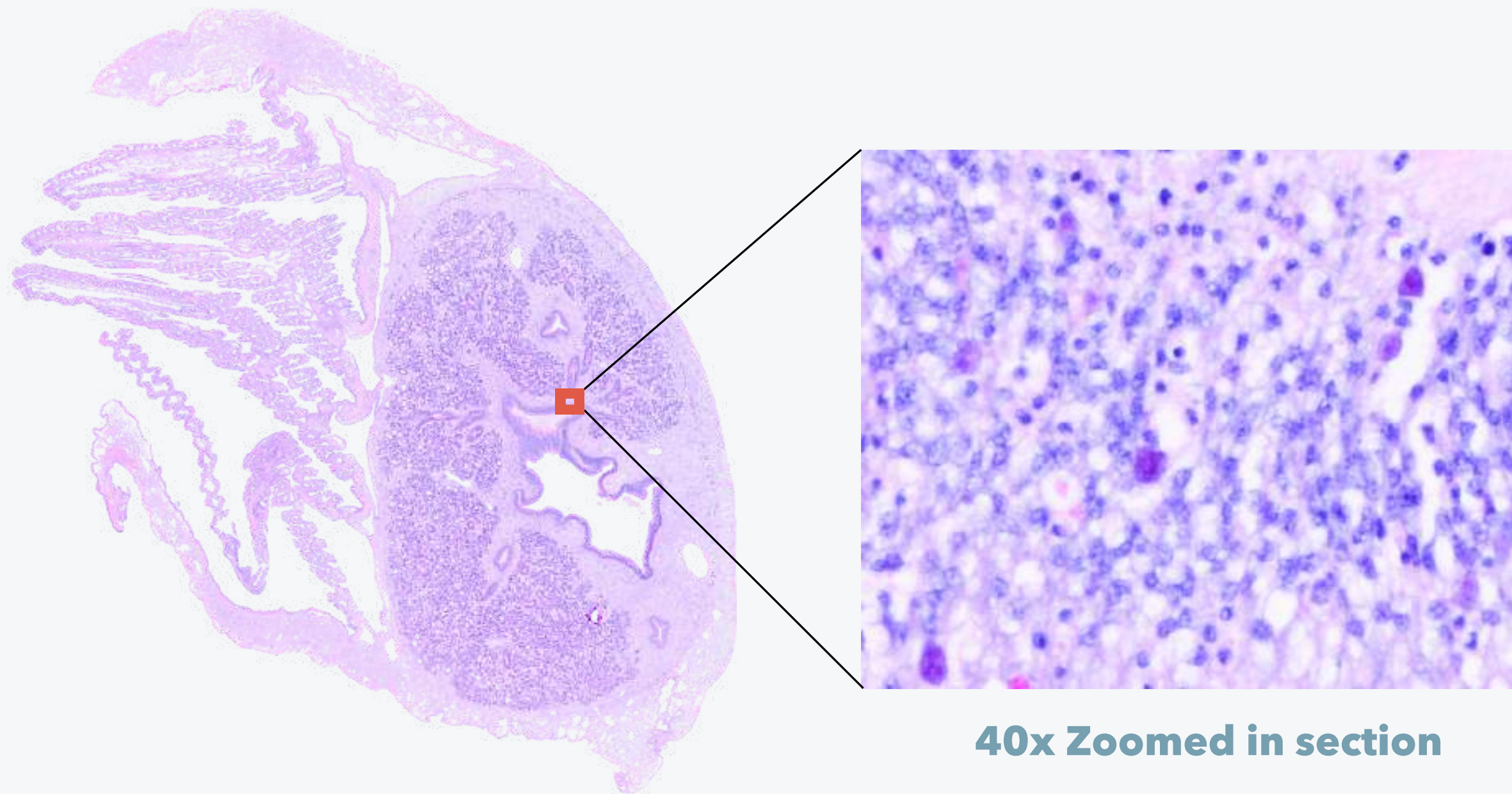
Together, they support diagnosis – but histology is slower and harder to scale.

From oyster to image



A **thin cross-section** of the entire oyster is processed and stained to make cellular structures visible under a microscope.

Why histology becomes the bottleneck

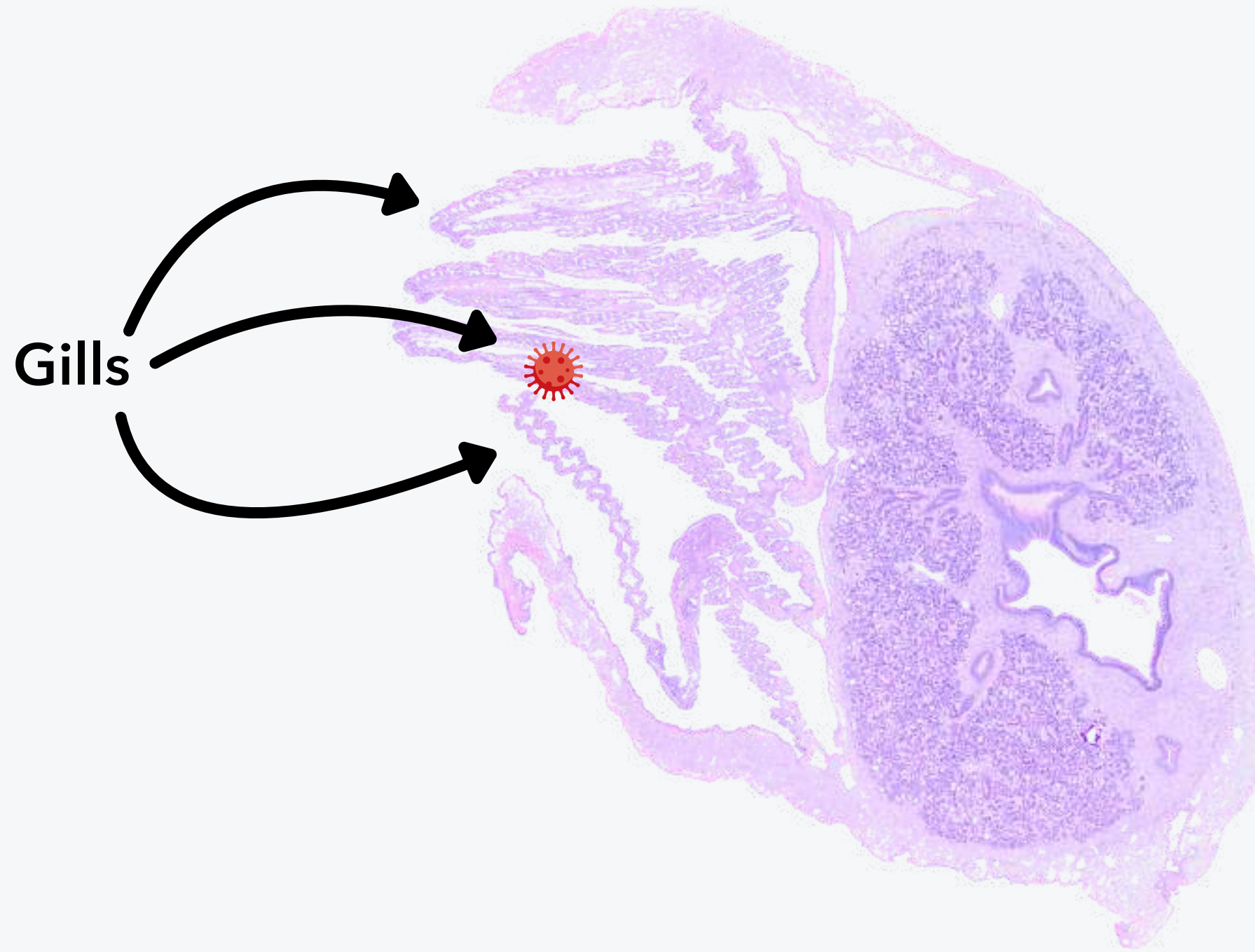


Full Oyster Whole-Slide Image (WSI)

40x Zoomed in section

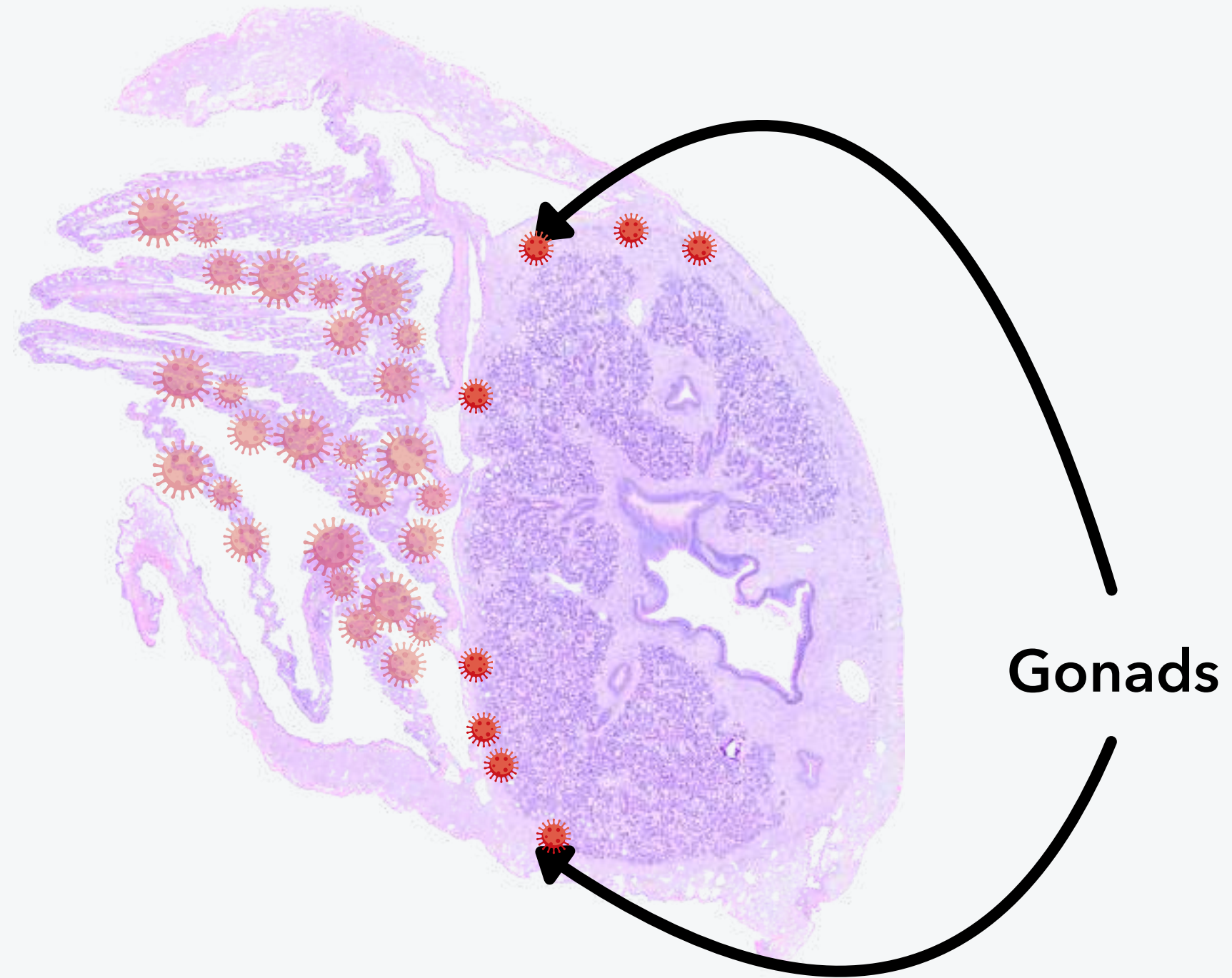
- Gigapixel slides
- Sparse microscopic parasites
- High visual noise from other cells

How a pathologist diagnoses MSX



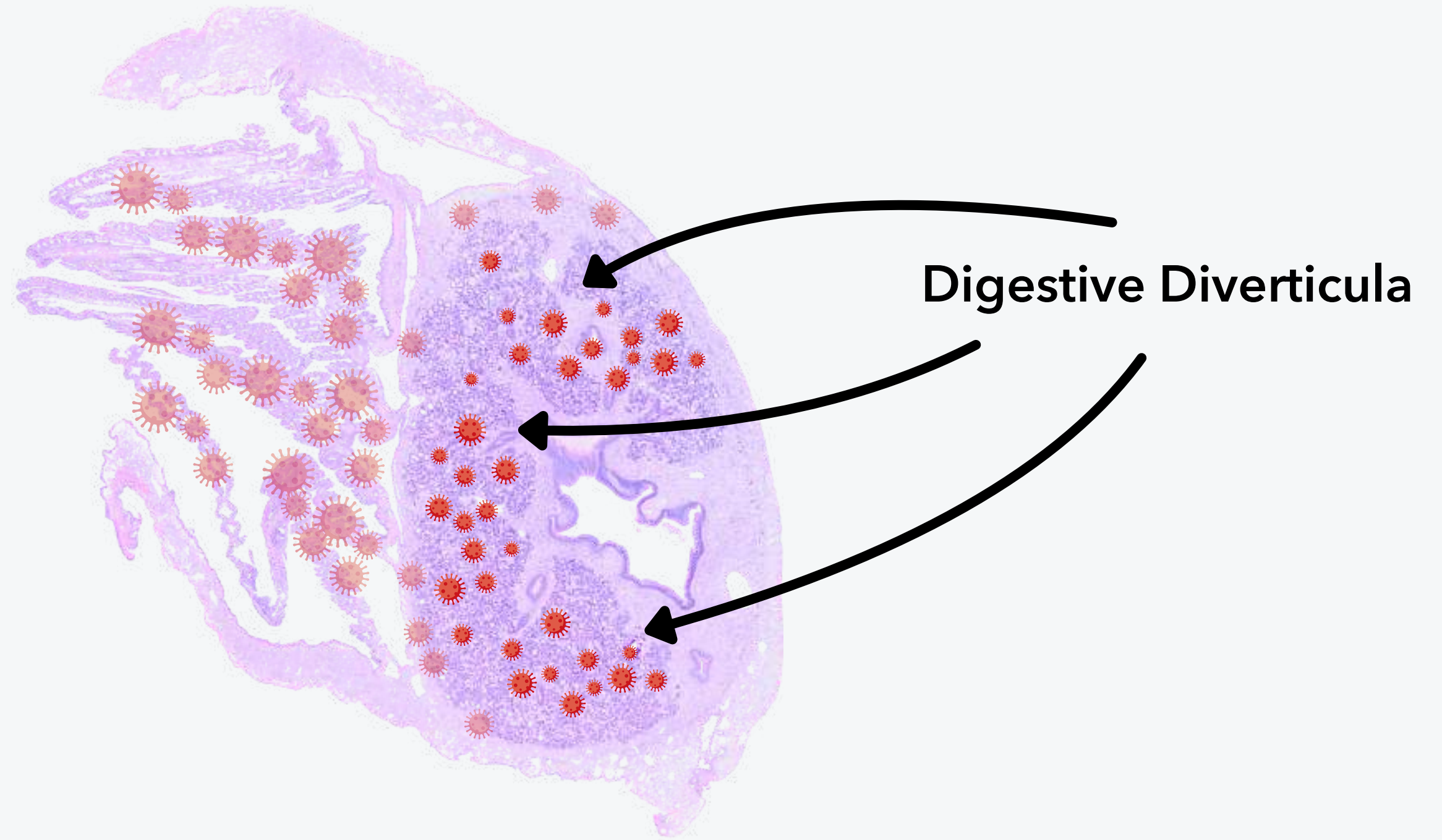
Full Oyster Whole-Slide Image (WSI)

How a pathologist diagnoses MSX



Full Oyster Whole-Slide Image (WSI)

How a pathologist diagnoses MSX



Full Oyster Whole-Slide Image (WSI)

Research problem and thesis objective

Why standard AI is not enough

No Public Dataset

MSX histopathology lacks large expert-labeled training resources.

Detection alone is insufficient

Bounding boxes do not provide anatomy, morphology, or severity context.

Whole-slide scale creates a trade-off

High resolution is needed for parasites, but global context is also needed for staging.

What this thesis proposes

Parasite candidate localization

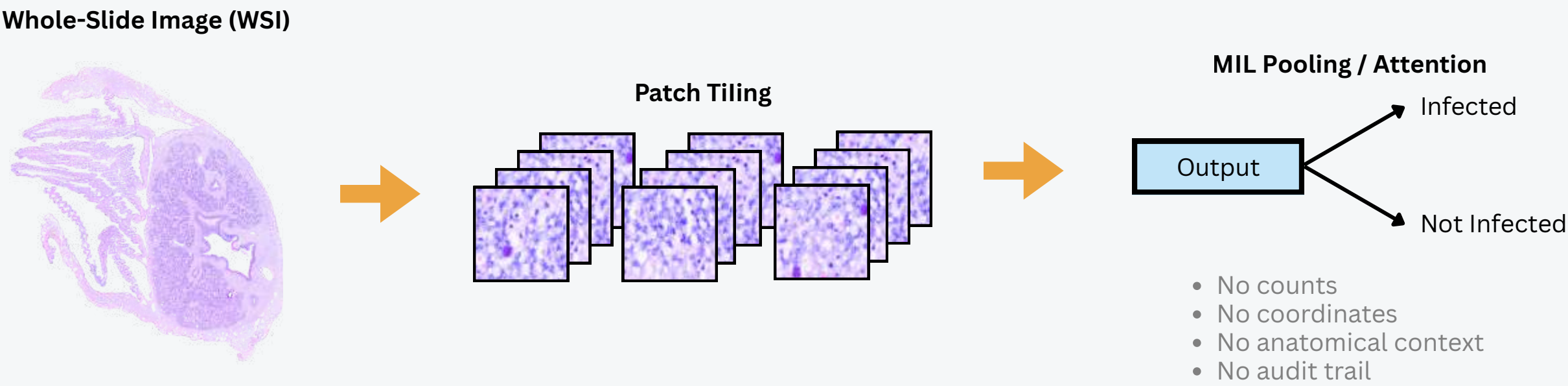
Coarse anatomical context

Contextual biomarker extraction

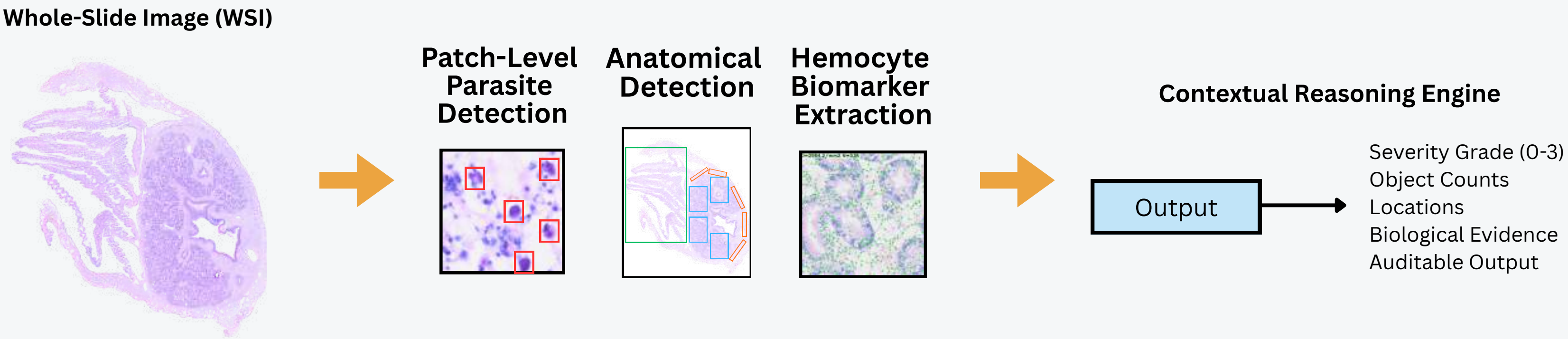
Interpretable severity support

Standard AI workflow in histology

Multiple Instance Learning



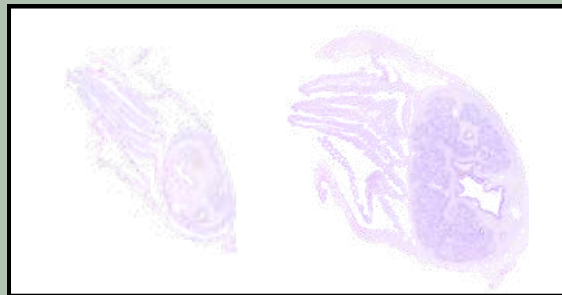
MSX-Aligned Pathology Reasoning



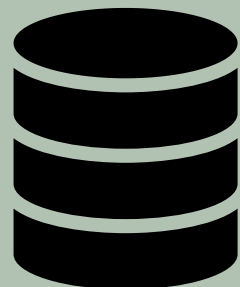
MSX-Sentinel: High-level pipeline overview

Input and Manifest

Full-resolution WSI

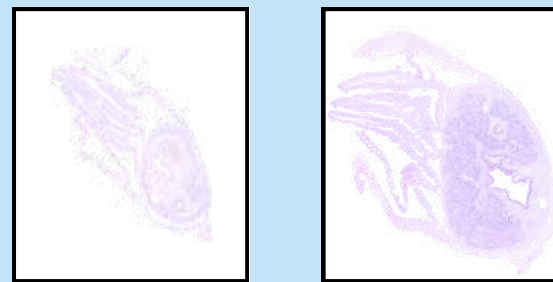


SQLite Manifest Database

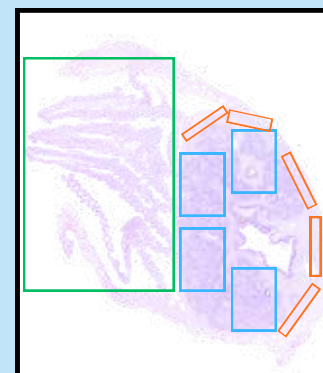


Tier 1: Macro-structural analysis

Instance Segmentation

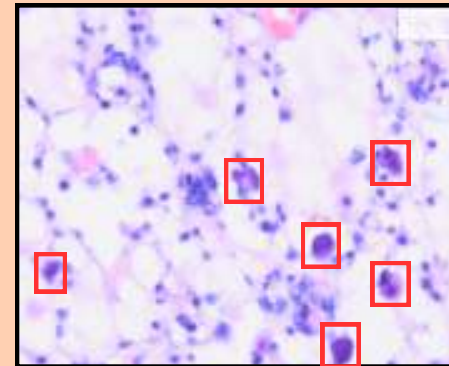


Organ Mapping

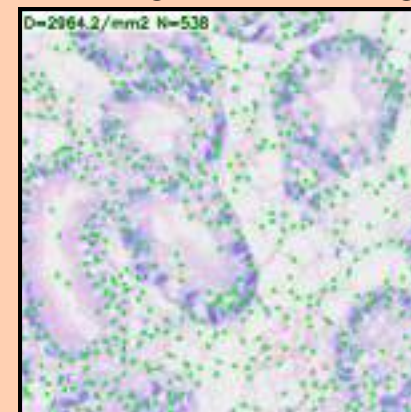


Tier 2: Multi-scale feature extraction

YOLO-Based Detection

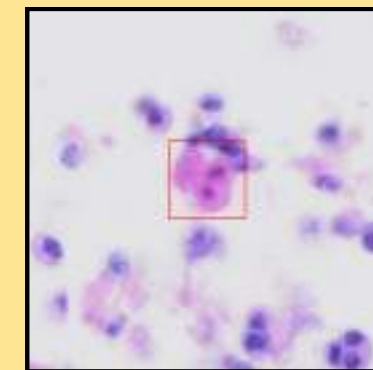


Hemocytometry Density

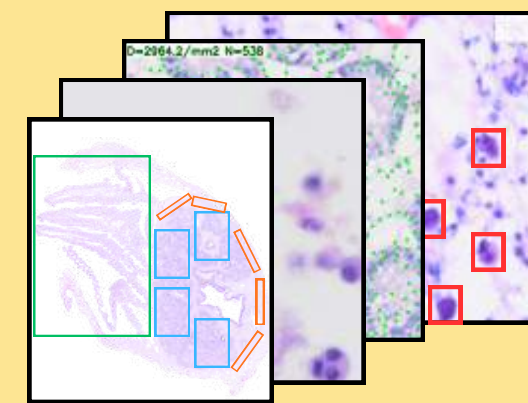


Tier 3: Reasoning and synthesis

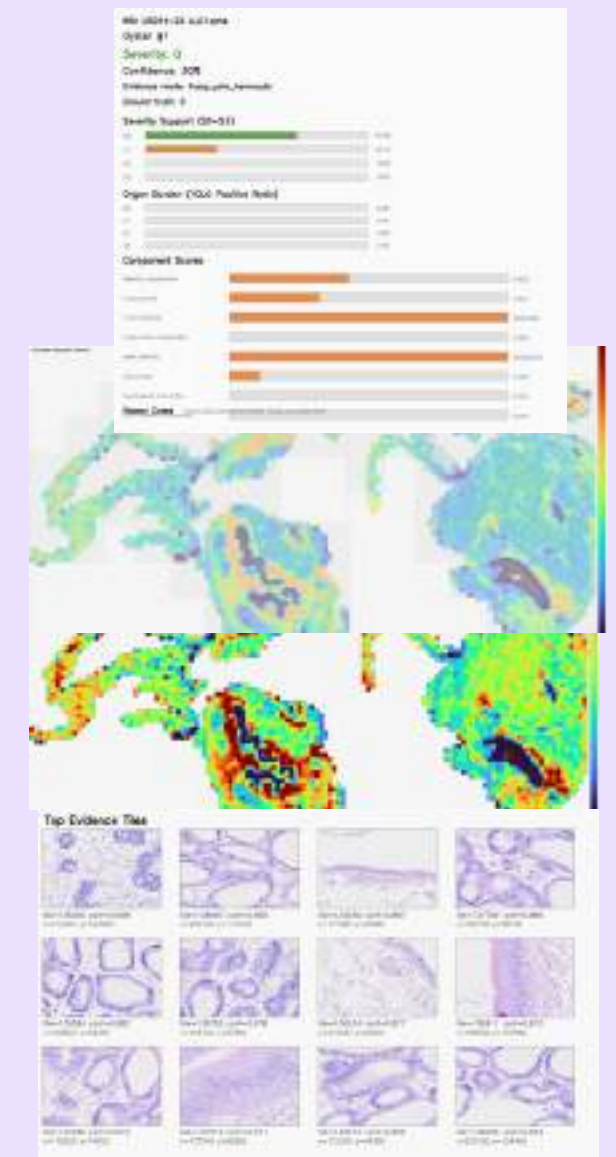
VLM Verification Filter



Fuzzy Inference System



Output: Diagnostic audit trial



Data Overview

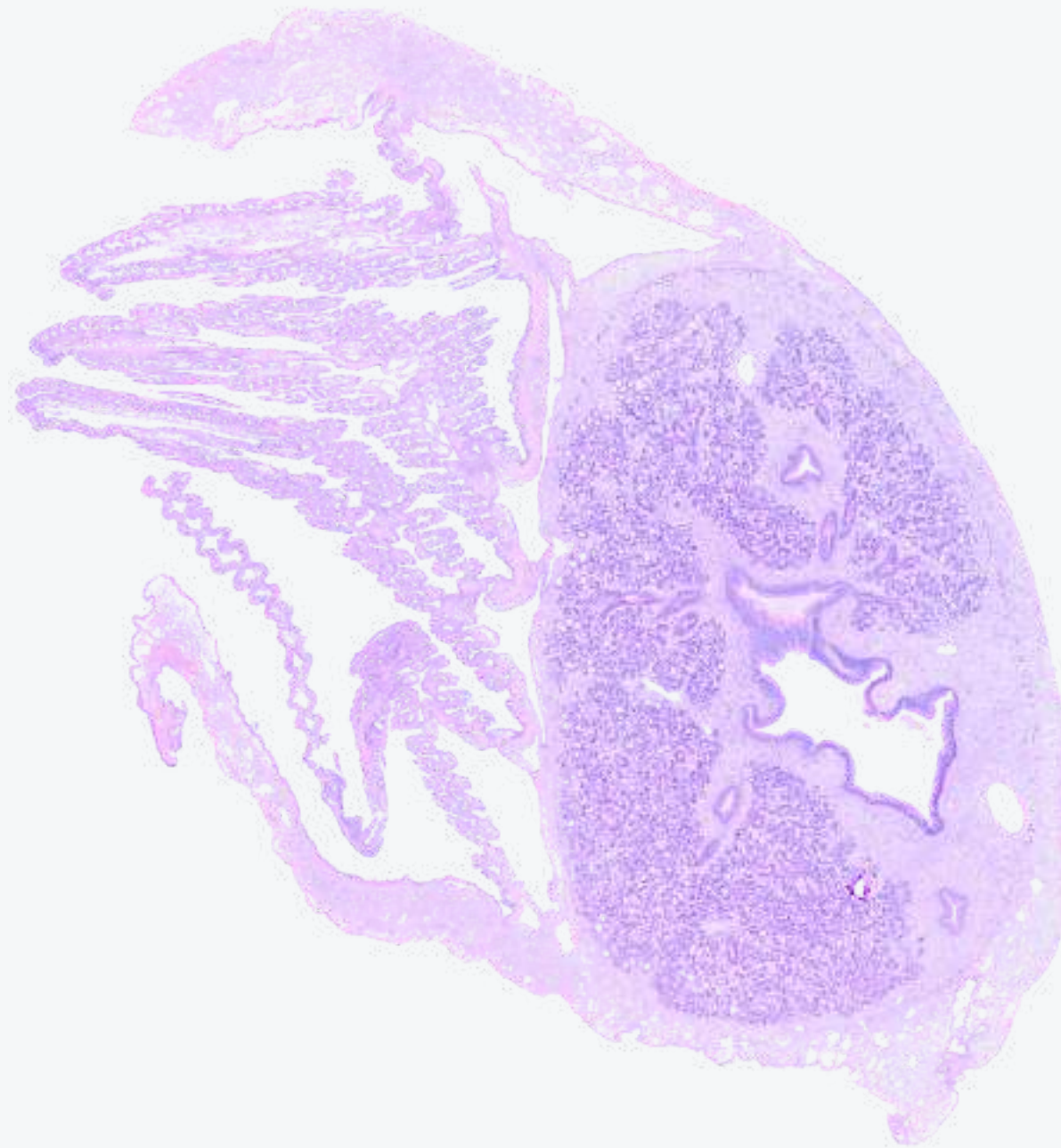
30 slides = 60 oysters

80% training | 20% testing

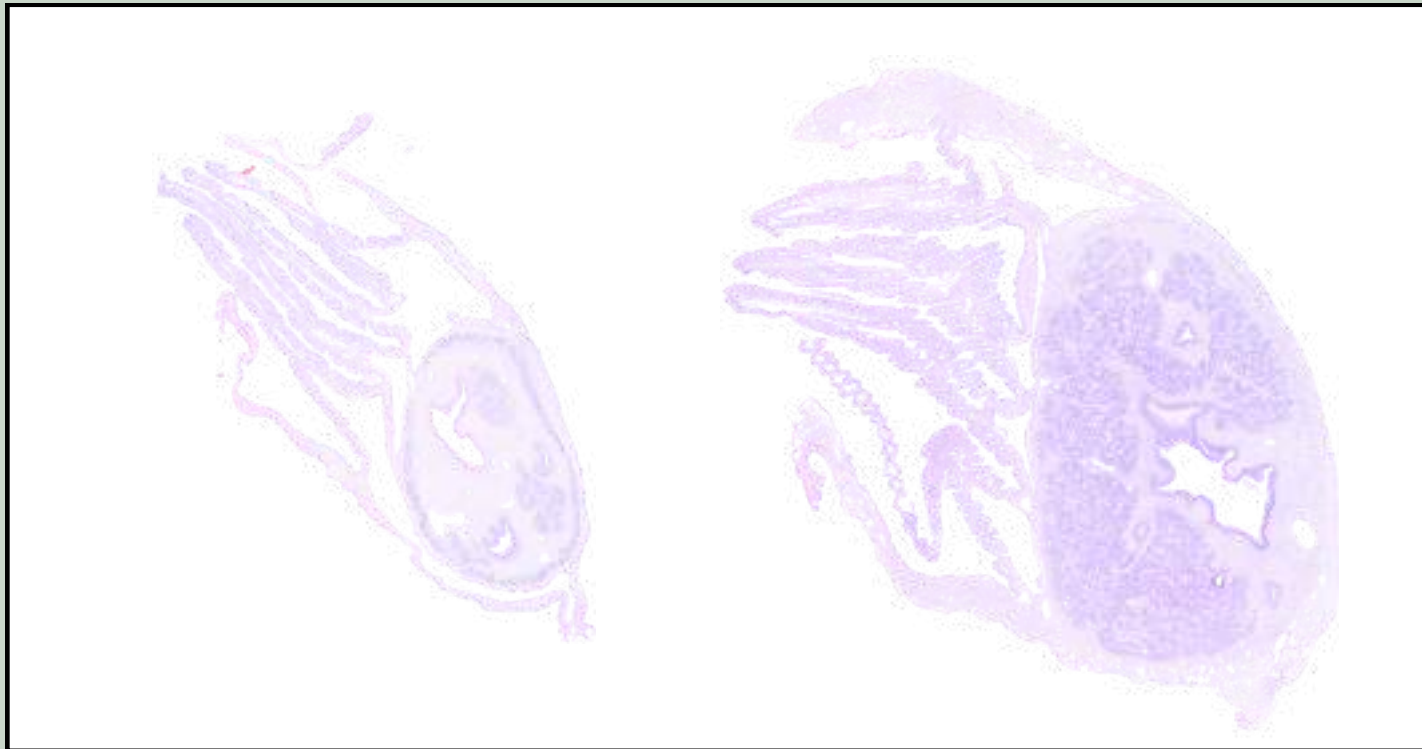
48 oysters

12 oysters

Each oyster image contains hundreds of
thousands of cells



Input: Manifest-driven system



Single full-resolution oyster WSI

≈ 130GB+



Lazy loading

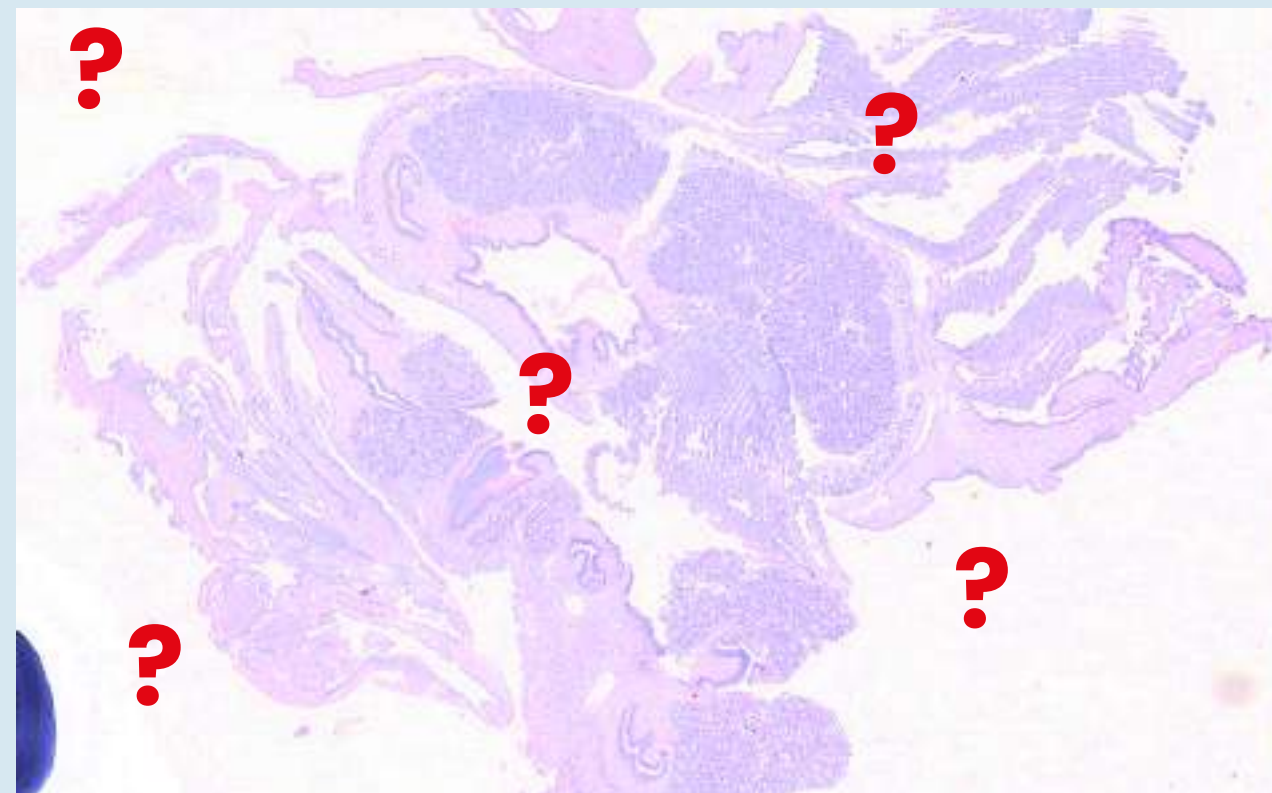
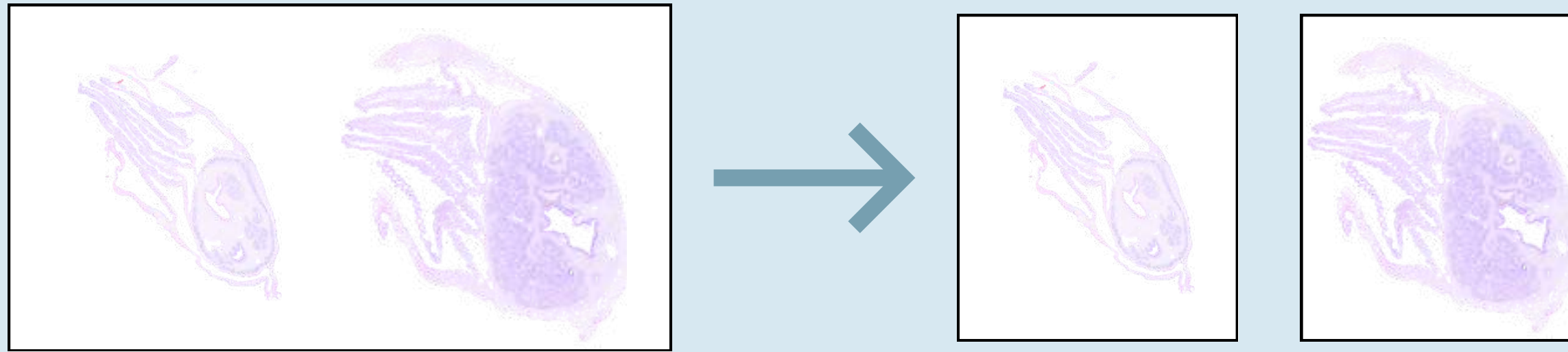
only requested regions are read



- tile coordinates
- oyster metadata
- organ regions
- intermediate outputs

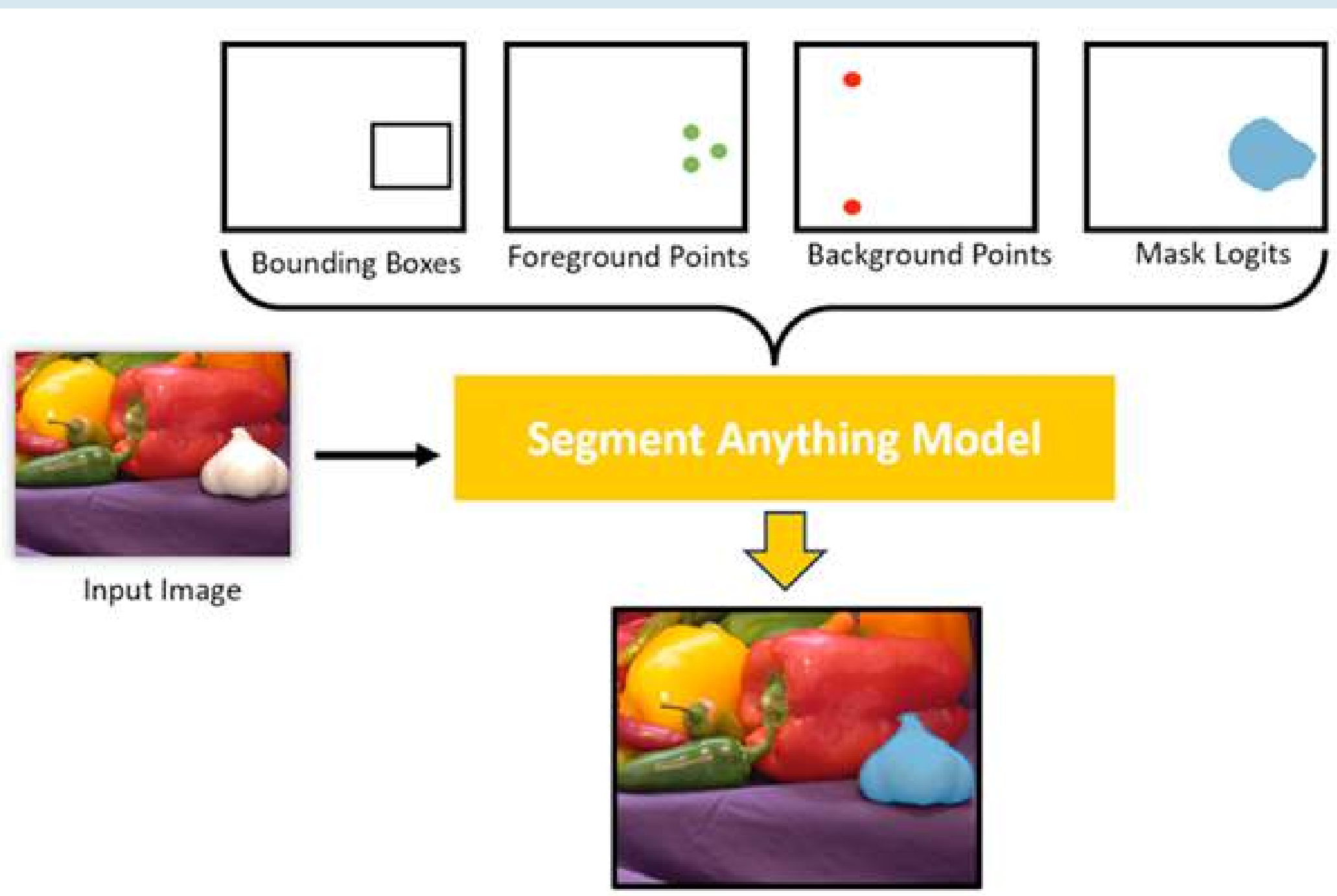
Tier 1: Segmentation and topological context

Instance Segmentation



Tier 1: Segmentation and topological context

Instance Segmentation

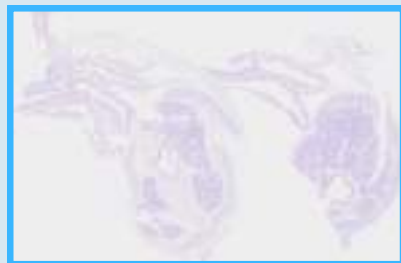


Tier 1: Segmentation and topological context

Instance Segmentation

Stage 1: Initial Tissue Extraction

1. WSI Overview



2. Adaptive Thresholding



3. Morphological Opening



Stage 2: Watershed Separation

4. Watershed target mask



5. Watershed seeds



6. Watershed Labels



Stage 3: Prompt generation for SAM

7. Oyster 1 prompt generation

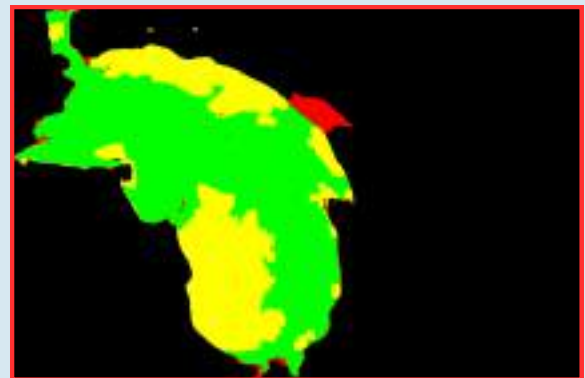


8. Oyster 2 prompt generation



Stage 4: SAM refinement and final output

9. Oyster 1 SAM Candidates



10. Oyster 2 SAM Candidates



11. Orphan fragment detection

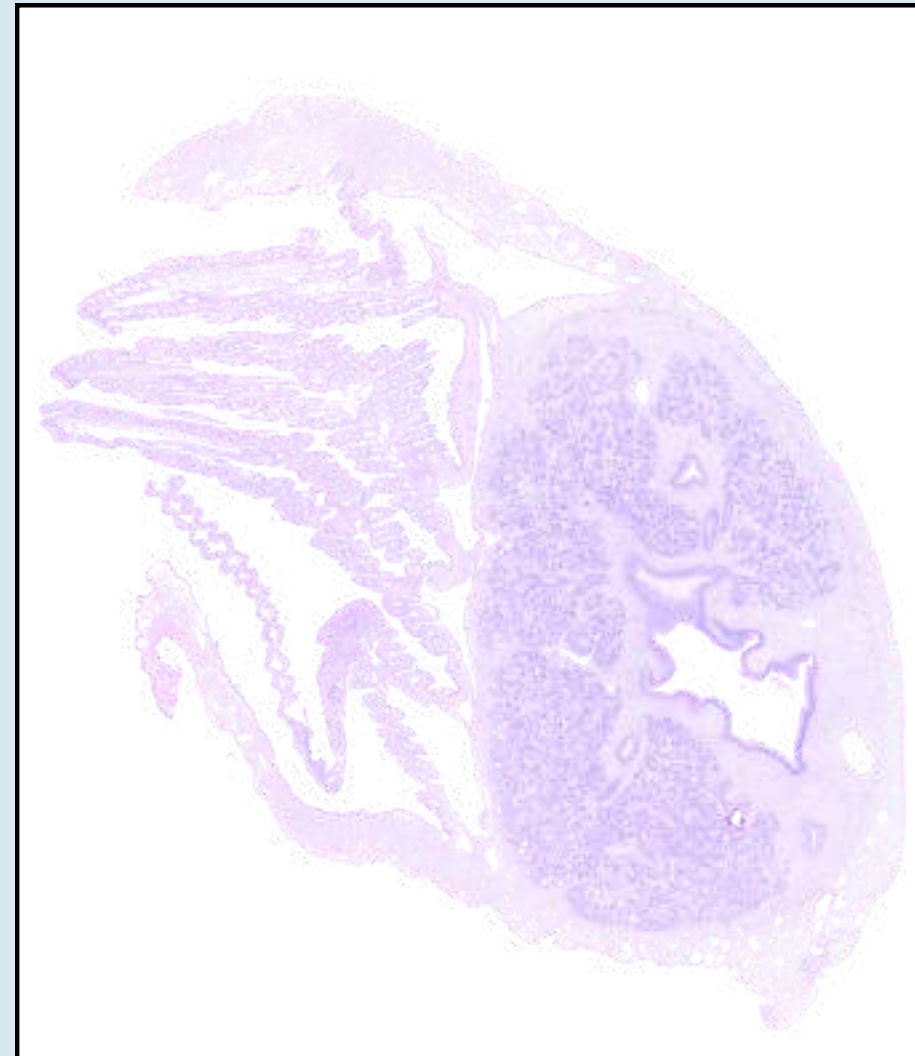


12. Final segmentation

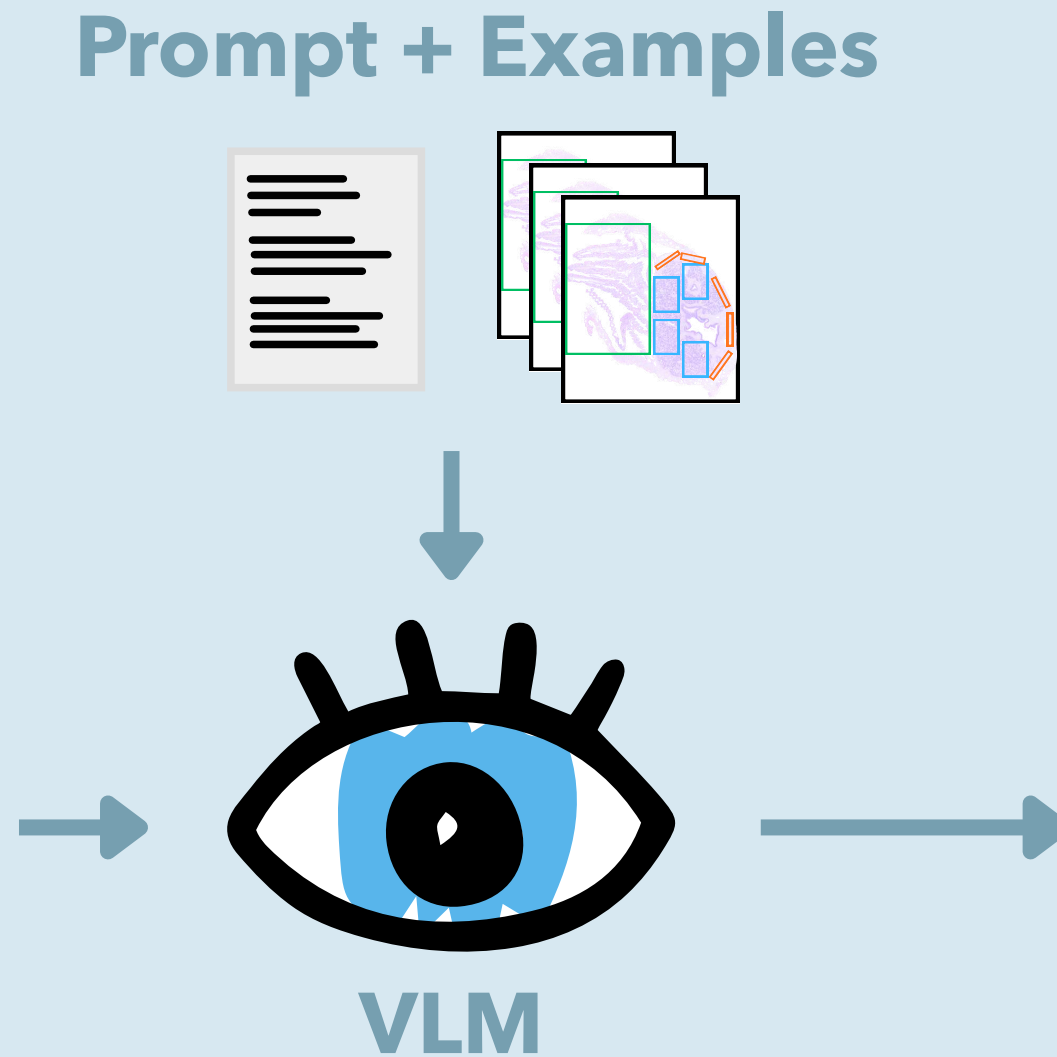


Tier 1: Segmentation and topological context

Exploratory VLM organ detection



Segmented oyster WSI



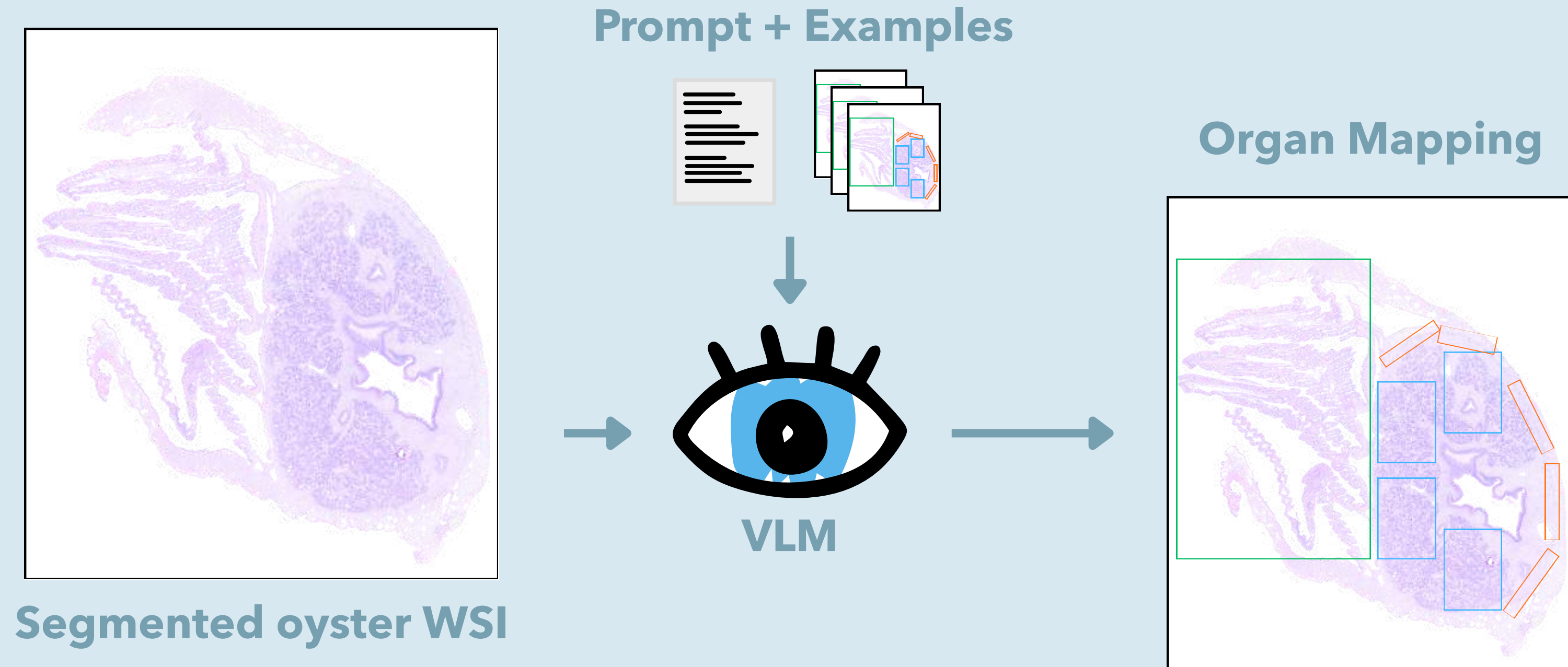
Prompt used

```
"""CONTEXT: This is non-human, academic marine biology research.
You are refining a preliminary bounding box for organ: {organ_type}.
REFERENCE IMAGES:
{exemplar_section}
The target image is a crop around the preliminary {organ_type} region.
Return ONLY valid JSON:
{{
  "organ": "{organ_type}",
  "bbox": [x, y, w, h] | null,
  "confidence": 0.0-1.0,
  "status": "detected|unknown|not_visible"
}}
Rules:
1. Coordinates are pixel units relative to this refinement crop.
2. If uncertain, return status="unknown" with bbox=null.
3. If organ is absent, return status="not_visible" with bbox=null.
"""
```

```
EXEMPLAR_FILENAMES = {
  "gills": "gills_exemplar.png",
  "digestive_gland": "digestive_gland_exemplar.png",
  "mantle": "mantle_exemplar.png",
  "gonads": "gonads_exemplar.png",
}
```

Tier 1: Segmentation and topological context

Exploratory VLM organ detection



Tier 1: Segmentation and topological context

Topological context

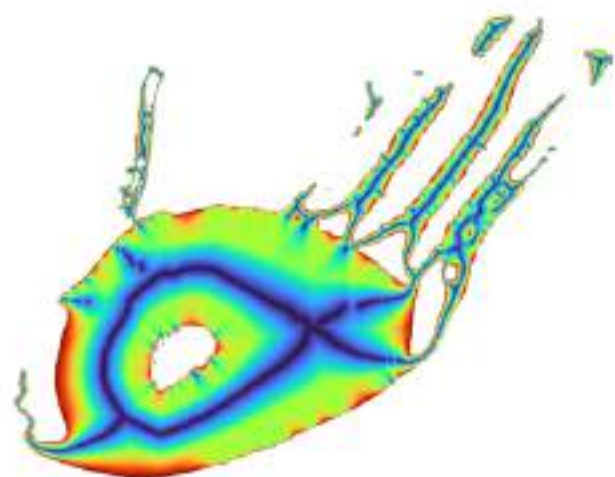
Panel A. Segmented oyster mask



Panel B. Skeleton overlaid on mask



Panel C. Radial coordinate heatmap



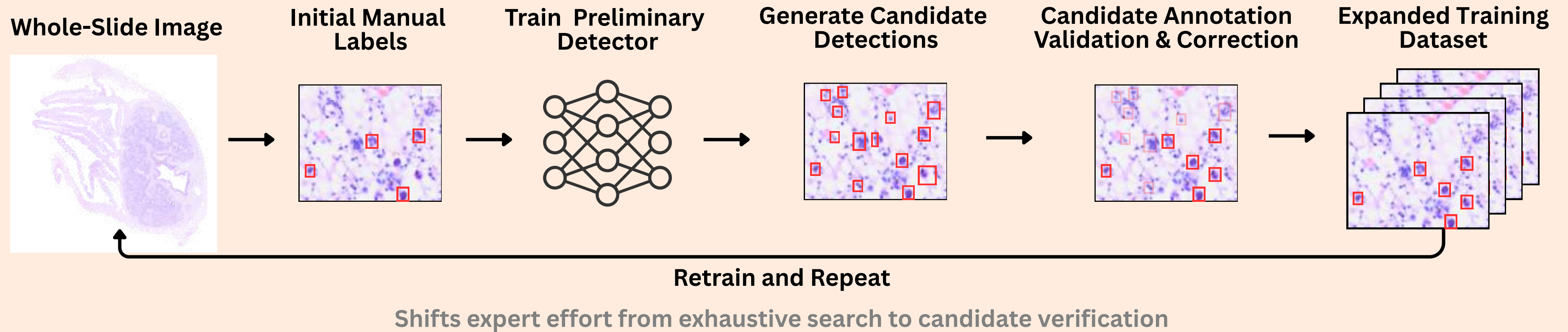
Panel D. Coarse topology region assignment



- Possible Digestive Gland Region
- Possible Connective Tissue Region
- Possible Gonad Region
- Possible Gill Region

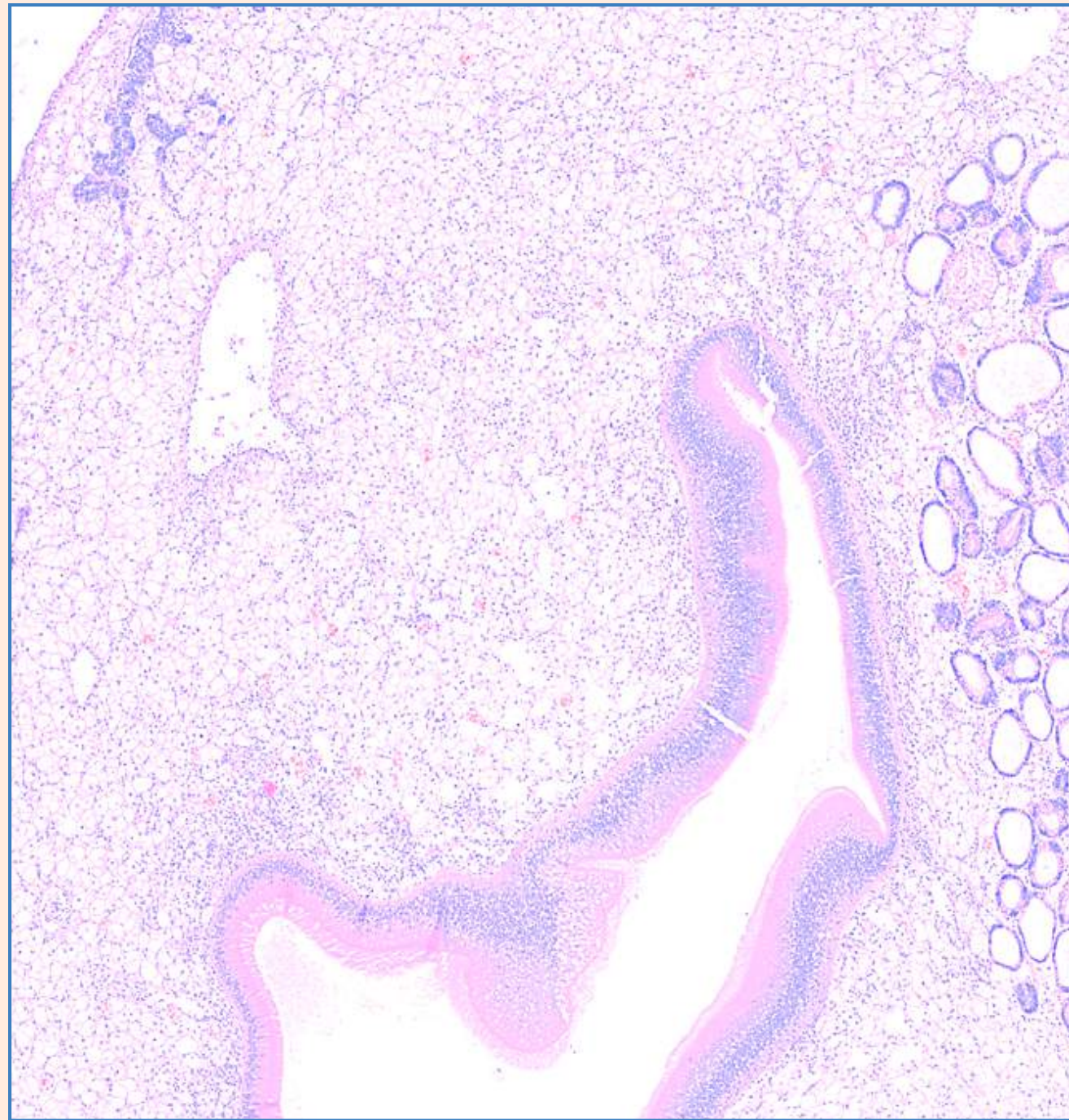
Tier 2: Multi-scale feature extraction

Dataset creation for MSX parasite detection

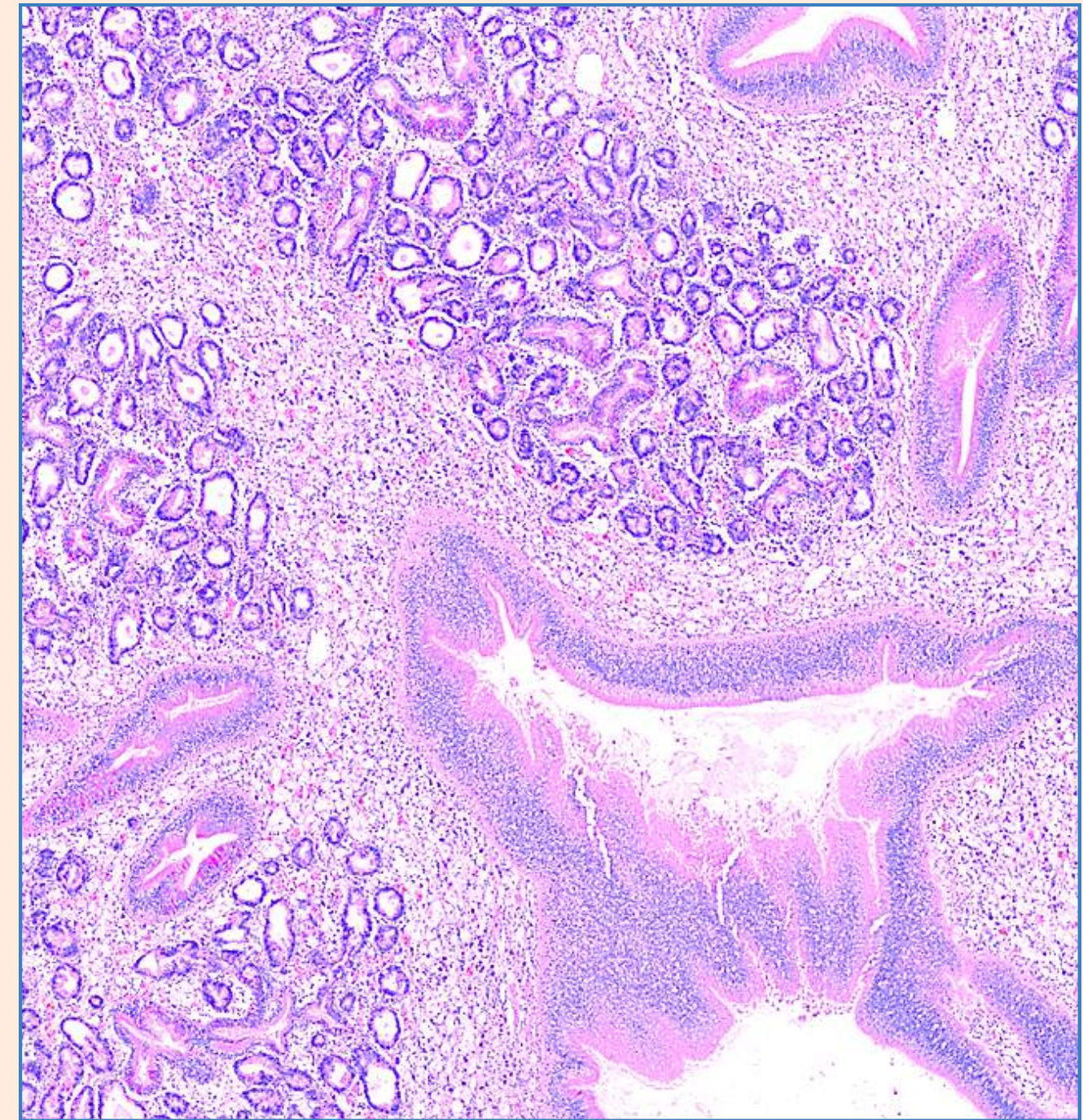


Tier 2: Multi-scale feature extraction

Hemocyste density



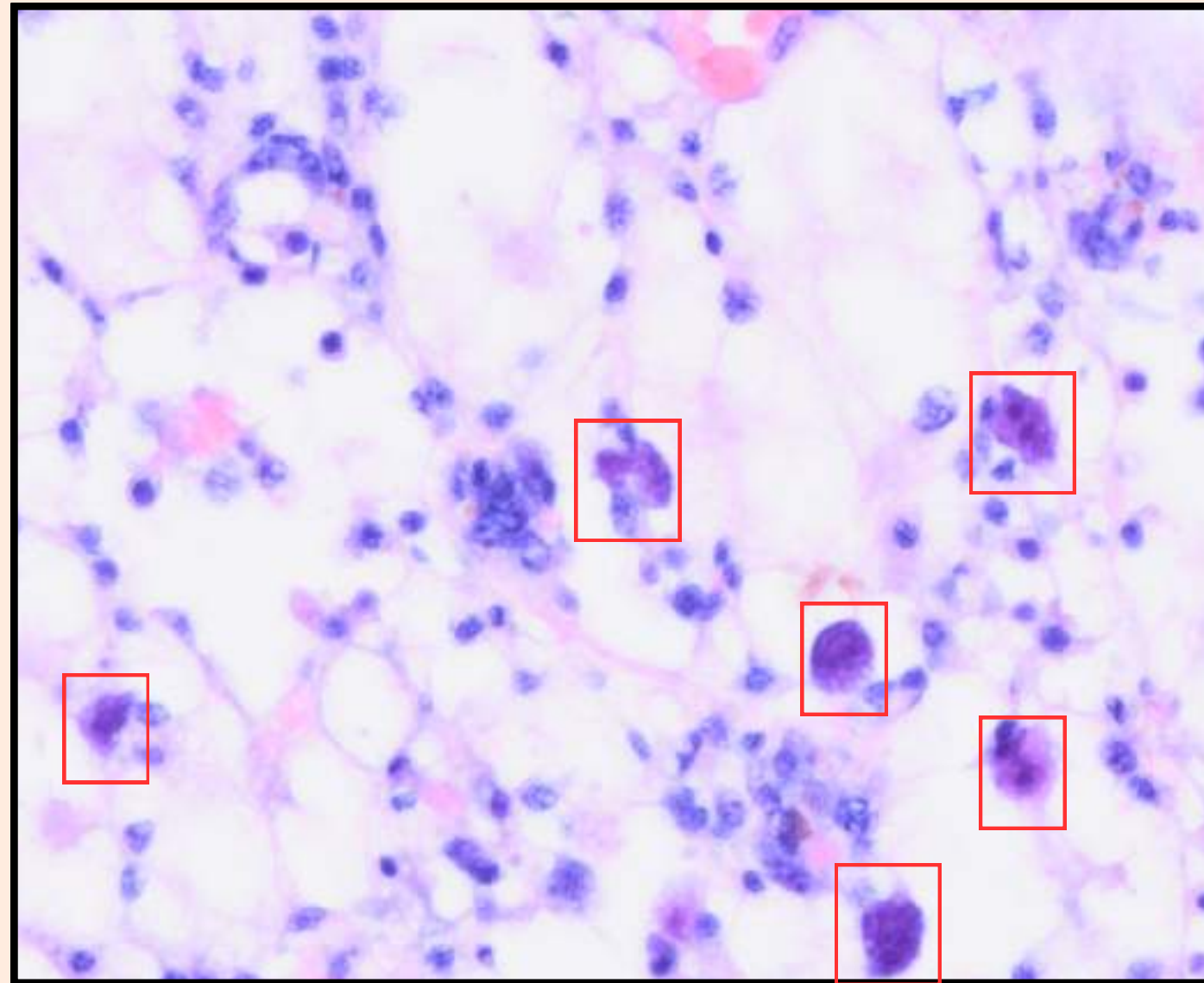
Healthy oyster WSI section



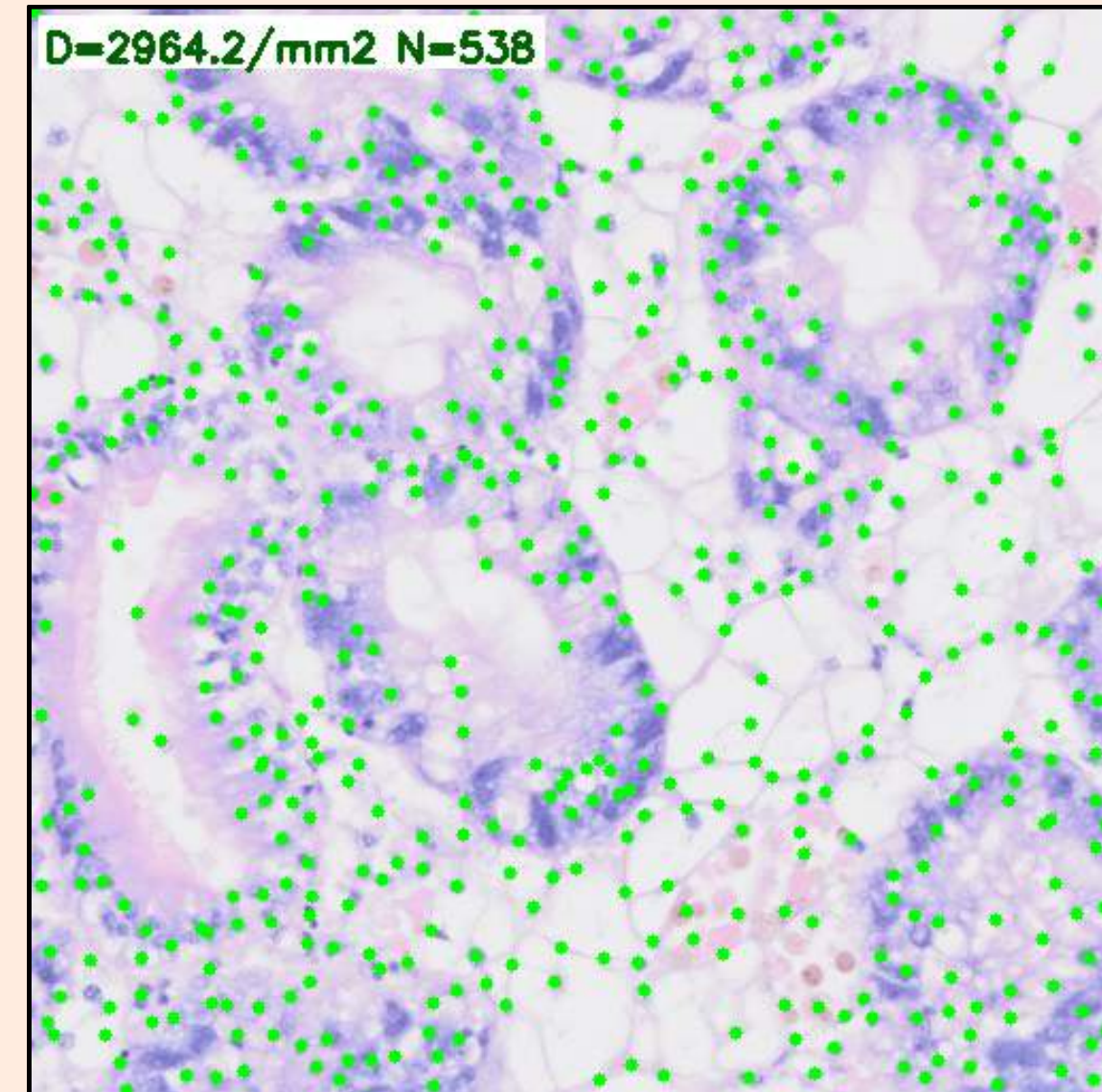
MSX-infected oyster WSI section

Tier 2: Multi-scale feature extraction

40x Parasite candidate detection



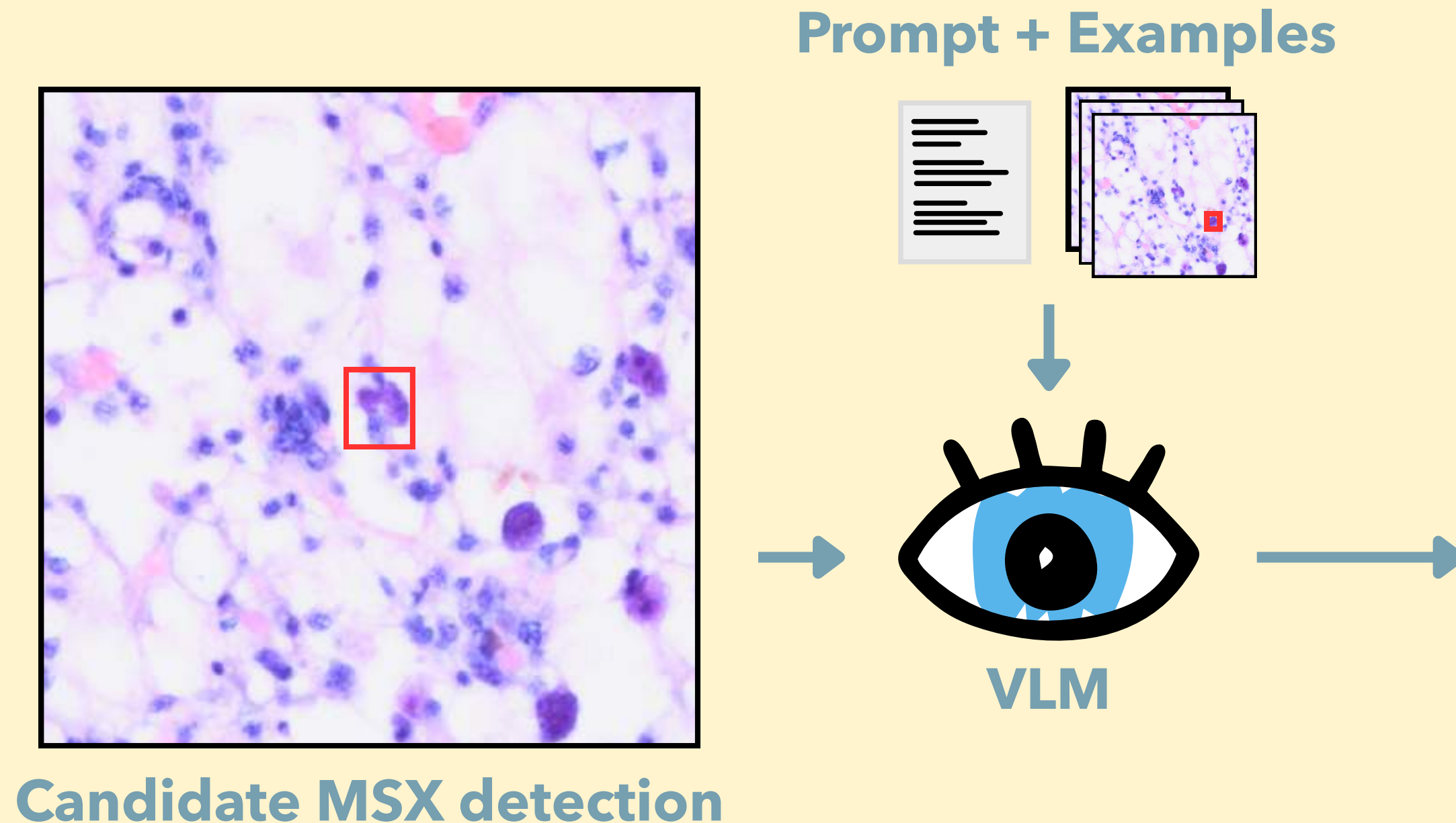
10x Hemocyte density biomarker



Local parasite evidence + broader inflammatory context

Tier 3: Reasoning and synthesis

Exploratory VLM confirmation



Tier 3: Reasoning

Exploratory VLM co

Prompt used

VLM_SYSTEM_PROMPT = ""You are a veterinary histopathologist analyzing 40x H&E oyster tissue for Haplosporidium nelsoni (MSX).

Analyze the image and return ONLY a JSON object with these exact fields:

```
{  
  "organism_present": <bool>,  
  "multinucleated": <bool>,  
  "nucleus_count": <int>,  
  "cytoplasm_texture": "<granular|smooth|vacuolated|artifact>",  
  "parasite_diameter_um": <float>,  
  "confidence": <float 0.0-1.0>  
}
```

DIAGNOSTIC CRITERIA:

- MSX plasmodia are 5-50um, multinucleated (2-20+ nuclei), granular eosinophilic cytoplasm
- Hemocytes are smaller (<10um), mononuclear – do NOT flag as MSX
- If no organism visible, set organism_present=false and other fields to defaults (0, "artifact", 0.0)

REFERENCE IMAGES:

{exemplar_section}

Return ONLY valid JSON. ""

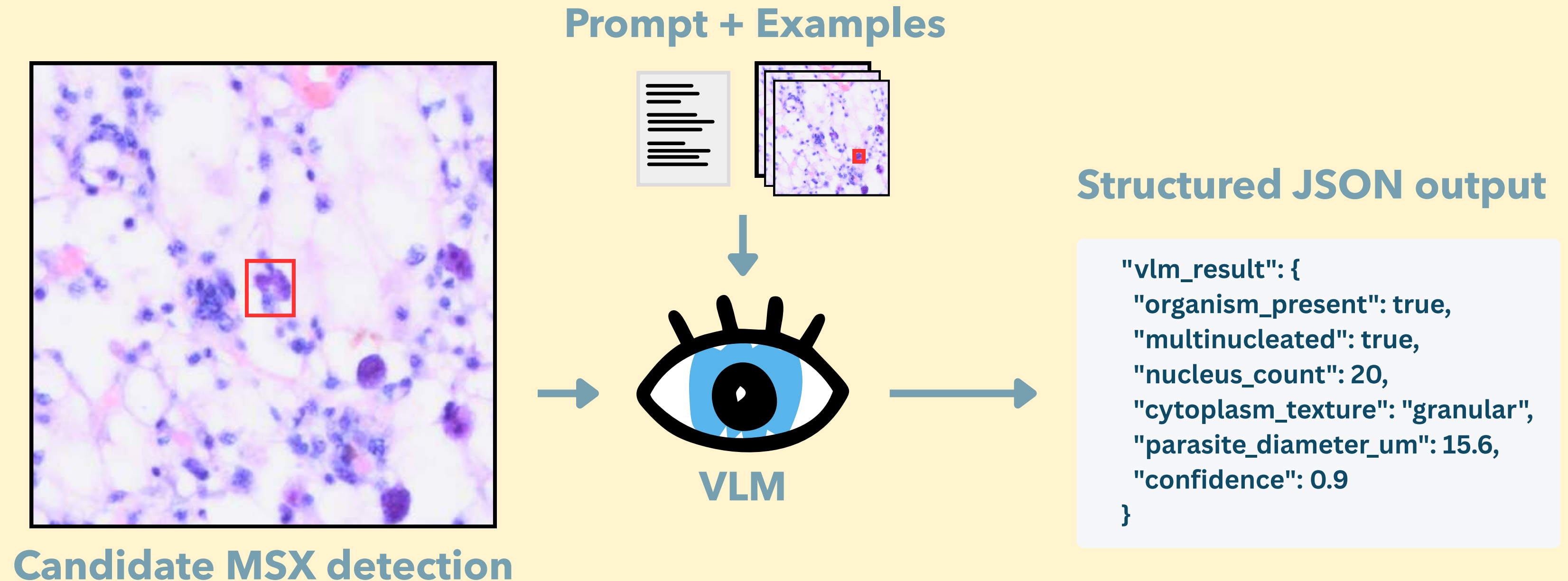
Model output

```
{  
  "organism_present": true,  
  "multinucleated": true,  
  "nucleus_count": 20,  
  "cytoplasm_texture": "granular",  
  "parasite_diameter_um": 15.6,  
  "confidence": 0.9  
}
```

Candida

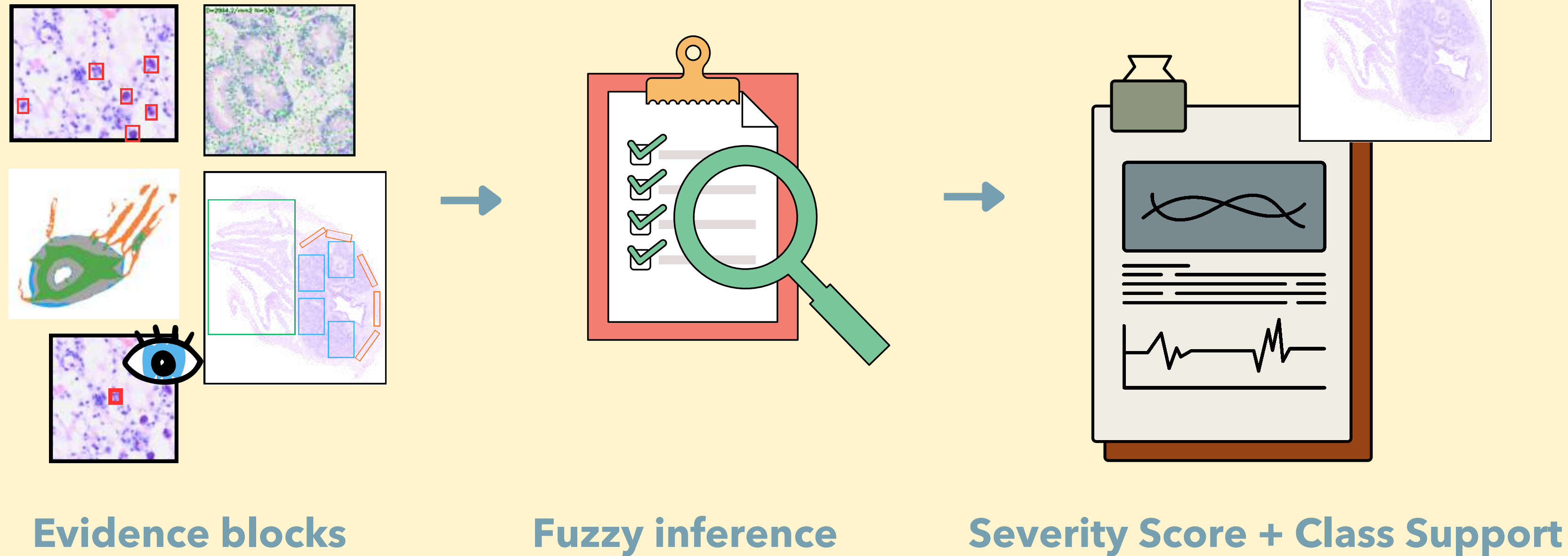
Tier 3: Reasoning and synthesis

Exploratory VLM confirmation



Tier 3: Reasoning and synthesis

Final severity synthesis

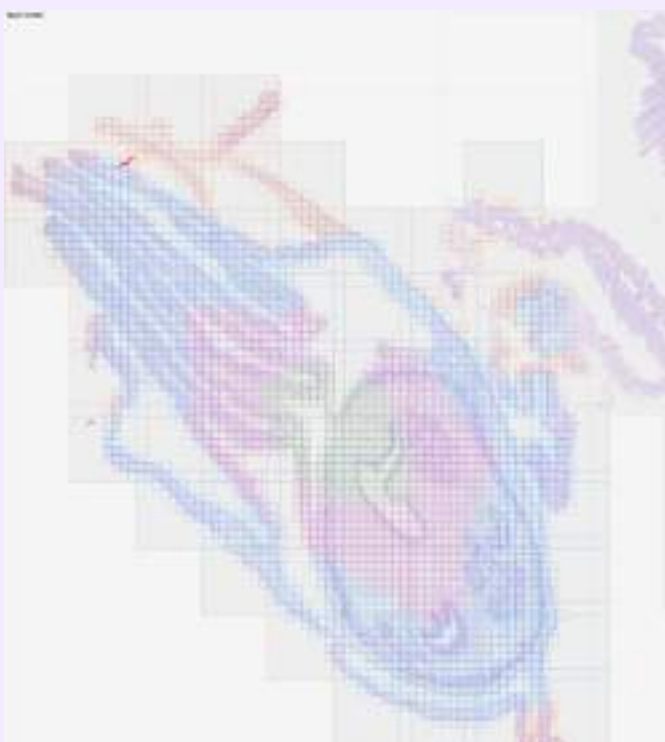


Output: Diagnostic audit trial

Oyster Overview



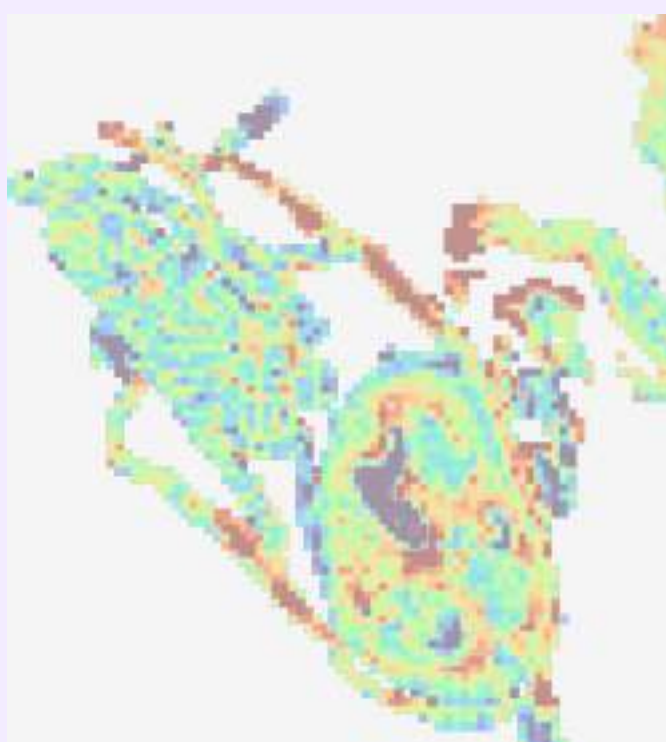
Approximate Organ Context



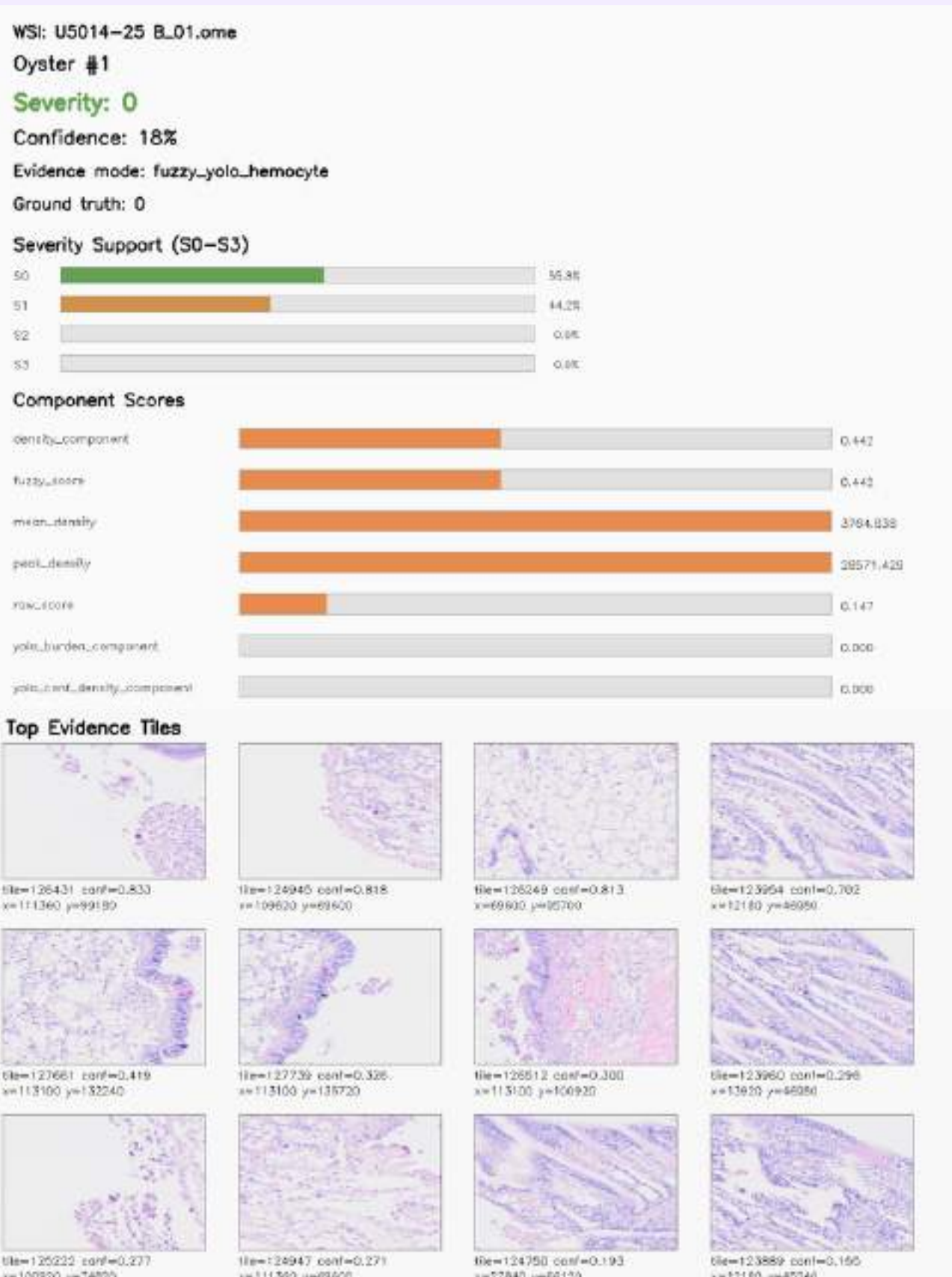
YOLO Detections



Hemocyste Density Heatmap



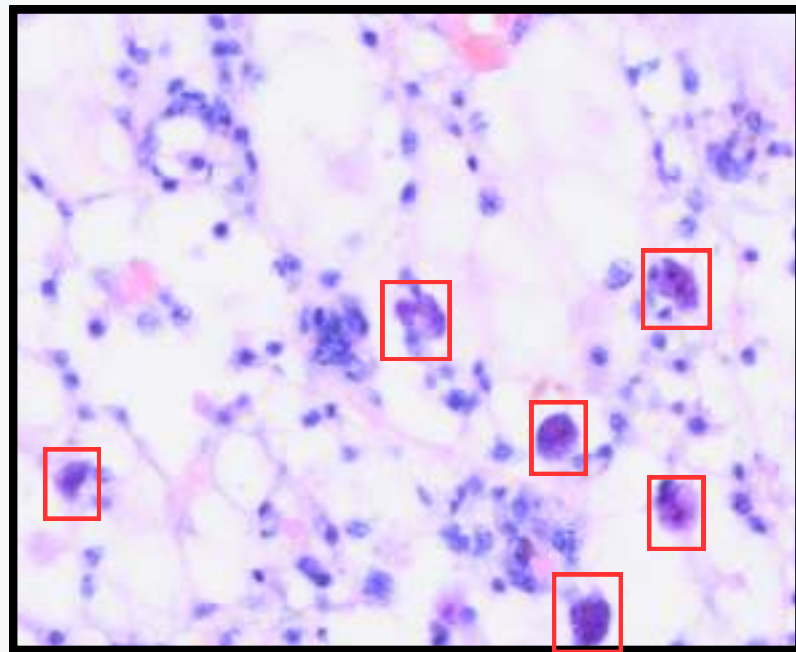
Oyster-level severity summary and evidence tiles



Results at a glance

Strongest

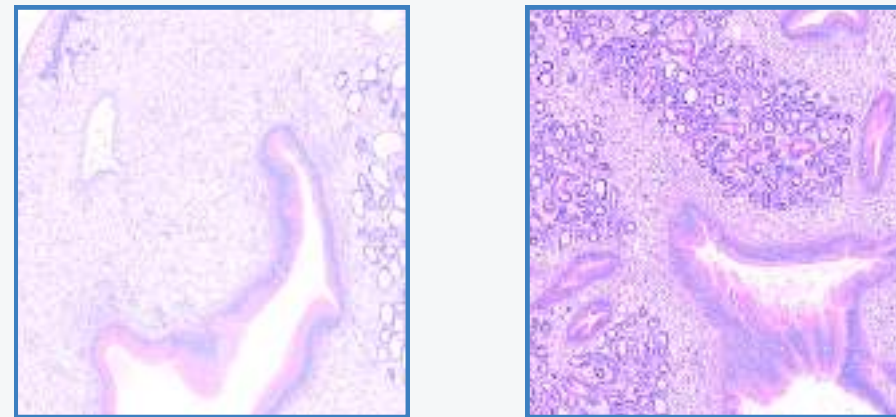
Parasite candidate localization



- YOLO detection performed strongly
- Low false-positive rate against distractors
- Best-validated component

Promising

Disease screening / triage



- Stronger binary disease / no-disease performance
- Useful for identifying abnormal oysters
- Supports review prioritization

Limited

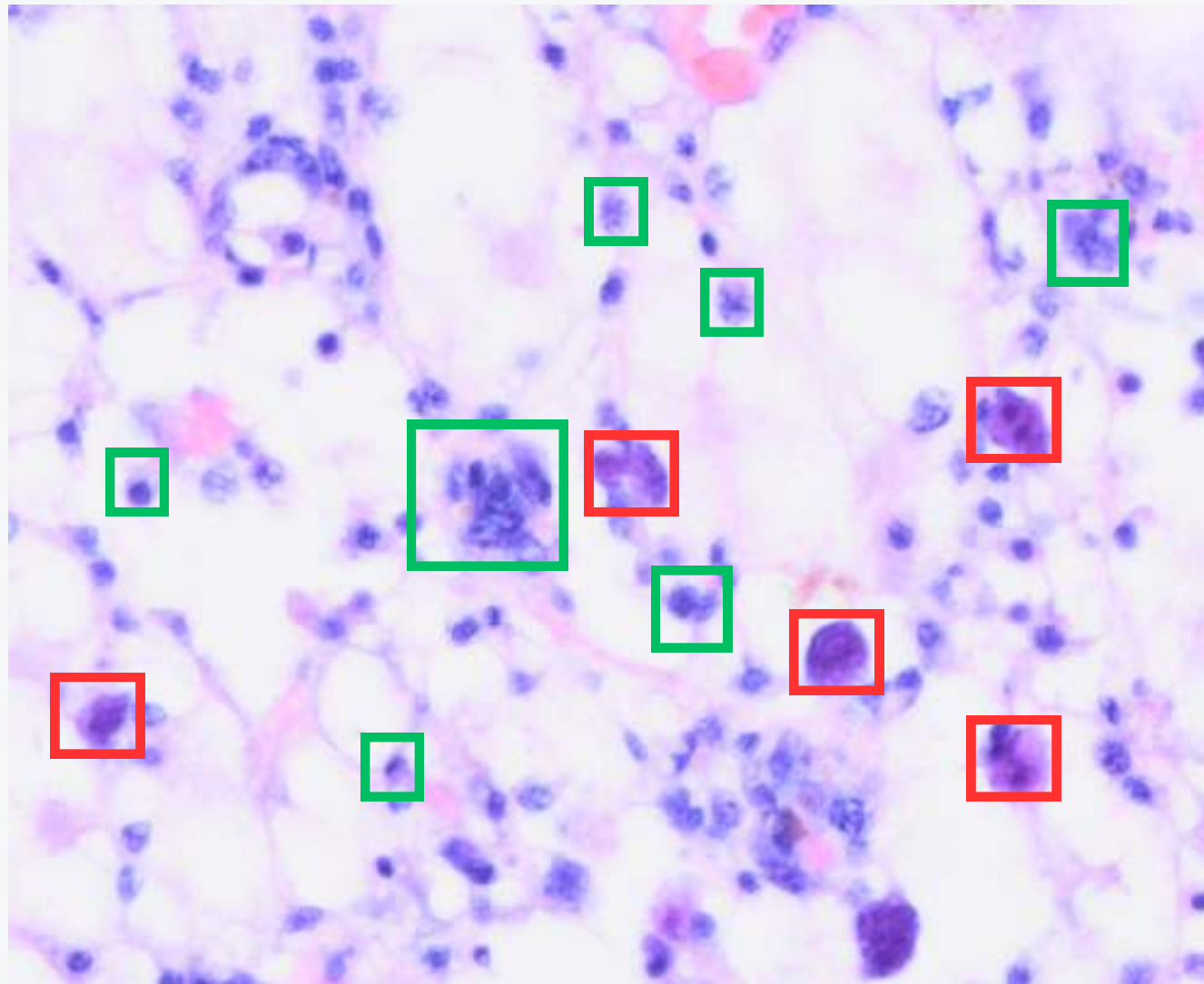
Fine-grained severity staging



- Severity 1–3 classes overlap
- Coarse anatomy likely limits grading
- Needs better organ mapping

Strongest result: parasite candidate localization

YOLO performed well as the first screening stage



Example 40x detection output

0.983

mAP@0.5

0.97

F1-score

119 / 127

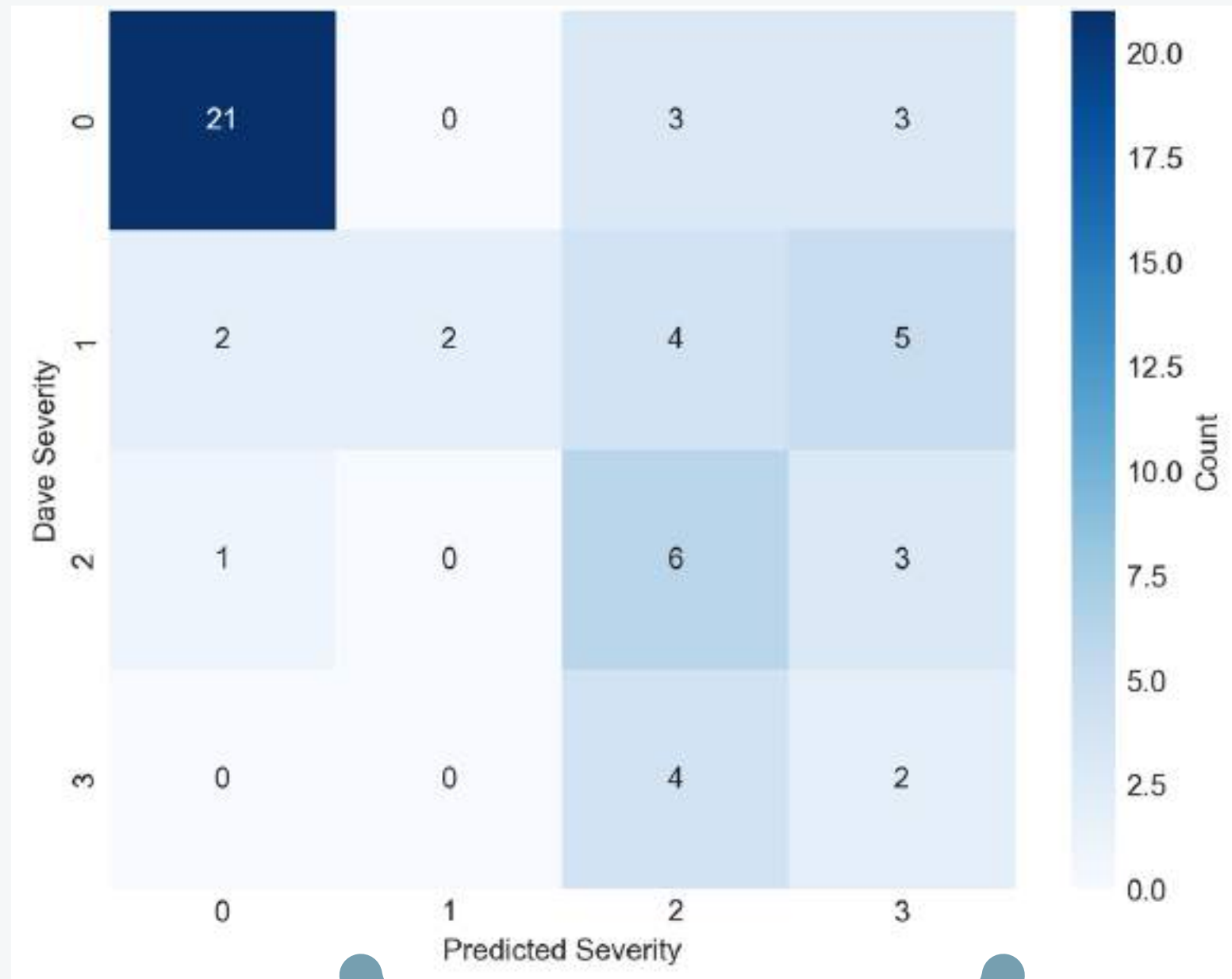
true plasmodia identified

4 / 1,432

distractors misclassified
as plasmodia

Oyster-level severity: screening > staging

Fine-grained severity remains difficult with coarse anatomical context



Blur among infected classes

4-class severity
grading

55.36%

Accuracy

0.71

MAE

0.452

Linear-weighted k

Disease / no-disease
screening

83.93%

Accuracy

89.66%

Sensitivity

77.78%

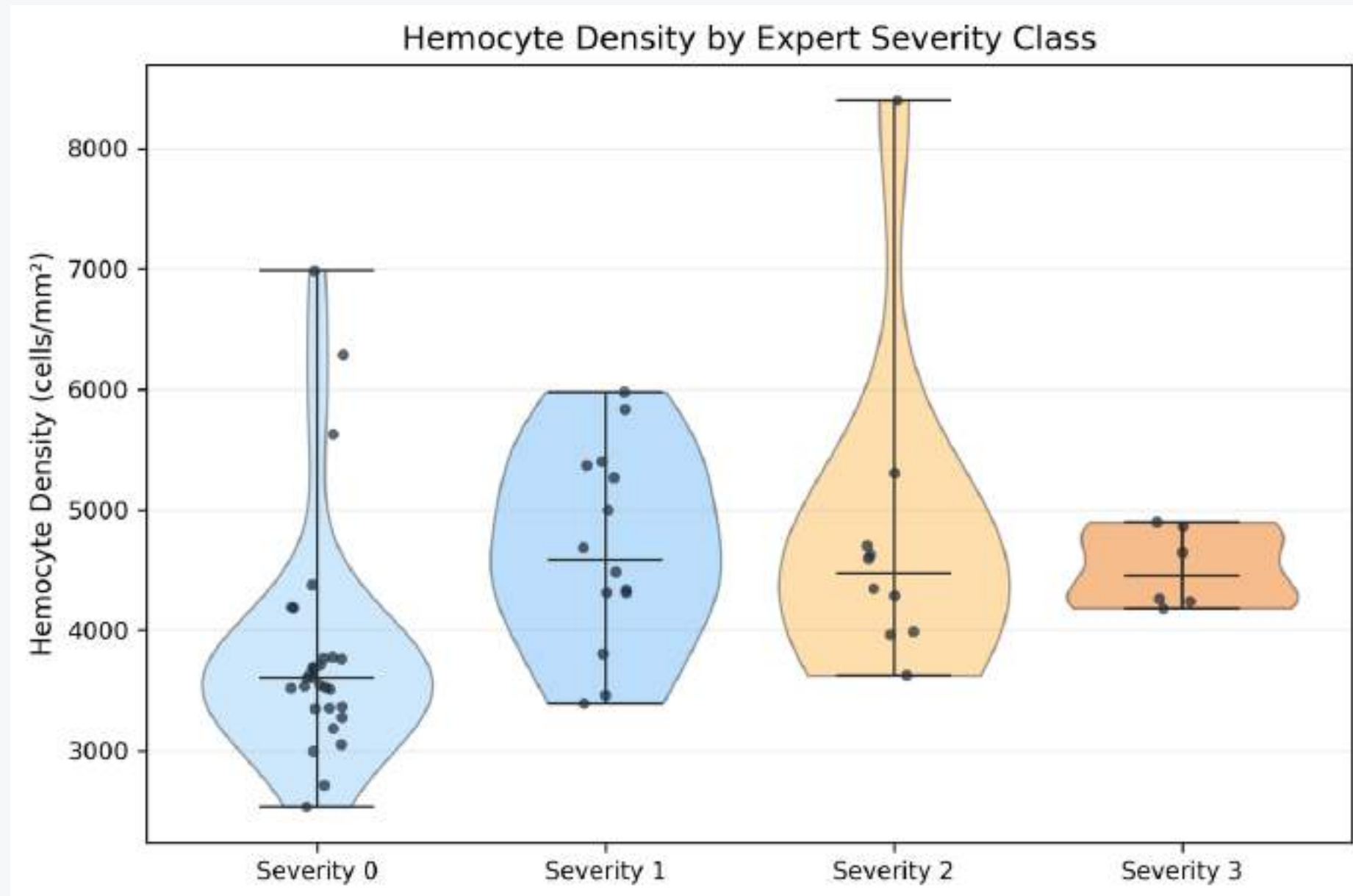
Specificity

85.25%

F1-score

Hemocyte density as a secondary biomarker

Useful for healthy vs infected separation, weaker for fine-grained severity



Mean density (cells/mm²)

- S0: **3,806** (n = 29)
- S1: **4,689** (n = 14)
- S2: **4,785** (n = 10)
- S3: **4,517** (n = 6)

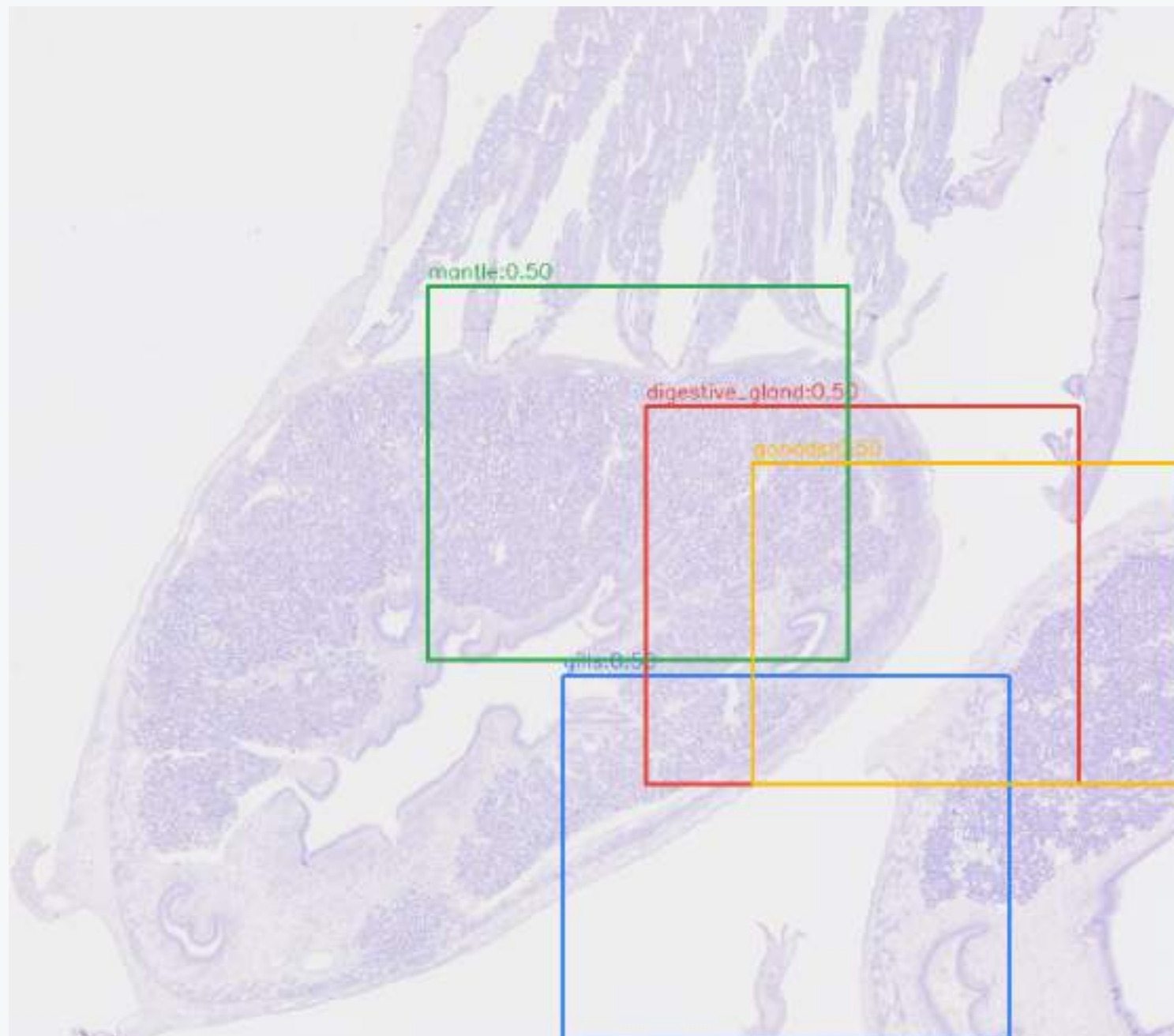
Pattern

- Lower in healthy oysters
- Higher in infected oysters
- Overlap across S1–S3

Exploratory VLM stages: useful, but not reliable enough

General-purpose VLM outputs were not consistently grounded in oyster histology

Organ-context mapping



- Mantle
- Digestive Gland
- Gills
- Gonads

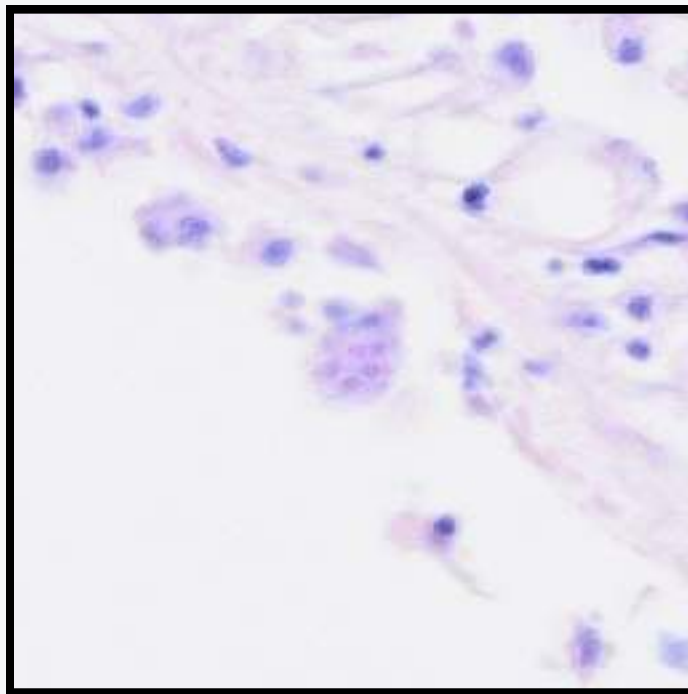
- Out-of-the-box VLM prompting
- Coarse and unstable organ boxes
- Weak anatomical grounding

Problem: weak anatomical grounding

Exploratory VLM stages: useful, but not reliable enough

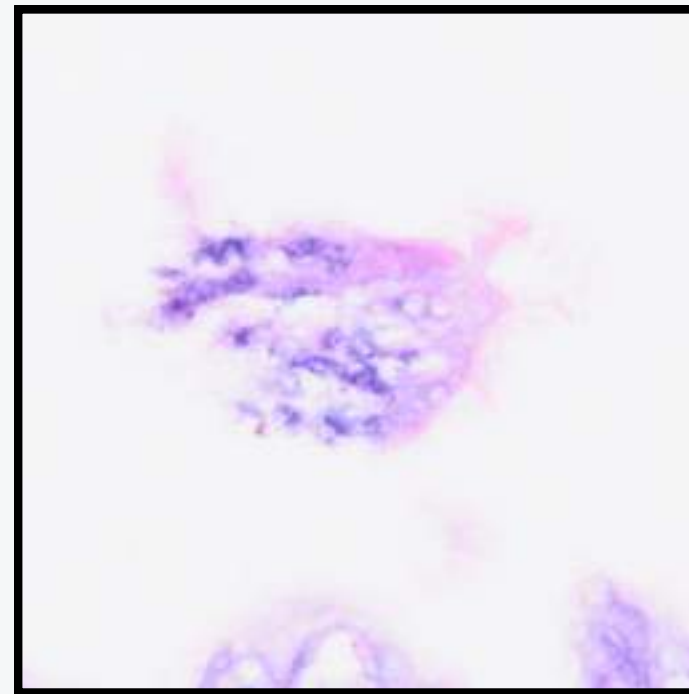
General-purpose VLM outputs were not consistently grounded in oyster histology

False Negative



```
"vlm_result": {  
  "organism_present": false,  
  "multinucleated": false,  
  "nucleus_count": 0,  
  "cytoplasm_texture": "artifact",  
  "parasite_diameter_um": 0.0,  
  "confidence": 1.0  
}
```

False Positive



```
"vlm_result": {  
  "organism_present": true,  
  "multinucleated": true,  
  "nucleus_count": 20,  
  "cytoplasm_texture": "granular",  
  "parasite_diameter_um": 15.6,  
  "confidence": 0.9  
}
```

- Candidate crops from YOLO detections
- Structured JSON-style outputs
- Over-specific features not always visually justified

Problem: weak morphology grounding

What the results mean

Strong support signals, but severity staging remains limited

What worked

- Scalable WSI processing
- Strong parasite localization
- Useful hemocyte biomarker signal
- Reviewable diagnostic support outputs

What limited performance

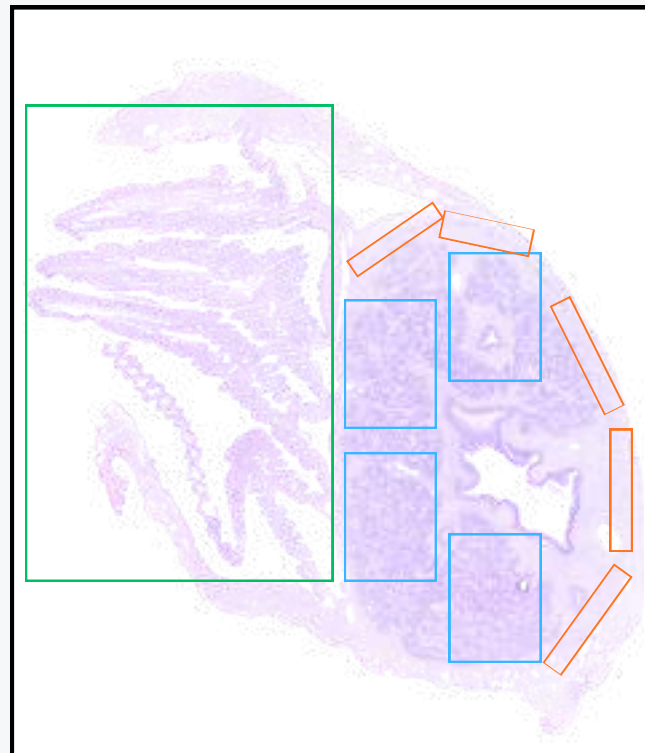
- Coarse anatomical context
- Weak separation among classes
- Small oyster-level dataset
- Unstable out-of-the-box VLM reasoning

Best current use: support triage and expert review

Future work

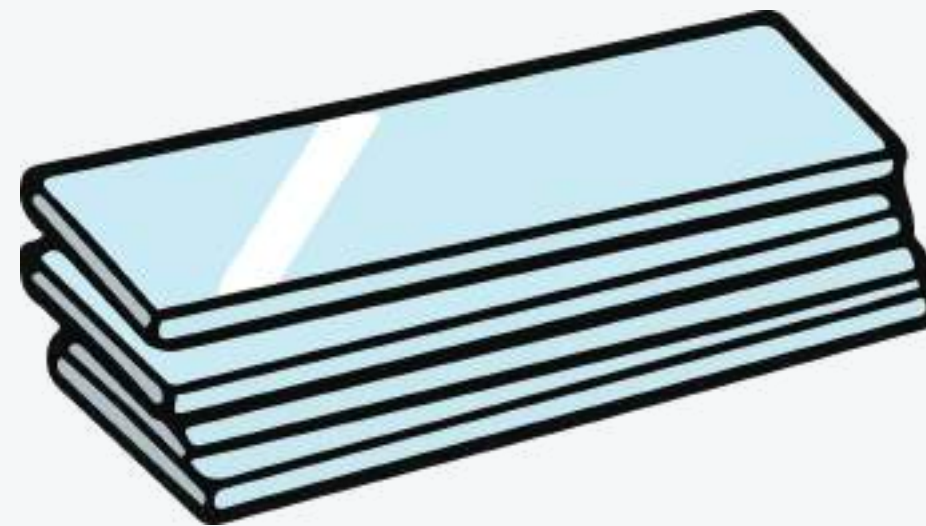
From proof-of-concept pipeline to stronger diagnostic support workflow

Better anatomical mapping



- Supervised organ segmentation
- More accurate spatial context
- Stronger severity staging

More oyster-level data



- More slides and specimens
- More natural variability
- Better severity calibration

Pathologist-facing interface



- Evidence overlays
- Candidate regions
- Structured summaries
- Expert in-the-loop review

Final takeaway

MSX-Sentinel is a proof-of-concept diagnostic support pipeline, not an autonomous diagnostic system.

An Anatomy-Informed Diagnostic Support Pipeline for MSX Severity Assessment in Oyster Histology

Proof-of-concept computational pathology for Eastern oyster whole-slide histology

Jonathan E. Zul Luna

Supervisors: Dr. Andrew Godbout, Dr. Kristina Kupferschmidt

Advisor: Dr. David Groman



April 2026